

EL ENG 226A

Random Processes in Systems

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Lecture 1

Probability Spaces

1.1 Elements of probability

Definition 1.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where

- Ω is the *sample space*, a set; an element $\omega \in \Omega$ is a *sample*;
- $\mathcal{F}$ is the set of *events*, a set of subsets of Ω (for instance $\mathcal{F} = \mathcal{P}(\Omega) = 2^\Omega$);
- $\mathbb{P}$ is a *probability measure*, $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$.

Definition 1.2 (σ -algebra). $\mathcal{F}$ is a σ -*algebra*, meaning

1. $\Omega \in \mathcal{F}$;
2. $\mathcal{F}$ is closed under complements: $A \in \mathcal{F} \implies A^c \in \mathcal{F}$;
3. $\mathcal{F}$ is closed under countable unions: $A_1, \dots, A_n, \dots \in \mathcal{F} \implies \bigcup_{i \geq 1} A_i \in \mathcal{F}$.

Definition 1.3 (Kolmogorov's axioms). A probability measure $\mathbb{P}$ satisfies, for every event $A \in \mathcal{F}$:

1. $\mathbb{P}(A) \geq 0$;
2. $\mathbb{P}(\Omega) = 1$;
3. if $A_1, A_2, \dots \in \mathcal{F}$ are disjoint, i.e. $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i) \quad (\sigma\text{-additivity}).$$

Remark 1.4. The prefix σ means “countable” in this context.

Remark 1.5. Countability in Axiom 3 is not decoration. If we tried to extend it to uncountable collections, take $X \sim \text{Unif}([0, 1])$: each singleton has probability 0, but $[0, 1]$ is an uncountable union of singletons and has probability 1, so the uncountable version of Axiom 3 fails. (Exercise.)

1.2 Discrete versus continuous sample spaces

Definition 1.6. Ω is *discrete* if it is countable.

Example 1.7. $\Omega = \{H, T\}$; $\Omega = \{1, \dots, 6\}$; $\Omega = \{0, 1, 2, \dots\}^2$.

Fact 1.8. For discrete Ω we often take $\mathcal{F} = 2^\Omega$, the discrete σ -algebra on Ω , and define $\mathbb{P}$ by specifying $\mathbb{P}(\{\omega\}) \in [0, 1]$ for every $\omega \in \Omega$ subject to

$$\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1.$$

Remark 1.9. Even for discrete Ω , $\mathcal{F} = 2^\Omega$ is not always the right choice.

Definition 1.10. If Ω is uncountable, we call it *continuous*.

Example 1.11. $\Omega = \mathbb{R}$, $[0, 1]$, $\dots$; $\Omega = \mathbb{R}^\mathbb{N}$.

Fact 1.12. Unlike the discrete case, one cannot construct a probability measure by assigning a probability to each sample. Instead $\mathbb{P}$ must be defined by assigning probabilities to events in such a way that Axioms 1–3 are satisfied.

Message. By judicious assignment of probabilities to sets $A \in \mathcal{G} \subset 2^\Omega$, we ensure that a probability measure compatible with those assignments is well defined on $(\Omega, \mathcal{F})$, where $\mathcal{F} = \sigma(\mathcal{G})$ is the smallest σ -algebra containing $\mathcal{G}$.

Example 1.13. Let $\Omega = [0, 1]$ and

$$\mathcal{G} = \{[0, x] : x \in [0, 1]\}, \quad p([0, x]) = x \quad \forall x \in [0, 1].$$

By the Carathéodory extension theorem there is a unique probability measure $\mathbb{P}$ on $(\Omega, \sigma(\mathcal{G}))$ with $\mathbb{P}(A) = p(A)$ for all $A \in \mathcal{G}$.

1.3 Basic properties of probability measures

Theorem 1.14. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, then $\mathbb{P}$ satisfies:

1. Monotonicity: if $A \subset B \in \mathcal{F}$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
2. Subadditivity: if $A_1, A_2, \dots \in \mathcal{F}$, then $\mathbb{P}(\bigcup_{i \geq 1} A_i) \leq \sum_{i \geq 1} \mathbb{P}(A_i)$.
3. Continuity from below: if $A_1 \subset A_2 \subset A_3 \subset \dots \in \mathcal{F}$ and $A = \bigcup_{i \geq 1} A_i$, then $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.
4. Continuity from above: if $A_1 \supset A_2 \supset A_3 \supset \dots \in \mathcal{F}$ and $A = \bigcap_{i \geq 1} A_i$, then $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.

Proof. (1) By Axiom 1 and then Axiom 3 applied to the disjoint pair $A, A^c \cap B$,

$$\mathbb{P}(A) \leq \mathbb{P}(A) + \mathbb{P}(A^c \cap B) = \mathbb{P}(B).$$

(2) Disjointify: set

$$E_1 = A_1, \quad E_i = A_i \cap \left(\bigcup_{j < i} A_j \right)^c \quad \forall i \geq 2.$$

Then the E_i are disjoint, $E_i \cap E_j = \emptyset$ for $i \neq j$, and $\bigcup_{i=1}^j E_i = \bigcup_{i=1}^j A_i$. Hence

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i \geq 1} E_i\right) = \sum_{i \geq 1} \mathbb{P}(E_i) \leq \sum_{i \geq 1} \mathbb{P}(A_i),$$

using $E_i \subset A_i$ and monotonicity.

(3) Define E_i as in (2). Then

$$\mathbb{P}(A) = \sum_{i \geq 1} \mathbb{P}(E_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(E_i) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

(4) Similar to (3), applied to the complements. □

1.4 Borel–Cantelli

Example 1.15 (Limsup and liminf of events). Given a collection of events $A_1, A_2, \dots \in \mathcal{F}$, define

$$\begin{aligned} \{A_i \text{ i.o.}\} &= \{A_i \text{ infinitely often}\} = \bigcap_{n \geq 1} \bigcup_{i \geq n} A_i \in \mathcal{F}, \\ \{A_i \text{ f.o.}\} &= \{A_i \text{ finitely often}\} = \{A_i \text{ i.o.}\}^c. \end{aligned}$$

Theorem 1.16 (Borel–Cantelli lemma). *Let $A_1, A_2, \dots \in \mathcal{F}$. If $\sum_i \mathbb{P}(A_i) < \infty$, then*

$$\mathbb{P}(\{A_i \text{ i.o.}\}) = 0.$$

Proof. Notice that $(\bigcup_{i \geq n} A_i)_{n \geq 1}$ is a decreasing sequence, so continuity from above applies:

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}\left(\bigcap_{n \geq 1} \bigcup_{i \geq n} A_i\right) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i \geq n} A_i\right) \leq \lim_{n \rightarrow \infty} \sum_{i \geq n} \mathbb{P}(A_i).$$

Since $\sum_i \mathbb{P}(A_i) < \infty$, the tail sums satisfy $\sum_{i \geq n} \mathbb{P}(A_i) \rightarrow 0$, and therefore $\mathbb{P}(A_n \text{ i.o.}) = 0$. □

Lecture 2

Borel–Cantelli, Random Variables, and Distributions

Recall: $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. Generally $(\Omega, \mathcal{F}, \mathbb{P})$ is handed to us, but we can also construct one from a reasonable model.

2.1 The Borel–Cantelli lemmas

Notation. For events $A_1, A_2, \dots$, write

$$\{A_n \text{ i.o.}\} := \bigcap_{i \geq 1} \bigcup_{n \geq i} A_n$$

for the event that infinitely many of the A_n occur.

Lemma 2.1 (Borel–Cantelli I). *If $A_1, A_2, \dots \in \mathcal{F}$ and $\sum_{i \geq 1} \mathbb{P}(A_i) < \infty$, then $\mathbb{P}(\{A_n \text{ i.o.}\}) = 0$.*

Proof. Set $B_n := \bigcup_{i \geq n} A_i$. These are nested downward,

$$B_1 \supset B_2 \supset \dots,$$

so continuity from above applies:

$$\mathbb{P}\left(\bigcap_{i \geq 1} \bigcup_{n \geq i} A_n\right) = \mathbb{P}\left(\bigcap_{n \geq 1} B_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i \geq n} A_i\right) \leq \lim_{n \rightarrow \infty} \sum_{i \geq n} \mathbb{P}(A_i) = 0.$$

□

Remark 2.2. The last step is just the statement that the tails of a convergent series vanish. If $\sum_{i \geq 1} \mathbb{P}(A_i) = S < \infty$, then for every $\varepsilon > 0$ there is $N > 0$ with

$$n \geq N \implies \left| \sum_{i < n} \mathbb{P}(A_i) - S \right| < \varepsilon, \quad \text{equivalently} \quad \sum_{i \geq n} \mathbb{P}(A_i) < \varepsilon.$$

Definition 2.3 (Independent events). Events $(A_i)_{i \geq 1}$ are *independent* if for every finite set of indices S ,

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i).$$

Lemma 2.4 (Homework). If $(A_i)_{i \geq 1}$ are independent, then so are $A_1^c, A_2, A_3, \dots$, and hence so are $A_1^c, A_2^c, \dots$.

Lemma 2.5 (Borel–Cantelli II). If $(A_i)_{i \geq 1}$ are independent events and $\sum_{i \geq 1} \mathbb{P}(A_i) = \infty$, then

$$\mathbb{P}(\{A_i \text{ i.o.}\}) = 1.$$

Proof. Using Lemma 2.4 and $1 - x \leq e^{-x}$,

$$\begin{aligned} \mathbb{P}(\{A_i \text{ i.o.}\}) &= \mathbb{P}\left(\bigcap_{i \geq 1} \bigcup_{n \geq i} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i \geq n} A_i\right) \\ &= \lim_{n \rightarrow \infty} \left(1 - \mathbb{P}\left(\bigcap_{i \geq n} A_i^c\right)\right) \\ &= \lim_{n \rightarrow \infty} \left(1 - \prod_{i \geq n} \mathbb{P}(A_i^c)\right) = \lim_{n \rightarrow \infty} \left(1 - \prod_{i \geq n} (1 - \mathbb{P}(A_i))\right) \\ &\geq \lim_{n \rightarrow \infty} \left(1 - e^{-\sum_{i \geq n} \mathbb{P}(A_i)}\right) = 1. \end{aligned}$$

□

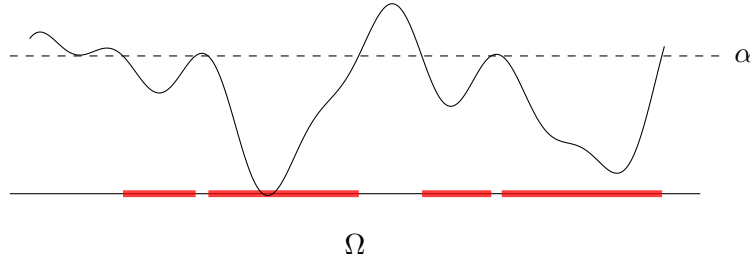
2.2 Random variables

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 2.6 (Random variable). A *random variable* is a function $X: \Omega \rightarrow \mathbb{R} \cup \{-\infty, \infty\} = \overline{\mathbb{R}}$ such that

$$\{\omega : X(\omega) \leq \alpha\} \in \mathcal{F} \quad \forall \alpha \in \overline{\mathbb{R}}, \quad (*)$$

i.e. X is $\mathcal{F}$ -measurable.



$$\{\omega : X(\omega) \leq \alpha\} \in \mathcal{F}$$

Two consequences of (*):

1. It lets us compute $\mathbb{P}(X \leq \alpha) \equiv \mathbb{P}(\{\omega : X(\omega) \leq \alpha\})$.
2. It further implies

$$\underbrace{X^{-1}(B)}_{\{\omega : X(\omega) \in B\}} \in \mathcal{F} \quad \forall B \in \sigma(\{[-\infty, \alpha] : \alpha \in \overline{\mathbb{R}}\}),$$

i.e. we can compute $\mathbb{P}(X \in B)$ for every Borel set B .

Example 2.7. $\{\omega : X(\omega) > \alpha\} \in \mathcal{F}$, being the complement of $\{X \leq \alpha\}$, and

$$\{\omega : X(\omega) \geq \alpha\} = \bigcap_{n \geq 1} \{\omega : X(\omega) > \alpha - \frac{1}{n}\} \in \mathcal{F}.$$

Remark 2.8. Random variables are well behaved under algebraic operations, limits, and so on. The next two propositions make this precise.

Definition 2.9. X is a *real-valued* random variable if $\mathbb{P}\{|X| < \infty\} = 1$.

Proposition 2.10. If X, Y are real-valued random variables, then

1. $|X|^p$ and ρX are random variables, for every $p > 0$ and $\rho \in \mathbb{R}$;
2. $X + Y$ is a random variable;
3. XY is a random variable.

Proof. (1) Exercise.

(2). Decompose over a rational cut point:

$$\{\omega : X(\omega) + Y(\omega) > \alpha\} = \bigcup_{q \in \mathbb{Q}} \left(\underbrace{\{X(\omega) > q\}}_{\in \mathcal{F}} \cap \underbrace{\{Y(\omega) > \alpha - q\}}_{\in \mathcal{F}} \right),$$

and $\mathcal{F}$ is closed under countable unions, so the left-hand side is in $\mathcal{F}$.

(3). Polarize:

$$XY = \frac{1}{4} \left((X + Y)^2 - (X - Y)^2 \right).$$

Here $X + Y$ and $X - Y$ are random variables by (2) — the second using $\rho = -1$ from (1) — then $(X + Y)^2$ and $(X - Y)^2$ are random variables by (1), their difference is one by (1) and (2), and finally the factor $\frac{1}{4}$ is handled by (1). $\square$

Q. Why $\mathbb{Q}$ in the proof of (2)?

A. $\overline{\mathbb{Q}} = \mathbb{R}$, so we can get within $\varepsilon > 0$ of every real number while keeping the union countable.

Remark 2.11. Part (1) is stated for $p > 0$. For $p < 0$ the function $|X|^p$ is only defined where $X \neq 0$, so one needs $\mathbb{P}\{X = 0\} = 0$ (or a convention at 0) for the statement to make sense.

Proposition 2.12. *If $(X_n)_{n \geq 1}$ is a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, then*

1. $\sup_{n \geq 1} X_n$ and $\inf_{n \geq 1} X_n$ are random variables;
2. $\limsup_{n \rightarrow \infty} X_n$ and $\liminf_{n \rightarrow \infty} X_n$ are random variables;
3. if $\lim_{n \rightarrow \infty} X_n(\omega)$ exists for every $\omega \in \Omega$, then $\lim_{n \rightarrow \infty} X_n$ is a random variable.

Proof. Take $Y(\omega) := \sup_{n \geq 1} X_n(\omega)$. We need to show that for $\alpha \in \overline{\mathbb{R}}$ we have $\{\omega : Y(\omega) \leq \alpha\} \in \mathcal{F}$:

$$\{\omega : Y(\omega) \leq \alpha\} = \{\omega : X_n(\omega) \leq \alpha \ \forall n \geq 1\} = \bigcap_{n \geq 1} \{\omega : X_n(\omega) \leq \alpha\} \in \mathcal{F}.$$

The remaining parts follow by writing $\inf$, $\limsup$, and $\liminf$ in terms of $\sup$ and $\inf$ over tails. $\square$

2.3 Almost sure equivalence

Definition 2.13. If X, Y are random variables and $\mathbb{P}\{\omega : X(\omega) = Y(\omega)\} = 1$, then $X = Y$ *almost surely*, written $X = Y$ a.s.

Almost surely $\iff$ with probability 1.

In particular, it may happen that $\lim_{n \rightarrow \infty} X_n(\omega)$ fails to exist for some $\omega \in \Omega$ while

$$\mathbb{P}\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) \text{ exists}\} = 1.$$

In that case $\lim_{n \rightarrow \infty} X_n$ is still well defined in an a.s. sense.

Example 2.14. Let $\mathbb{P} \sim \text{Unif}[0, 1]$ and let X be a random variable. Define

$$X_n(\omega) = \begin{cases} (-1)^n & \text{if } \omega \in \mathbb{Q}, \\ X(\omega) + \frac{1}{n} & \text{if } \omega \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

The limit fails to exist on $\mathbb{Q} \cap [0, 1]$, but that set has probability 0, so $X_n \xrightarrow{\text{a.s.}} X$.

2.4 Distributions as probability measures

Associated to any random variable is a *distribution*. It is a probability measure on $(\overline{\mathbb{R}}, \mathcal{B})$, where $\mathcal{B}$ is the Borel σ -algebra.

Definition 2.15 (Law of a random variable). For a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, the distribution (or *law*) of X is the probability measure

$$\mathcal{L}_X(B) := \mathbb{P}\{X \in B\} \quad \forall B \in \mathcal{B}.$$

A distribution $\mathcal{L}_X$ is characterized by its distribution function F_X :

$$F_X(x) = \mathcal{L}_X([-\infty, x]) = \mathbb{P}(X \leq x) \quad \forall x \in \mathbb{R}.$$

Lecture 3

More on Random Variables and Expectation

Recap. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A map $X: \Omega \rightarrow \overline{\mathbb{R}}$ is a *random variable* if it is $\mathcal{F}$ -measurable, i.e. $\{X \in B\} \in \mathcal{F}$ for every reasonable Borel set $B \subset \overline{\mathbb{R}}$. Attached to X are two objects:

$$\begin{aligned} L_X(B) &:= \mathbb{P}(X \in B), & \text{a probability measure on } (\overline{\mathbb{R}}, \mathcal{B}); \\ F_X(x) &= \mathbb{P}\{X \leq x\}, & \text{the distribution function of } X. \end{aligned}$$

3.1 Distribution functions

Theorem 3.1. *A function $F: \mathbb{R} \rightarrow [0, 1]$ is the distribution function of a random variable if and only if*

1. *F is non-decreasing;*
2. *F is right-continuous, i.e. $\lim_{y \downarrow x} F(y) = F(x)$.*

Proof. ($\Rightarrow$) Suppose F is the distribution function of a random variable X .

1. For $x \leq y$ we have the inclusion $\{\omega : X(\omega) \leq x\} \subset \{\omega : X(\omega) \leq y\}$, and monotonicity of $\mathbb{P}$ gives $F(x) \leq F(y)$.

2. Take any sequence $(x_n)_{n \geq 1}$ with $x_n \downarrow x$ (so $x_{n+1} \leq x_n$, approaching x from the right). Setting $A_n = \{\omega : X(\omega) \leq x_n\}$, we get $A_{n+1} \subset A_n$ for all n and

$$\bigcap_{n \geq 1} A_n = \{\omega : X(\omega) \leq x\}.$$

Continuity of $\mathbb{P}$ from above now gives $F(x_n) = \mathbb{P}(A_n) \rightarrow \mathbb{P}(\bigcap_n A_n) = F(x)$. Since the sequence was arbitrary, F is right-continuous.

($\Leftarrow$) Let $\mathbb{P} = \text{Unif}([0, 1])$ on $\Omega = [0, 1]$ and define

$$X(\omega) = \sup\{y \in \mathbb{R} : F(y) < \omega\},$$

with the conventions $\sup \emptyset = -\infty$ and $\sup \mathbb{R} = +\infty$. One can show $F_X(x) = F(x)$. (*Exercise.*) $\square$

Remark 3.2. No boundary conditions appear in Theorem 3.1, which is correct *because we allow $\overline{\mathbb{R}}$ -valued random variables*. The construction above puts mass $\lim_{x \rightarrow -\infty} F(x)$ at $-\infty$ and mass $1 - \lim_{x \rightarrow +\infty} F(x)$ at $+\infty$. If one insists that X be real-valued almost surely, the two extra conditions

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1$$

must be added.

Remark 3.3 (Checking the exercise). The identity behind it is

$$X(\omega) \leq x \iff F(y) \geq \omega \quad \forall y > x \iff \omega \leq \lim_{y \downarrow x} F(y) = F(x),$$

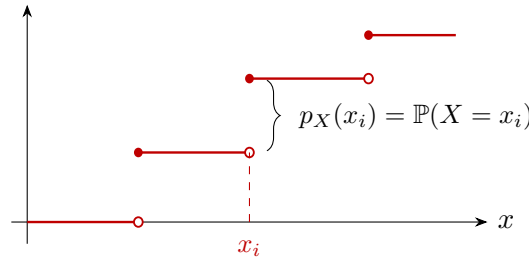
where the last step is exactly right-continuity. Hence $\mathbb{P}\{X \leq x\} = \mathbb{P}\{\omega \leq F(x)\} = F(x)$.

3.2 Two special cases

In undergraduate probability one learns about discrete and continuous random variables. It turns out there are other random variables, but these two cases cover most of what we need.

Definition 3.4 (Discrete random variable). X is a *DRV* if its distribution is discrete, i.e. F_X is a step function supported on some countable set $(x_i)_{i \geq 1} \subset \overline{\mathbb{R}}$:

$$F_X(x) := \sum_{\substack{i \geq 1 \\ x_i \leq x}} \mathbb{P}(X = x_i) = \sum_{i \geq 1} p_X(x_i) \mathbf{1}_{[x_i, +\infty)}(x), \quad p_X(x_i) := \mathbb{P}(X = x_i).$$



Definition 3.5 (Continuous random variable). X is a *CRV* if its distribution function F_X is absolutely continuous, i.e. there exists f_X — the *density* of X — with

$$F_X(x) = \int_{-\infty}^x f_X(u) du \quad \forall x \in \mathbb{R}.$$

3.3 Random vectors

Definition 3.6 (Random vector). A *random vector* is a collection of random variables on the same probability space: $X = (X_1, \dots, X_n)$ on $(\Omega, \mathcal{F}, \mathbb{P})$. Its joint distribution function is

$$F_X(x_1, \dots, x_n) = \mathbb{P}\{X_1 \leq x_1, \dots, X_n \leq x_n\} = \mathbb{P}\left(\bigcap_{i=1}^n \{\omega : X_i(\omega) \leq x_i\}\right).$$

The law of a random vector is $L_X(B) = \mathbb{P}(X \in B)$ for

$$B \in \mathcal{B}_{\overline{\mathbb{R}}^n} = \sigma\left(\prod_{i=1}^n [-\infty, x_i] : x_i \in \overline{\mathbb{R}}, i \in \{1, \dots, n\}\right).$$

Marginalization

If the X_i are real-valued (equivalently, $\mathbb{P}(X_i = \pm\infty) = 0$), then

$$F_{X_1, \dots, X_{n-1}}(x_1, \dots, x_{n-1}) = \lim_{x_n \rightarrow \infty} F_{X_1, \dots, X_n}(x_1, \dots, x_n).$$

Remark 3.7. Finiteness is needed here. For a genuinely $\overline{\mathbb{R}}$ -valued X_n the limit computes

$$\mathbb{P}\{X_1 \leq x_1, \dots, X_{n-1} \leq x_{n-1}, X_n < \infty\},$$

which undershoots the marginal by exactly the mass X_n places at $+\infty$.

Independence

Definition 3.8 (Independence). A collection $(X_i)_{i=1}^n$ is *independent* if

$$\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \prod_{i=1}^n \mathbb{P}(X_i \in B_i) \quad \forall (B_i)_{i=1}^n \subset \mathcal{B},$$

equivalently

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n \mathbb{P}(X_i \leq x_i) = \prod_{i=1}^n F_{X_i}(x_i).$$

Definition 3.9. A collection $(X_\alpha)_{\alpha \in A}$ of random variables — such a collection is what we will call a *random process* — is independent if all finite subcollections are independent.

3.4 Expectation

From undergraduate probability, a DRV X has $\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i)$. We now build $\mathbb{E}$ from the ground up, in three steps.

Step 1: indicators

Definition 3.10. The *indicator* random variable of $A \in \mathcal{F}$ is

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A, \end{cases} \quad \text{and we define} \quad \mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A).$$

Step 2: simple random variables

Simple random variables are finite linear combinations of indicators. Let

$$X = \sum_{i=1}^n a_i \mathbf{1}_{A_i}, \quad (a_i) \subset \mathbb{R}, (A_i) \subset \mathcal{F}.$$

We define

$$\mathbb{E}(X) = \sum_{i=1}^n a_i \mathbb{E}[\mathbf{1}_{A_i}] = \sum_{i=1}^n a_i \mathbb{P}(A_i).$$

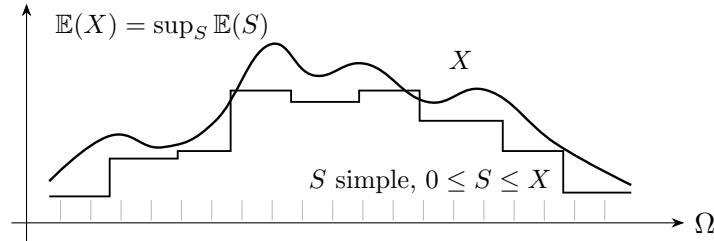
Remark 3.11. This is well defined, in the sense that

$$\sum_{i=1}^n a_i \mathbb{P}(A_i) = \sum_{j=1}^m b_j \mathbb{P}(B_j) \quad \text{whenever} \quad \sum_i a_i \mathbf{1}_{A_i} = \sum_j b_j \mathbf{1}_{B_j}.$$

Step 3: non-negative random variables

Definition 3.12. Let $X \geq 0$ be a random variable. We define

$$\mathbb{E}(X) = \sup\{\mathbb{E}(S) : 0 \leq S \leq X, S \text{ a simple random variable}\}.$$



This seems abstract, but it is the right definition.

Consequences of the definition

Lemma 3.13 (Fatou's lemma). Let $(X_n)_{n \geq 1}$ be non-negative random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\liminf_{n \rightarrow \infty} \mathbb{E}(X_n) \geq \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n\right).$$

Proposition 3.14. Let $X, Y \geq 0$ be random variables.

1. If $X = Y$ almost surely, then $\mathbb{E}(X) = \mathbb{E}(Y)$. (uniqueness)
2. If $X \leq Y$, then $\mathbb{E}(X) \leq \mathbb{E}(Y)$. (monotonicity)

Proof. ② If $X \leq Y$ then every simple S with $0 \leq S \leq X$ also satisfies $0 \leq S \leq Y$, so the supremum defining $\mathbb{E}(Y)$ is over a larger set:

$$\mathbb{E}(X) = \sup\{\mathbb{E}(S) : S \text{ simple}, S \in [0, X]\} \leq \mathbb{E}(Y).$$

① Define $E = \{X \neq Y\}$, so $\mathbb{P}(E) = 0$. For any simple $S = \sum_i a_i \mathbf{1}_{A_i}$ with $0 \leq S \leq X$,

$$\begin{aligned} \mathbb{E}(S) &= \sum_i a_i \mathbb{P}(A_i) = \sum_i a_i (\mathbb{P}(A_i \cap E) + \mathbb{P}(A_i \cap E^c)) \\ &= \sum_i a_i \mathbb{P}(A_i \cap E^c) = \mathbb{E}\left(\sum_i a_i \mathbf{1}_{A_i \cap E^c}\right) =: \mathbb{E}(S'). \end{aligned}$$

Now $S' = S \mathbf{1}_{E^c}$ is simple and, on E^c , $S' \leq X = Y$; on E , $S' = 0 \leq Y$. So $S' \leq Y$ everywhere and $\mathbb{E}(Y) \geq \mathbb{E}(S') = \mathbb{E}(S)$. Taking the supremum over S gives $\mathbb{E}(Y) \geq \mathbb{E}(X)$, and the same argument with the roles of X and Y exchanged gives the reverse inequality. $\square$

Corollary 3.15 (Linearity). If $X, Y \geq 0$ are random variables and $a, b \geq 0$ are real, then

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y),$$

provided we adopt the convention $0 \times \infty = 0$.

Proof. $\mathbb{E}(aX) = a\mathbb{E}(X)$ follows directly from the definition, so assume $a = b = 1$. Define the simple random variables

$$S_n^X = \min\left\{\frac{\lfloor 2^n X \rfloor}{2^n}, n\right\}, \quad S_n^Y = \min\left\{\frac{\lfloor 2^n Y \rfloor}{2^n}, n\right\},$$

which satisfy

$$S_n^X \leq S_{n+1}^X \leq \cdots \leq X, \quad \lim_{n \rightarrow \infty} S_n^X = X,$$

and likewise for Y . Linearity holds for simple random variables, so $\mathbb{E}(S_n^X + S_n^Y) = \mathbb{E}(S_n^X) + \mathbb{E}(S_n^Y)$ for every n ; passing to the limit on both sides is exactly the monotone convergence theorem. *(Continued next lecture.)* $\square$

Lecture 4

Expectation

Last time, Step 3: nonnegative random variables $X \geq 0$,

$$\mathbb{E}[X] = \sup\{\mathbb{E}(S) : 0 \leq S \leq X, S \text{ simple}\}.$$

4.1 Nonnegative random variables

Proposition 4.1.

(a) If $X = Y \geq 0$ a.s., then $\mathbb{E}(X) = \mathbb{E}(Y)$.

(b) If $0 \leq X \leq Y$, then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.

Lemma 4.2 (Fatou). If $(X_n)_{n \geq 1}$ is a sequence of nonnegative random variables, then

$$\liminf_{n \rightarrow \infty} \mathbb{E}(X_n) \geq \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n\right).$$

Theorem 4.3 (Monotone convergence theorem, MCT). Let $0 \leq X_1 \leq X_2 \leq \dots$ be a sequence of random variables and $X = \lim_{n \rightarrow \infty} X_n$. Then

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X).$$

Example 4.4 (Monotonicity is not optional). Let $\mathbb{P} \sim \text{Unif}[0, 1]$ and $X_n = n\mathbf{1}_{[0, 1/n]}$. Then $\lim_{n \rightarrow \infty} X_n = 0$ a.s., but $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = 1$. ($\times$)

The mass escapes upward as the interval shrinks. Nothing here is monotone, so MCT does not apply — and Fatou's inequality is strict.

Proof of Theorem 22.1. By Fatou's lemma,

$$\underbrace{\liminf_{n \rightarrow \infty} \mathbb{E}(X_n)}_{= \lim_{n \rightarrow \infty} \mathbb{E}(X_n)} \geq \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n\right) = \mathbb{E}(X).$$

Conversely, by monotonicity $X_n \leq X$ for all n , so $\mathbb{E}(X_n) \leq \mathbb{E}(X)$ and therefore $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) \leq \mathbb{E}(X)$. $\square$

Corollary 4.5 (Linearity on nonnegatives). *If $X, Y \geq 0$ are random variables and $a, b \geq 0$, then*

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y),$$

with the convention $0 \cdot \infty = 0$.

Proof. It suffices to consider $a = b = 1$. Construct simple random variables S_n^X, S_n^Y with

$$0 \leq S_1^X \leq S_2^X \leq \dots \leq \lim_{n \rightarrow \infty} S_n^X = X, \quad 0 \leq S_1^Y \leq S_2^Y \leq \dots \leq \lim_{n \rightarrow \infty} S_n^Y = Y.$$

Then $S_1^X + S_1^Y \leq S_2^X + S_2^Y \leq \dots$ and $S_n^X + S_n^Y \rightarrow X + Y$. For simple random variables linearity is already known:

$$\mathbb{E}(S_n^X + S_n^Y) = \mathbb{E}(S_n^X) + \mathbb{E}(S_n^Y). \quad (4.1)$$

Applying MCT to both sides of (4.1) gives $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$. $\square$

4.2 Step 4: signed random variables

If X is a random variable, decompose $X = X^+ - X^-$, where

$$X^+ = \max\{X, 0\}, \quad X^- = -\min\{X, 0\},$$

and define

$$\mathbb{E}(X) = \mathbb{E}(X^+) - \mathbb{E}(X^-),$$

provided $\mathbb{E}(X^+)$ and $\mathbb{E}(X^-)$ are not both $+\infty$.

A sufficient condition for $\mathbb{E}(X)$ to exist is

$$\mathbb{E}(|X|) < \infty \iff (\mathbb{E}(X^+) < \infty \text{ and } \mathbb{E}(X^-) < \infty),$$

in which case we say X is integrable.

Remark 4.6. If $X = X_1 - X_2$ for integrable $X_1, X_2 \geq 0$, then $\mathbb{E}(X) = \mathbb{E}(X_1) - \mathbb{E}(X_2)$; the particular decomposition does not matter.

4.3 Properties of expectation

Let X, Y be integrable random variables, so their expectations exist.

- 1). *Uniqueness:* if $X = Y$ a.s., then $\mathbb{E}(X) = \mathbb{E}(Y)$.
- 2). If $X \leq Y$, then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.
- 3). *Linearity:* for $a, b \in \mathbb{R}$, $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$.
- 4). *Cauchy-Schwarz:* $(\mathbb{E}(XY))^2 \leq \mathbb{E}(|X|^2) \mathbb{E}(|Y|^2)$.
- 5). *Jensen's inequality:* if $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ is convex and $\varphi(X)$ is integrable, then $\varphi(\mathbb{E}(X)) \leq \mathbb{E}(\varphi(X))$.

Proof of linearity. Without loss of generality $a, b \geq 0$. Then

$$aX + bY = (aX^+ + bY^+) - (aX^- + bY^-),$$

so by Corollary 4.5,

$$\begin{aligned} \mathbb{E}(aX + bY) &= \mathbb{E}(aX^+ + bY^+) - \mathbb{E}(aX^- + bY^-) \\ &= a\mathbb{E}(X^+) + b\mathbb{E}(Y^+) - a\mathbb{E}(X^-) - b\mathbb{E}(Y^-) \\ &= a\mathbb{E}(X) + b\mathbb{E}(Y). \end{aligned}$$

□

Theorem 4.7 (Dominated convergence theorem, DCT). *Let $(X_n)_{n \geq 1}$ be a sequence of random variables with $|X_n| \leq Y$ for all n , where Y is an integrable random variable. If $X_n \rightarrow X$ a.s., then*

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X).$$

Proof. Note $Y + X_n \geq 0$ for all n . By Fatou's lemma,

$$\liminf_{n \rightarrow \infty} \mathbb{E}(Y + X_n) \geq \mathbb{E}\left(\liminf_{n \rightarrow \infty} (Y + X_n)\right) = \mathbb{E}(X + Y),$$

and the left side equals $\mathbb{E}(Y) + \liminf_{n \rightarrow \infty} \mathbb{E}(X_n)$, while the right side equals $\mathbb{E}(Y) + \mathbb{E}(X)$. Cancelling the finite quantity $\mathbb{E}(Y)$,

$$\liminf_{n \rightarrow \infty} \mathbb{E}(X_n) \geq \mathbb{E}(X).$$

Applying the same argument to $Y - X_n \geq 0$ gives $\liminf_{n \rightarrow \infty} -\mathbb{E}(X_n) \geq -\mathbb{E}(X)$, i.e.

$$\limsup_{n \rightarrow \infty} \mathbb{E}(X_n) \leq \mathbb{E}(X).$$

Since $\liminf \geq \limsup$ always forces equality,

$$\limsup_{n \rightarrow \infty} \mathbb{E}(X_n) = \liminf_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n).$$

□

Example 4.8 (Products of independent random variables). Let X, Y be independent random variables. If either (i) $X, Y \geq 0$, or (ii) both are integrable, then $\mathbb{E}(XY)$ exists and equals $\mathbb{E}(X)\mathbb{E}(Y)$.

Proof. (i) For $X, Y \geq 0$, let S_n^X, S_n^Y be the simple random variables constructed above. Then

$$\mathbb{E}(XY) \stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \mathbb{E}(S_n^X S_n^Y) = \lim_{n \rightarrow \infty} \mathbb{E}(S_n^X) \mathbb{E}(S_n^Y) = \mathbb{E}(X) \mathbb{E}(Y),$$

where the middle equality uses simplicity and independence.

(ii) By part (i) applied to the nonnegative variables $|X|, |Y|$,

$$\mathbb{E}(|XY|) = \mathbb{E}(|X|) \mathbb{E}(|Y|) < \infty,$$

and since $|XY|$ dominates each of X^+Y^+ , X^-Y^+ , X^+Y^- , X^-Y^- , all four are integrable. Hence

$$\begin{aligned}\mathbb{E}(XY) &= \mathbb{E}((X^+ - X^-)(Y^+ - Y^-)) \\ &= \mathbb{E}(X^+Y^+ - X^-Y^+ - X^+Y^- + X^-Y^-) \\ &= \mathbb{E}(X^+ - X^-)\mathbb{E}(Y^+ - Y^-) = \mathbb{E}(X)\mathbb{E}(Y),\end{aligned}$$

regrouping by part (i) applied to each of the four cross terms. $\square$

4.4 Worked examples

Example 4.9 (Markov's inequality). If $X \geq 0$ is a random variable, then

$$\mathbb{P}\{X \geq \lambda\} \leq \frac{\mathbb{E}(X)}{\lambda}, \quad \lambda > 0.$$

Proof. $X \geq \lambda \mathbf{1}_{\{X \geq \lambda\}}$, so $\mathbb{E}(\lambda \mathbf{1}_{\{X \geq \lambda\}}) \leq \mathbb{E}(X)$, i.e. $\lambda \mathbb{P}\{X \geq \lambda\} \leq \mathbb{E}(X)$. $\square$

Example 4.10 (Monotonicity of moments). For $1 \leq p \leq q < \infty$ and X a random variable,

$$\mathbb{E}(|X|^p)^{1/p} \leq \mathbb{E}(|X|^q)^{1/q}.$$

Proof. Let $X_n = \min\{|X|, n\} \geq 0$, which is bounded and hence has all moments. Apply Jensen's inequality to the convex function $t \mapsto t^{q/p}$ on $t \geq 0$:

$$\mathbb{E}(|X_n|^p)^{q/p} \leq \mathbb{E}(|X_n|^{p \cdot q/p}) = \mathbb{E}(|X_n|^q),$$

so $\mathbb{E}(|X_n|^p)^{1/p} \leq \mathbb{E}(|X_n|^q)^{1/q}$. Letting $n \rightarrow \infty$ and applying MCT to both sides gives the claim. $\square$

Example 4.11. Let $\mathbb{P} \sim \text{Unif}([0, 1])$ and let $(q_n)_{n \geq 1}$ be an enumeration of $\mathbb{Q} \cap [0, \frac{1}{2})$. Define

$$X_n = \sum_{k=1}^n k \mathbf{1}_{[q_k, q_{k+2^{-k}}]}, \quad n \geq 1.$$

The sequence is nondecreasing, so by MCT

$$\mathbb{E}(X) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n k 2^{-k} = \sum_{k \geq 1} k 2^{-k} = 2.$$

4.5 Conditional probability

For $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Then $\mathbb{P}(\cdot | B)$ is a probability measure on $(\Omega, \mathcal{F}_B)$, where

$$\mathcal{F}_B = \{F \cap B : F \in \mathcal{F}\}.$$

Lecture 5

Conditional Expectation and Stochastic Processes

5.1 Recall: the undergraduate picture

If X, Y are discrete with joint pmf p_{XY} , then the conditional probability is

$$\mathbb{P}(X = x \mid Y = y) = \frac{p_{XY}(x, y)}{p_Y(y)} \quad \text{if } p_Y(y) > 0,$$

and the conditional expectation is defined by

$$\mathbb{E}(X \mid Y = y) = \sum_x x p_{X|Y}(x \mid y).$$

Evaluated at $Y \sim p_Y$, the quantity $\mathbb{E}(X \mid Y)$ can be regarded as a random variable, and it satisfies the tower property

$$\mathbb{E}(\mathbb{E}(X \mid Y) g(Y)) = \mathbb{E}(X g(Y)) \quad \text{for every bounded measurable } g. \quad (5.1)$$

Q. How do we understand conditional expectation in general? Working through explicit conditional distributions is a bad idea, since the conditional distribution may not exist.

The answer is to promote (5.1) from a property to a definition.

5.2 The general definition

Definition 5.1 (Conditional expectation). Let X, Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We say a measurable function $h(Y)$ is a *conditional expectation of X given Y* if

$$\mathbb{E}(h(Y) g(Y)) = \mathbb{E}(X g(Y)) \quad \text{for every bounded measurable } g.$$

Theorem 5.2. Let X, Y be random variables with X integrable. Then a conditional expectation of X given Y exists, is integrable, and is unique almost surely.

Proof (existence: see the book). **Integrability.** Suppose $h(Y)$ is a conditional expectation of X given Y . Take the bounded measurable function

$$g(Y) := \mathbf{1}_{\{h(Y) > 0\}} - \mathbf{1}_{\{h(Y) < 0\}},$$

so that $h(Y)g(Y) = |h(Y)|$. Then

$$\mathbb{E}(|h(Y)|) = \mathbb{E}(h(Y)g(Y)) = \mathbb{E}(X \underbrace{g(Y)}_{\text{bounded}}) \leq \mathbb{E}(|X|) < \infty$$

since X is integrable.

Uniqueness. Suppose $h_1(Y)$ and $h_2(Y)$ are both conditional expectations. Then

$$\mathbb{E}(h_1(Y)g(Y)) = \mathbb{E}(Xg(Y)) = \mathbb{E}(h_2(Y)g(Y)) \quad \forall \text{ bounded measurable } g,$$

so by linearity of the expectation of integrable random variables,

$$\mathbb{E}((h_1(Y) - h_2(Y))g(Y)) = 0.$$

Choosing the indicator combination $g = \mathbf{1}_{\{h_1(Y) > h_2(Y)\}} - \mathbf{1}_{\{h_1(Y) < h_2(Y)\}}$ turns the integrand into an absolute value and forces

$$\mathbb{E}(|h_1(Y) - h_2(Y)|) = 0 \implies h_1(Y) = h_2(Y) \text{ a.s.}$$

□

Remark 5.3. This motivates writing $\mathbb{E}(X | Y)$ for any representative of the conditional expectation of X given Y .

Remark 5.4. Y may be a random vector or an entire random process — nothing in Definition 5.1 used that Y was scalar.

Corollary 5.5 (Tower property). *Let X be integrable and let Y, Z be random variables. Then*

$$\mathbb{E}(\mathbb{E}(X | Y, Z) | Y) = \mathbb{E}(X | Y) \quad \text{a.s.}$$

Proof. Since $\mathbb{E}(X | Y, Z)$ is integrable, $\mathbb{E}(\mathbb{E}(X | Y, Z) | Y)$ exists, is integrable, and is a.s. unique. For every bounded measurable g ,

$$\mathbb{E}(\mathbb{E}(\mathbb{E}(X | Y, Z) | Y)g(Y)) = \mathbb{E}(\mathbb{E}(X | Y, Z)g(Y)) = \mathbb{E}(Xg(Y)),$$

where each step is Definition 5.1 — the second one using that $g(Y)$ is in particular a bounded measurable function of (Y, Z) . So $\mathbb{E}(\mathbb{E}(X | Y, Z) | Y)$ is a conditional expectation of X given Y , and therefore equals $\mathbb{E}(X | Y)$ a.s. □

5.3 Properties of conditional expectation

Let X_1, X_2, Y be random variables with X_1, X_2 integrable.

Proposition 5.6.

1. (Linearity.) For all $a, b \in \mathbb{R}$,

$$\mathbb{E}(aX_1 + bX_2 \mid Y) = a\mathbb{E}(X_1 \mid Y) + b\mathbb{E}(X_2 \mid Y).$$

2. (Monotonicity.) If $X_1 \leq X_2$ a.s., then $\mathbb{E}(X_1 \mid Y) \leq \mathbb{E}(X_2 \mid Y)$ a.s.

3. (Cauchy–Schwarz.) $(\mathbb{E}(X_1 X_2 \mid Y))^2 \leq \mathbb{E}(X_1^2 \mid Y) \mathbb{E}(X_2^2 \mid Y)$ a.s.

4. (Jensen’s inequality.) If $\psi: \mathbb{R} \rightarrow \mathbb{R}$ is convex and $\psi(X_1)$ is integrable, then

$$\psi(\mathbb{E}(X_1 \mid Y)) \leq \mathbb{E}(\psi(X_1) \mid Y) \quad \text{a.s.}$$

Proof of 1. For every bounded measurable g ,

$$\begin{aligned} \mathbb{E}\left(\mathbb{E}(aX_1 + bX_2 \mid Y) g(Y)\right) &= \mathbb{E}((aX_1 + bX_2)g(Y)) \\ &= a\mathbb{E}(X_1 g(Y)) + b\mathbb{E}(X_2 g(Y)) \\ &= a\mathbb{E}(\mathbb{E}(X_1 \mid Y)g(Y)) + b\mathbb{E}(\mathbb{E}(X_2 \mid Y)g(Y)) \\ &= \mathbb{E}\left((a\mathbb{E}(X_1 \mid Y) + b\mathbb{E}(X_2 \mid Y))g(Y)\right). \end{aligned}$$

So $a\mathbb{E}(X_1 \mid Y) + b\mathbb{E}(X_2 \mid Y)$ is a conditional expectation of $aX_1 + bX_2$ given Y , hence equals it a.s. $\square$

Remark 5.7. Fatou’s lemma, the monotone convergence theorem, and the dominated convergence theorem all have versions for conditional expectation.

Example 5.8 (Pulling out what is known). One can use the DCT for conditional expectation to show

$$\mathbb{E}(Xf(Y) \mid Y) = \mathbb{E}(X \mid Y)f(Y) \quad \text{a.s.}$$

for X and $f(Y)$ integrable (with $Xf(Y)$ integrable). The idea is to check the defining identity for bounded f first, then approximate.

5.4 Conditional expectation as an MMSE estimator

Claim 5.9. If $\mathbb{E}(X^2) < \infty$, then for every measurable function f ,

$$\mathbb{E}\left((X - \mathbb{E}(X \mid Y))^2\right) \leq \mathbb{E}\left((X - f(Y))^2\right).$$

Proof. Add and subtract $\mathbb{E}(X | Y)$ and expand:

$$\begin{aligned} \mathbb{E}\left((X - f(Y))^2\right) &= \mathbb{E}\left((X - \mathbb{E}(X | Y) + \mathbb{E}(X | Y) - f(Y))^2\right) \\ &= \mathbb{E}\left((X - \mathbb{E}(X | Y))^2\right) + \mathbb{E}\left((\mathbb{E}(X | Y) - f(Y))^2\right) \\ &\quad + 2 \mathbb{E}\left((X - \mathbb{E}(X | Y)) \underbrace{(\mathbb{E}(X | Y) - f(Y))}_{=: g(Y)}\right) \\ &\geq \mathbb{E}\left((X - \mathbb{E}(X | Y))^2\right). \end{aligned}$$

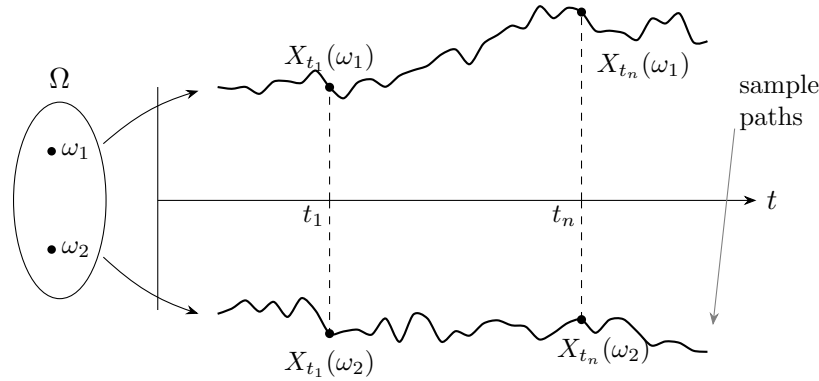
The cross term vanishes because $\mathbb{E}((X - \mathbb{E}(X | Y))g(Y)) = \mathbb{E}(Xg(Y)) - \mathbb{E}(\mathbb{E}(X | Y)g(Y)) = 0$ by Definition 5.1, and the remaining discarded term is a square. $\square$

Remark 5.10. So the conditional expectation is the minimum mean square error estimate of X from Y among *all* measurable functions of Y — not just the linear ones.

5.5 Stochastic processes

Definition 5.11 (Stochastic process). A *stochastic process* is an indexed collection $(X_t)_{t \in \tau}$ of random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Usually $\tau := \mathbb{N}, \mathbb{Z}, \mathbb{R}$, or $\mathbb{R}^n$ — discrete time, continuous time, space, and so on. For each fixed ω , the map $t \mapsto X_t(\omega)$ is a *sample path*.



Finite-dimensional distributions

Stochastic processes are typically defined entirely in terms of their *finite-dimensional distributions* (FDDs).

Example 5.12 (Infinite sequence of coin flips).

$$\mathbb{P}\{X_1 = x_1, \dots, X_n = x_n\} = 2^{-n} \quad \forall x_1, \dots, x_n \in \{0, 1\}, \quad \forall n \geq 1.$$

Remark 5.13. This sequence of FDDs is *consistent* in the sense of marginalization: summing the n -dimensional FDD over x_n recovers the $(n - 1)$ -dimensional one.

Example 5.14 (Markov process). A *Markov process* is a process whose FDDs are generated by an initial distribution together with a transition kernel: for all $n \geq 1$,

$$\mathbb{P}(X_{n+1} \in \cdot \mid X_1, \dots, X_n) = \mathbb{P}(X_{n+1} \in \cdot \mid X_n),$$

and when the process is temporally homogeneous this common value is $P(\cdot \mid x)$ on $\{X_n = x\}$, independent of n .

Theorem 5.15 (Kolmogorov extension theorem). *Given a consistent sequence of FDDs, there exist a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a process $(X_n)_{n \geq 1}$ on it that is consistent with the model.*

5.6 Limit theorems and modes of convergence

Theorem 5.16 (Weak law of large numbers). *Let $(X_n)_{n \geq 1}$ be i.i.d. with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) < \infty$. Then for every $\varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| > \varepsilon \right) = 0.$$

Writing $S_n := \frac{1}{n} \sum_{i=1}^n X_i$ for the sample mean, $(S_n)_{n \geq 1}$ is itself a stochastic process. (Note that here S_n denotes the sample *mean*, not the partial sum.)

Proof. By Markov's inequality applied to the squared deviation,

$$\mathbb{P} \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right)^2 > \varepsilon^2 \right) \leq \frac{\mathbb{E} \left(\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right)^2 \right)}{\varepsilon^2} = \frac{\text{Var}(X_1)}{n\varepsilon^2} \xrightarrow{n \rightarrow \infty} 0.$$

□

Remark 5.17. The conclusion only requires $\mathbb{E}(|X_1|) < \infty$, which is weaker than $\mathbb{E}(X_1^2) < \infty$; the second moment is used here purely for the Chebyshev bound.

Theorem 5.18 (Strong law of large numbers). *Let $(X_n)_{n \geq 1}$ be i.i.d. integrable random variables with $\mathbb{E}(X_1) = \mu$. Then*

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu \right) = 1.$$

Lecture 6

The Strong Law and Modes of Convergence

Recall. Let $(X_n)_{n \geq 1}$ be i.i.d. and integrable, with

$$S_n := \sum_{i=1}^n X_i, \quad \mathbb{E}(X_1) = \mu.$$

Theorem 6.1 (WLLN). For every $\varepsilon > 0$, $\mathbb{P}(|\frac{1}{n}S_n - \mu| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 6.2 (SLLN). $\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{1}{n}S_n = \mu\right) = 1$.

6.1 Proof of the strong law

Remark 6.3 (Standing assumption). The argument below assumes a finite fourth moment, $\mathbb{E}(X_1^4) < \infty$. The SLLN holds under integrability alone, but that proof is harder; this is the classical fourth-moment argument.

Set

$$A_n(\varepsilon) := \{\omega : |\frac{1}{n}S_n - \mu| > \varepsilon\}.$$

Just like in the proof of the WLLN, we use Markov — but at the fourth moment rather than the second, and with $\mu_4 := \mathbb{E}((X_1 - \mu)^4)$, $\sigma^2 := \text{Var}(X_1)$:

$$\begin{aligned} \mathbb{P}(A_n(\varepsilon)) &\leq \frac{\mathbb{E}((\frac{1}{n}S_n - \mu)^4)}{\varepsilon^4} \\ &= \frac{1}{\varepsilon^4 n^4} \sum_{i,j,k,\ell} \mathbb{E}((X_i - \mu)(X_j - \mu)(X_k - \mu)(X_\ell - \mu)). \end{aligned}$$

Any term in which some index appears exactly once factors, by independence, into something containing $\mathbb{E}(X_i - \mu) = 0$. So only the terms with all four indices equal, or with two distinct indices each appearing twice, survive:

$$= \frac{1}{\varepsilon^4 n^4} \left(n\mu_4 + \binom{n}{2} \binom{4}{2} \underbrace{\text{Var}(X_1)^2}_{=\sigma^4} \right) \leq \frac{C \max(\mu_4, \sigma^4)}{n^2 \varepsilon^4} \implies \sum_{n \geq 1} \mathbb{P}(A_n(\varepsilon)) < \infty.$$

By Borel–Cantelli, $\mathbb{P}(A_n(\varepsilon) \text{ i.o.}) = 0$ for every $\varepsilon > 0$. Now

$$\{\omega : \tfrac{1}{n}S_n(\omega) \not\rightarrow \mu\} = \bigcup_{k \geq 1} \{A_n(2^{-k}) \text{ i.o.}\},$$

and this union is increasing in k , so by continuity of $\mathbb{P}$ from below,

$$\mathbb{P}(\tfrac{1}{n}S_n \not\rightarrow \mu) = \mathbb{P}\left(\bigcup_{k \geq 1} \{A_n(2^{-k}) \text{ i.o.}\}\right) = \lim_{k \rightarrow \infty} \mathbb{P}(A_n(2^{-k}) \text{ i.o.}) = 0.$$

Hence $\mathbb{P}(\tfrac{1}{n}S_n \rightarrow \mu) = 1$. □

Remark 6.4. The SLLN connects the frequentist and axiomatic viewpoints. Suppose the outcome of an experiment has distribution function F_X . By the Kolmogorov extension theorem there exists $(X_n)_{n \geq 1}$ *i.i.d.* on some $(\Omega, \mathcal{F}, \mathbb{P})$ with $F_{X_i} = F_X$ for all i . By the SLLN,

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq x\}} \longrightarrow F_X(x) \quad \forall x, \text{ as } n \rightarrow \infty.$$

The empirical frequency of the event $\{X \leq x\}$ converges to its probability.

Example 6.5 (Monte Carlo integration). Let $f: [0, 1] \rightarrow \mathbb{R}$ be integrable, and suppose we want $\int_0^1 f dx$. Let $(U_n)_{n \geq 1}$ be *i.i.d.* $\text{Unif}([0, 1])$ and consider

$$\frac{1}{n} \sum_{i=1}^n f(U_i) \xrightarrow{\text{SLLN}} \int_0^1 f dx \quad \text{a.s.}$$

6.2 Modes of convergence

Random variables are functions. As such, they converge in many different ways.

Definition 6.6 (Convergence in probability). We say $X_n \rightarrow X$ *in probability* if for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$.

Example 6.7. The WLLN says $\tfrac{1}{n}S_n \rightarrow \mu$ in probability (i.p.).

Definition 6.8 (Almost sure convergence). We say $X_n \rightarrow X$ *almost surely* if $\lim_{n \rightarrow \infty} X_n = X$ with probability 1.

Example 6.9. The SLLN says $\tfrac{1}{n}S_n \rightarrow \mu$ a.s.

Definition 6.10 (Convergence in distribution). $X_n \rightarrow X$ *in distribution* if $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$ at every x where F_X is continuous.

Example 6.11. $X_n \sim \text{Unif}(-\frac{1}{n}, \frac{1}{n})$ converges to 0 in distribution. Note the restriction to continuity points is doing real work here: at $x = 0$ we have $F_{X_n}(0) = \frac{1}{2}$ for every n , while $F_0(0) = 1$.

Remark 6.12 (Equivalent formulation). $X_n \rightarrow X$ in distribution if and only if, for every continuous bounded g ,

$$\mathbb{E}(g(X_n)) \longrightarrow \mathbb{E}(g(X)) \quad \text{as } n \rightarrow \infty.$$

Theorem 6.13.

$$X_n \rightarrow X \text{ a.s.} \implies X_n \rightarrow X \text{ i.p.} \implies X_n \rightarrow X \text{ in dist.}$$

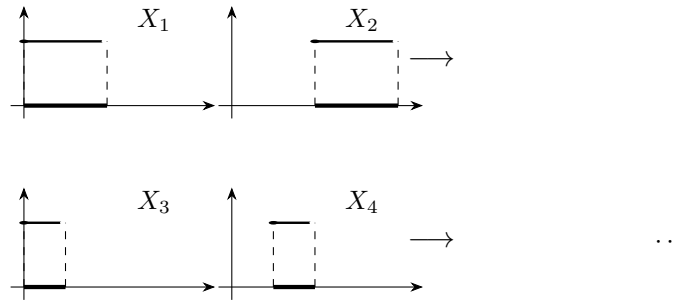
Proof. See the book. □

Remark 6.14. Even the strongest mode above — almost sure convergence — is not enough to ensure statements like $\mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$. For that you need the dominated or monotone convergence theorems.

Neither implication in Theorem 6.13 reverses

Example 6.15 (The gliding hump: i.p. but not a.s.). Take $\mathbb{P} = \text{Unif}([0, 1])$ and, writing $n = 2^k + j$ with $0 \leq j < 2^k$, set

$$X_n = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k})}.$$



The humps get narrower and narrower:

$$\mathbb{P}(X_n = 1) = 2^{-\lfloor \log_2 n \rfloor} \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

so $X_n \rightarrow 0$ i.p. But the hump also keeps sliding across $[0, 1]$, so for every ω you sample $X_n(\omega) = 1$ infinitely often. Hence

$$X_n \rightarrow 0 \text{ i.p.}, \quad X_n \not\rightarrow 0 \text{ a.s.}$$

Nevertheless, one can always extract a good subsequence.

Proposition 6.16. *If $X_n \rightarrow X$ i.p., there is a subsequence $(X_{n_k})_{k \geq 1} \subset (X_n)_{n \geq 1}$ with $X_{n_k} \rightarrow X$ a.s.*

Proof. Since $X_n \rightarrow X$ i.p., for every $k \geq 1$ there is n_k with

$$\mathbb{P}(|X_{n_k} - X| > \tfrac{1}{k}) \leq 2^{-k}, \quad \text{so} \quad \sum_{k \geq 1} \mathbb{P}\{|X_{n_k} - X| > \tfrac{1}{k}\} < \infty.$$

Borel–Cantelli gives $\mathbb{P}\{|X_{n_k} - X| > \tfrac{1}{k} \text{ i.o.}\} = 0$, hence $X_{n_k} \rightarrow X$ a.s. □

6.3 Convergence in L^p

Definition 6.17. Let $p \geq 1$. We say $X_n \rightarrow X$ in p th moment, or in L^p , if $\mathbb{E}(|X_n - X|^p) \rightarrow 0$.

Proposition 6.18. Let $1 \leq p \leq q < \infty$ and suppose $X_n \rightarrow X$ in L^q . Then

1. $X_n \rightarrow X$ in L^p ; (monotonicity of moments)
2. $X_n \rightarrow X$ i.p.; (Markov)
3. $\mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$, provided X_n, X are integrable, since

$$|\mathbb{E}(X_n) - \mathbb{E}(X)| = |\mathbb{E}(X_n - X)| \leq \mathbb{E}|X_n - X| \leq (\mathbb{E}|X_n - X|^p)^{1/p} \rightarrow 0.$$

Example 6.19 (In distribution does not imply in probability). Take $(X_n)_{n \geq 1}$ i.i.d. $\text{Bern}(\frac{1}{2})$; trivially $X_n \rightarrow \text{Bern}(\frac{1}{2})$ in distribution. But suppose Z is any random variable. For $n \neq m$,

$$\begin{aligned} \mathbb{P}(|X_n - Z| > \tfrac{1}{4}) + \mathbb{P}(|X_m - Z| > \tfrac{1}{4}) &\geq \mathbb{P}(\{|X_n - Z| > \tfrac{1}{4}\} \cup \{|X_m - Z| > \tfrac{1}{4}\}) \\ &\geq \mathbb{P}\{|X_n - X_m| > \tfrac{1}{2}\} = \mathbb{P}\{X_n \neq X_m\} = \tfrac{1}{2}. \end{aligned}$$

If $X_n \rightarrow Z$ in probability, the left-hand side would have to vanish as $n, m \rightarrow \infty$. So $(X_n)_{n \geq 1}$ converges in probability to nothing at all.

Lecture 7

Characteristic Functions and the CLT

Last time: modes of convergence, laws of large numbers.

7.1 Statement

Theorem 7.1 (Central Limit Theorem). *Let $(X_n)_{n \geq 1}$ be i.i.d. with $\text{Var}(X_1) = \sigma^2 < \infty$ and $\mathbb{E}(X_1) = \mu$. Define*

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma}.$$

Then $Z_n \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$.

The proof rests on three ingredients, developed in the next three sections.

7.2 Key idea 1: convergence in distribution

$$X_n \xrightarrow{d} X \iff \mathbb{E}(g(X_n)) \longrightarrow \mathbb{E}(g(X)) \quad \text{for every continuous bounded } g.$$

7.3 Key idea 2: tightness

Definition 7.2 (Tightness). A sequence of random variables $(X_n)_{n \geq 1}$ is *tight* if

$$\lim_{x \rightarrow \infty} \sup_{n \geq 1} \mathbb{P}(|X_n| > x) = 0.$$

This is uniform control on tail probabilities: no X_n can escape to infinity.

Example 7.3. $X_n = n$ for all $n \geq 1$ is not tight.

Remark 7.4 (Sufficient condition). $\sup_{n \geq 1} \mathbb{E}(|X_n|) < \infty$ implies tightness, by Markov's inequality.

Theorem 7.5 (Prokhorov). *If $(X_n)_{n \geq 1}$ is a tight sequence of random variables, then there is a subsequence $(X_{n_k})_{k \geq 1} \subset (X_n)_{n \geq 1}$ and a real-valued random variable X such that $X_{n_k} \xrightarrow{d} X$.*

Remark 7.6. This is the probabilistic counterpart of the Bolzano–Weierstrass theorem: tightness plays the role of boundedness.

7.4 Key idea 3: characteristic functions

Definition 7.7 (Characteristic function). For a real random variable X , the *characteristic function* φ_X is

$$\varphi_X(t) = \mathbb{E}(e^{jtX}) = \underbrace{\mathbb{E}(\cos(tX))}_{< \infty} + j \underbrace{\mathbb{E}(\sin(tX))}_{< \infty}.$$

Both pieces are finite because $\cos$ and $\sin$ are bounded, so φ_X always exists.

Fact 7.8. *Characteristic functions are unique: every distribution on $\mathbb{R}$ has its own characteristic function, and the map is injective.*

Remark 7.9. Contrast with the moment generating function, which in general does *not* exist: $\mathbb{E}(e^{tX})$ can be infinite for heavy-tailed distributions.

Example 7.10. If X, Y are independent, then

$$\varphi_{aX+bY}(t) = \mathbb{E}(e^{jt(aX+bY)}) = \mathbb{E}(e^{j(at)X}) \mathbb{E}(e^{j(bt)Y}) = \varphi_X(at) \varphi_Y(bt).$$

Remark 7.11. Characteristic functions are Fourier transforms of measures.

Theorem 7.12 (Lévy’s continuity theorem). *Let $(X_n)_{n \geq 1}$ and X be real-valued random variables. The following are equivalent:*

1. $X_n \xrightarrow{d} X$;
2. $\varphi_{X_n}(t) \rightarrow \varphi_X(t)$ for every t .

Lemma 7.13.

1. $t \mapsto \varphi_X(t)$ is uniformly continuous.
2. For every $u > 0$,

$$\mathbb{P}\left(|X| > \frac{2}{u}\right) \leq \frac{1}{u} \int_{-u}^u (1 - \varphi_X(t)) dt.$$

Proof of Theorem 7.12. (1 $\Rightarrow$ 2). Write

$$\varphi_{X_n}(t) = \underbrace{\mathbb{E}(\cos(tX_n))}_{\text{bounded}} + j \underbrace{\mathbb{E}(\sin(tX_n))}_{\text{bounded}}.$$

Since $\cos(t \cdot)$ and $\sin(t \cdot)$ are continuous and bounded, convergence in distribution gives

$$\varphi_{X_n}(t) \rightarrow \mathbb{E}(\cos(tX)) + j \mathbb{E}(\sin(tX)) = \varphi_X(t).$$

(2 $\Rightarrow$ 1). Fix $\varepsilon > 0$, and note $\varphi_X(0) = 1$ and $\varphi_X(-t) = \varphi_X^*(t)$. By continuity of φ_X at 0, there is u_ε with

$$\frac{1}{2u_\varepsilon} \int_{-u_\varepsilon}^{u_\varepsilon} (1 - \varphi_X(t)) dt < \varepsilon.$$

Since $\varphi_{X_n}(t) \rightarrow \varphi_X(t)$ for all t , dominated convergence gives

$$\lim_{n \rightarrow \infty} \frac{1}{2u_\varepsilon} \int_{-u_\varepsilon}^{u_\varepsilon} (1 - \varphi_{X_n}(t)) dt < \varepsilon,$$

so by Lemma 7.13(2),

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(|X_n| > \frac{2}{u_\varepsilon}\right) \leq 2\varepsilon.$$

Hence $(X_n)_{n \geq 1}$ is tight. By Prokhorov (Theorem 7.5) there is a subsequence $(X_{n_k})_{k \geq 1}$ and a real random variable X_∞ with $X_{n_k} \xrightarrow{d} X_\infty$. Then $\varphi_{X_{n_k}} \rightarrow \varphi_{X_\infty}$ pointwise, so $\varphi_{X_\infty} = \varphi_X$, and by uniqueness of characteristic functions (Fact 7.8) $X \stackrel{d}{=} X_\infty$.

So far: if $\varphi_{X_n} \rightarrow \varphi_X$ pointwise, then *some* subsequence satisfies $X_{n_k} \xrightarrow{d} X$. To upgrade this to the full sequence, suppose $X_n \not\xrightarrow{d} X$. Then there exist $\delta > 0$ and a subsequence $(X_{n_k})_{k \geq 1}$ with

$$\liminf_{k \rightarrow \infty} |F_{X_{n_k}}(x) - F_X(x)| > \delta \quad (7.1)$$

at some continuity point x of F_X . Applying the argument above to $(X_{n_k})_{k \geq 1}$ produces a further subsequence $(X_{n_{k_m}})_{m \geq 1}$ with $X_{n_{k_m}} \xrightarrow{d} X$, contradicting (7.1). $\square$

7.5 Proof of the CLT

Lemma 7.14 (Moment expansion). *If $\mathbb{E}(|X|^n) < \infty$, then*

$$\varphi_X(t) = \sum_{k=0}^n \frac{(jt)^k \mathbb{E}(X^k)}{k!} + o(|t|^n).$$

Proof of Theorem 7.1. Without loss of generality take $\mathbb{E}(X_1) = 0$ and $\text{Var}(X_1) = \mathbb{E}(X_1^2) = 1$, with $(X_n)_{n \geq 1}$ i.i.d.. Define

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i, \quad n \geq 1.$$

We want $Z_n \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$, and we use the fact that $\varphi_Z(t) = e^{-t^2/2}$.

By independence and Example 7.10,

$$\varphi_{Z_n}(t) = \varphi_{\frac{1}{\sqrt{n}}(X_1 + \dots + X_n)}(t) = \prod_{i=1}^n \varphi_{X_i}\left(\frac{t}{\sqrt{n}}\right) = \varphi_{X_1}\left(\frac{t}{\sqrt{n}}\right)^n.$$

Applying Lemma 7.14 with $n = 2$ and using $\mathbb{E}(X_1) = 0$, $\mathbb{E}(X_1^2) = 1$,

$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n.$$

Since

$$\lim_{n \rightarrow \infty} \left(1 - \frac{t^2}{2n}\right)^n = e^{-t^2/2} = \varphi_Z(t),$$

Lévy's continuity theorem gives $Z_n \xrightarrow{d} Z$. $\square$

7.6 Aside: tightness in optimization

In optimization one often has a collection of probability distributions

$$\Pi = (P_\alpha)_{\alpha \in A}$$

— repayment strategies, asset allocations, and so on — together with the objective

$$\inf_{P \in \Pi} \mathbb{E}_P(\ell) = V^*,$$

where ℓ is the loss (objective) function. Suppose ℓ is lower semicontinuous and bounded from below. If Π is tight, then there exists $P^* \in \overline{\Pi}$ attaining the value:

$$\mathbb{E}_{P^*}(\ell) = V^*.$$

Lecture 8

Gaussian Random Vectors and Processes

8.1 Gaussian random variables

Write $\mathcal{N}(0, 1)$ for the standard normal distribution. If $X \sim \mathcal{N}(0, 1)$ then

$$\varphi_X(t) = e^{-t^2/2}, \quad f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

If $X = \sigma Z + \mu$ with $Z \sim \mathcal{N}(0, 1)$, then we say $X \sim \mathcal{N}(\mu, \sigma^2)$, and

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}.$$

8.2 Gaussian random vectors

Definition 8.1 (Gaussian vector). A collection of random variables $(X_1, \dots, X_k)$ is a *Gaussian vector* — equivalently, the X_i are *jointly Gaussian* — if X can be written as an affine transformation of *i.i.d.* $\mathcal{N}(0, 1)$ variables:

$$X = AZ + \mu \quad \text{for some } \mu \in \mathbb{R}^k \text{ and } A \in \mathbb{R}^{k \times m},$$

where $Z = (Z_1, \dots, Z_m)$ are *i.i.d.* $\mathcal{N}(0, 1)$.

The first two moments come straight out of the definition:

$$\mathbb{E}(X) = \mu,$$

$$\text{Cov}(X) = \mathbb{E}((X - \mathbb{E}X)(X - \mathbb{E}X)^\top) = \mathbb{E}(AZZ^\top A^\top) \stackrel{\text{linearity}}{=} \underbrace{A \mathbb{E}(ZZ^\top)}_{=I} A^\top = \underbrace{AA^\top}_{=: \Sigma}.$$

Remark 8.2 (On assuming A has full rank). It is convenient to assume A has full row rank, so that $\Sigma = AA^\top$ is nonsingular — this is exactly what the density below requires. It is not automatic: A may be rank-deficient, in which case X is a *degenerate* Gaussian supported on a proper affine subspace of $\mathbb{R}^k$, with no density on $\mathbb{R}^k$. Everything except the density formula goes through regardless.

For a Gaussian vector $X \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$,

$$f_X(x) = \frac{1}{(2\pi)^{k/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{(x - \mu)^\top \Sigma^{-1} (x - \mu)}{2}\right).$$

Remark 8.3. The important takeaway is that Gaussian distributions are completely parametrized by mean and covariance — and any positive semidefinite matrix is a valid covariance.

Fact 8.4. If X, Y are jointly Gaussian, then

$$\text{Cov}(X, Y) := \mathbb{E}((X - \mathbb{E}X)(Y - \mathbb{E}Y)) = 0 \iff X, Y \text{ independent}.$$

Remark 8.5. The “joint” part is essential. The implication independent $\Rightarrow$ uncorrelated is free; the converse is what needs joint Gaussianity.

Example 8.6 (Uncorrelated, Gaussian marginals, not independent). Let $X \sim \mathcal{N}(0, 1)$, and let B be independent of X , with $B \sim \text{Unif}\{-1, 1\}$. Set $Y = BX$. Then $Y \sim \mathcal{N}(0, 1)$, and

$$\text{Cov}(X, Y) = \mathbb{E}(BX^2) = \mathbb{E}(B) \mathbb{E}(X^2) = 0,$$

yet X and Y are plainly dependent, since $|X| = |Y|$ always. The resolution is that (X, Y) is not a Gaussian vector, even though each marginal is Gaussian.

Partitioning the covariance

Let (X, Y) be a Gaussian random vector, so that X and Y are each Gaussian vectors in their own right. We typically partition the covariance matrix Σ of (X, Y) as

$$\begin{aligned} \Sigma &= \mathbb{E} \left[\begin{bmatrix} X - \mu_x \\ Y - \mu_y \end{bmatrix} \begin{bmatrix} (X - \mu_x)^\top & (Y - \mu_y)^\top \end{bmatrix} \right] \\ &= \mathbb{E} \begin{bmatrix} (X - \mu_x)(X - \mu_x)^\top & (X - \mu_x)(Y - \mu_y)^\top \\ (Y - \mu_y)(X - \mu_x)^\top & (Y - \mu_y)(Y - \mu_y)^\top \end{bmatrix} = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix}. \end{aligned}$$

8.3 Conditioning

Theorem 8.7. If X, Y are jointly Gaussian and $\Sigma_y \succ 0$, then we can express

$$X = \mu_x + \Sigma_{xy} \Sigma_y^{-1} (Y - \mu_y) + V,$$

where $V \sim \mathcal{N}(0, \Sigma_x - \Sigma_{xy} \Sigma_y^{-1} \Sigma_{yx})$ is independent of Y . In words,

$$X = A(Y - \mu_y) + \text{“independent Gaussian noise”} + \mu_x.$$

Two lemmas make the statement well posed.

Lemma 8.8. If $\Sigma \succeq 0$, then there exists $Z \sim \mathcal{N}(0, \Sigma)$.

Proof. Write $\Sigma = \Sigma^{1/2} \Sigma^{1/2}$ and take $Z = \Sigma^{1/2} W$ with $W \sim \mathcal{N}(0, I)$. □

Lemma 8.9. $\Sigma_x - \Sigma_{xy} \Sigma_y^{-1} \Sigma_{yx} \succeq 0$, by the Schur complement characterization of positive semidefiniteness.

Proof of Theorem 8.7. It is enough to build a pair with the right joint law and appeal to Remark 8.3: a Gaussian vector is pinned down by its mean and covariance, so if the construction is jointly Gaussian with the correct mean and covariance, it *is* (X, Y) in distribution.

Let $Y \sim \mathcal{N}(\mu_y, \Sigma_y)$ and define the random vector

$$X = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(Y - \mu_y) + V,$$

with V as in the statement, independent of Y . Then (X, Y) is jointly Gaussian, being an affine function of the Gaussian vector (V, Y) . Now check the moments:

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mu_x)(Y - \mu_y)^\top) = \mathbb{E}\left((\Sigma_{xy}\Sigma_y^{-1}(Y - \mu_y) + V)(Y - \mu_y)^\top\right) = \Sigma_{xy}, \quad \checkmark$$

using $\mathbb{E}(V(Y - \mu_y)^\top) = 0$ by independence, and

$$\begin{aligned} \text{Cov}(X) &= \mathbb{E}((X - \mu_x)(X - \mu_x)^\top) \\ &= \mathbb{E}\left((\Sigma_{xy}\Sigma_y^{-1}(Y - \mu_y) + V)(\Sigma_{xy}\Sigma_y^{-1}(Y - \mu_y) + V)^\top\right) \\ &= \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} + \text{Cov}(V) = \Sigma_x. \quad \checkmark \end{aligned} \quad \square$$

Corollary 8.10. *If (X, Y) is jointly Gaussian with covariance Σ , then*

$$\mathbb{E}(X \mid Y) = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(Y - \mu_y).$$

Remark 8.11. This is the fact that makes the linear theory worth so much: for jointly Gaussian data the conditional expectation — the optimal estimator under squared loss — is already an affine function of the observation, so the best linear estimator is the best estimator, full stop.

Remark 8.12. If (X, Y) is jointly Gaussian, then $A\begin{pmatrix} X \\ Y \end{pmatrix} + \mu$ is again a Gaussian vector.

8.4 Gaussian processes

Definition 8.13 (Gaussian process). A process $(X_t)_{t \in \mathcal{T}}$ is a *Gaussian process* if and only if every finite subcollection $(X_{t_i})_{i=1}^k \subset (X_t)_{t \in \mathcal{T}}$ is a Gaussian vector.

Example 8.14. Let $(X_n)_{n \geq 1}$ be a Gaussian process. We can model

$$X_{n+1} = \sum_{i=1}^n a_i X_i + V, \quad a_i \in \mathbb{R},$$

with V Gaussian and independent of $(X_i)_{i=1}^n$.

Brownian motion

Brownian motion is the continuous-time Gaussian process.

Definition 8.15 (Brownian motion). A process $(X_t)_{t \geq 0}$ is a *Brownian motion* if

1. it has independent, stationary increments;
2. it is a zero-mean Gaussian process with $X_0 = 0$;

3. its sample paths are continuous a.s., i.e.

$$\mathbb{P}(\omega : t \mapsto X_t(\omega) \text{ is continuous on } [0, \infty)) = 1.$$

Notation. Unpacking the two conditions in property 1:

- *Independent increments:* for all $t_1 < t_2 < \dots < t_n$, the increments $(X_{t_2} - X_{t_1}), (X_{t_3} - X_{t_2}), \dots, (X_{t_n} - X_{t_{n-1}})$ are independent.
- *Stationary increments:* $(X_{t+\tau} - X_\tau) \stackrel{d}{=} (X_t - X_0) = X_t$ for all $t, \tau \geq 0$.

Theorem 8.16. *Brownian motion exists as a process.*

Lecture 9

Brownian Motion and Linear Estimation

9.1 Brownian motion

Definition 9.1 (Brownian motion). A random process $(X_t)_{t \geq 0}$ is a *Brownian motion* if

1. it has independent and stationary increments;
2. it is a mean-zero Gaussian process with $X_0 = 0$;
3. it has a.s. continuous sample paths, i.e.

$$\mathbb{P}(\omega : t \mapsto X_t(\omega) \text{ is continuous}) = 1.$$

Theorem 9.2. *Brownian motion exists (in the sense of a random process).*

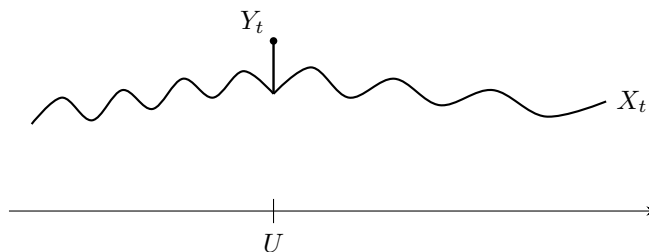
Proposition 9.3. *A Brownian motion $(X_t)_{t \geq 0}$ is a mean-zero Gaussian process with a.s. continuous sample paths and*

$$\text{Cov}(X_s, X_t) = \text{Var}(X_1) \min(s, t) = \text{Var}(X_1) (s \wedge t).$$

Remark 9.4 (Continuity is not a finite-dimensional property). Let $(X_t)_{t \geq 0}$ be a Brownian motion and $U \sim \text{Unif}(0, 1)$ independent of X , and define

$$Y_t = X_t + \mathbf{1}_{\{U=t\}}.$$

For each fixed t , $\mathbb{P}(U = t) = 0$, so $Y_t = X_t$ a.s.; hence Y has exactly the same finite-dimensional distributions as X and satisfies properties 1 and 2. But every sample path of Y has a jump at U , so Y fails property 3. Sample-path continuity has to be imposed separately — it is invisible to the finite-dimensional distributions.



Proof of Proposition 9.3. Split $[0, 1]$ into k equal pieces. By independent and stationary increments,

$$\text{Var}(X_1) = \text{Var}\left((X_1 - X_{1-1/k}) + (X_{1-1/k} - X_{1-2/k}) + \cdots + (X_{1/k} - X_0)\right) = k \text{Var}(X_{1/k}).$$

The same argument gives $\text{Var}(X_{n/k}) = \frac{n}{k} \text{Var}(X_1)$, i.e.

$$\text{Var}(X_q) = q \text{Var}(X_1) \quad \forall q \in \mathbb{Q}_{\geq 0}. \quad (9.1)$$

For irrational $t \geq 0$, pick $(q_n)_{n \geq 1} \subset \mathbb{Q}_{\geq 0}$ with $q_n \rightarrow t$; by sample-path continuity $X_{q_n} \rightarrow X_t$ a.s. Fatou then gives the upper bound

$$\text{Var}(X_t) \leq \liminf_{n \rightarrow \infty} \text{Var}(X_{q_n}) = \liminf_{n \rightarrow \infty} q_n \text{Var}(X_1) = t \text{Var}(X_1),$$

while for rational $q < t$, independence of increments gives the lower bound

$$\text{Var}(X_t) = \text{Var}((X_t - X_q) + (X_q - X_0)) \geq \text{Var}(X_q) = q \text{Var}(X_1),$$

and letting $q \uparrow t$ yields $\text{Var}(X_t) = t \text{Var}(X_1)$.

Finally take $t \leq s$. Then

$$\text{Cov}(X_s, X_t) = \text{Cov}(X_s - X_t + X_t, X_t) = \underbrace{\text{Cov}(X_s - X_t, X_t - X_0)}_{= 0 \text{ (indep. increments)}} + t \text{Var}(X_1) = t \text{Var}(X_1). \quad \square$$

9.2 Linear estimation

Motivation

The general estimation problem: estimate some unknown X given Y . As an optimization problem,

$$\inf_f \mathbb{E}\left(\underbrace{\ell(X, f(Y))}_{\text{loss fn.}}\right).$$

A popular choice is $\ell(x, x') = |x - x'|^2$, for which the optimal estimator is $f(Y) = \mathbb{E}(X | Y)$ — but this is intractable in practice.

In practice things are often approximately Gaussian, and when things are Gaussian, $\mathbb{E}(X | Y)$ is a *linear* function of Y . That is what makes the linear theory worth developing on its own.

The Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$

Analysis of linear estimation is best done in a Hilbert space. Fix $(\Omega, \mathcal{F}, \mathbb{P})$ and let $L^2(\Omega, \mathcal{F}, \mathbb{P}) =: L^2$ denote the set of random variables X on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}|X|^2 < \infty$.

Claim 9.5. L^2 is a vector space over the real field:

1. $X \in L^2 \implies \alpha X \in L^2$ for all $\alpha \in \mathbb{R}$;
2. $X, Y \in L^2 \implies X + Y \in L^2$, since $|a + b|^2 \leq 2a^2 + 2b^2$.

More than this, L^2 is a Hilbert space with inner product

$$\langle X, Y \rangle = \mathbb{E}(XY).$$

Definition 9.6 (Hilbert space). A *Hilbert space* is a vector space with an inner product which is *complete* with respect to the norm $\|X\| := \sqrt{\langle X, X \rangle}$ — that is, we can take limits without leaving the space.

Definition 9.7 (Cauchy sequence). A sequence $(X_n)_{n \geq 1}$ in a normed vector space $\mathcal{X}$ is a *Cauchy sequence* if for every $\varepsilon > 0$ there is $N > 0$ with $\|X_n - X_m\| < \varepsilon$ for all $n, m \geq N$.

Remark 9.8. If $\|X_n - X\| \rightarrow 0$, then $(X_n)_{n \geq 1}$ is Cauchy.

Definition 9.9 (Complete). $\mathcal{X}$ is *complete* if every Cauchy sequence has a limit $L \in \mathcal{X}$. Completeness is the converse to Remark 9.8.

Theorem 9.10. $L^2(\Omega, \mathcal{F}, \mathbb{P})$ equipped with $\langle X, Y \rangle = \mathbb{E}(XY)$ is a Hilbert space.

Proof. We have shown L^2 is a vector space and that $\langle X, Y \rangle$ is a valid inner product. It remains to show L^2 is complete.

Let $(X_n)_{n \geq 1}$ be Cauchy. Then we can extract a subsequence $(X_{n_k})_{k \geq 1} \subset (X_n)_{n \geq 1}$ with

$$\mathbb{E}(|X_{n_k} - X_{n_{k+1}}|^2) < 2^{-3k}$$

(the exponent is arbitrary — any summable-enough rate works). By Markov,

$$\mathbb{P}\{|X_{n_k} - X_{n_{k+1}}| > 2^{-k}\} \leq \frac{2^{-3k}}{2^{-2k}} = 2^{-k},$$

so by Borel–Cantelli $\mathbb{P}\{|X_{n_k} - X_{n_{k+1}}| > 2^{-k} \text{ i.o.}\} = 0$. Hence

$$\sum_{k \geq 1} |X_{n_k} - X_{n_{k+1}}| < \infty \quad \text{a.s.,}$$

so the telescoping series converges and the a.s. limit $X := \lim_{k \rightarrow \infty} X_{n_k}$ exists. By Fatou,

$$\mathbb{E}(|X_n - X|^2) = \mathbb{E}\left(\lim_{k \rightarrow \infty} |X_n - X_{n_k}|^2\right) \leq \liminf_{k \rightarrow \infty} \mathbb{E}(|X_n - X_{n_k}|^2) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

the last step because $(X_n)_{n \geq 1}$ is Cauchy. So $X_n \rightarrow X$ in L^2 and $X \in L^2$. $\square$

Definition 9.11. We say $X_n \rightarrow X$ in L^2 if $\mathbb{E}(|X_n - X|^2) \rightarrow 0$.

Remark 9.12. Hilbert spaces matter because they preserve our intuitive notion of geometry.

Theorem 9.13 (Hilbert projection theorem). Let $\mathcal{H}$ be a Hilbert space and $C \subset \mathcal{H}$ a closed subspace. For every $x \in \mathcal{H}$ there is a unique $y \in C$ with

$$\|y - x\| = \inf_{z \in C} \|z - x\|.$$

Moreover, y is the closest point in C to x if and only if

$$\langle y - x, z \rangle = 0 \quad \forall z \in C.$$

9.3 Best linear estimators

Let X be the random variable of interest in L^2 , and let $Y = (Y_i)_{i \in \mathcal{I}} \subset L^2$ be the observations. Define

$$\mathcal{L}(Y) := \{\text{linear estimators of } X \text{ given } Y\}$$

to be the L^2 -closure of the subspace of random variables of the form

$$L(Y) = a + \sum_{i \in S} a_i Y_i, \quad a, a_i \in \mathbb{R}, \quad S \subset \mathcal{I} \text{ finite.}$$

Theorem 9.14 (Existence and uniqueness of the best linear estimator). *There is an almost surely unique random variable $L(X | Y) \in \mathcal{L}(Y)$ with*

$$\mathbb{E}(|X - L(X | Y)|^2) \leq \mathbb{E}(|X - L(Y)|^2) \quad \forall L(Y) \in \mathcal{L}(Y).$$

Moreover $L = L(X | Y)$ a.s. if and only if $L \in \mathcal{L}(Y)$ and

1. $\mathbb{E}(L) = \mathbb{E}(X)$;
2. $\mathbb{E}((X - L)Y_i) = 0$ for all $i \in \mathcal{I}$.

The proof, and the consequences of this characterization, are taken up next lecture.

Lecture 10

Best Linear Estimation and the Kalman Filter

10.1 L^2 as a Hilbert space

$L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a Hilbert space with inner product

$$\langle X, Y \rangle := \mathbb{E}(XY), \quad \|X\|_{L^2} = \sqrt{\mathbb{E}(X^2)}.$$

Remark 10.1 (Why a Hilbert space?). Because it preserves the usual notion of geometry — in particular, projections.

Theorem 10.2 (Hilbert projection theorem). *Let H be a Hilbert space and $C \subset H$ a closed subspace. For every $x \in H$ there is a unique $y \in C$ such that*

$$\|y - x\| \leq \|c - x\| \quad \forall c \in C.$$

Moreover, y is the closest point in C to x if and only if

$$\langle y - x, c \rangle = 0 \quad \forall c \in C.$$

10.2 Best linear estimators

Given X and observations $Y = (Y_i)_{i \in I} \subset L^2$, let

$\mathcal{L}(Y) := \text{set of linear estimators} = L^2\text{-closure of all finite affine combinations of the } Y_i.$

Theorem 10.3 (Best linear estimator, BLE). *There is an almost surely unique random variable $L(X | Y) \in \mathcal{L}(Y)$ with*

$$\mathbb{E}(|X - L(X | Y)|^2) \leq \mathbb{E}(|X - L|^2) \quad \forall L \in \mathcal{L}(Y).$$

Moreover $L = L(X | Y)$ a.s. if and only if $L \in \mathcal{L}(Y)$ and

1. $\mathbb{E}(L) = \mathbb{E}(X)$ (unbiased)
2. $\mathbb{E}((X - L)Y_i) = 0 \quad \forall i \in I$ (uncorrelated)

Together, (1) and (2) are the orthogonality principle.

Proof. By the projection theorem (Theorem 10.2) the BLE exists and is unique, and it is characterized by

$$\mathbb{E}((X - L(X|Y))L) = 0 \quad \forall L \in \mathcal{L}(Y).$$

It suffices to check this against a general element of the closure. Take $\tilde{L} \in \mathcal{L}(Y)$ and pick finite affine combinations $(L_n)_{n \geq 1}$ with $L_n \rightarrow \tilde{L}$ in L^2 . Writing $L = L(X|Y)$,

$$\begin{aligned} \mathbb{E}((X - L)\tilde{L}) &= \mathbb{E}((X - L)(\tilde{L} + L_n - L_n)) \\ &= \underbrace{\mathbb{E}((X - L)L_n)}_{=0} + \mathbb{E}((X - L)(\tilde{L} - L_n)) \\ &\leq \sqrt{\mathbb{E}((X - L)^2) \underbrace{\mathbb{E}((\tilde{L} - L_n)^2)}_{\rightarrow 0 \text{ as } n \rightarrow \infty}}. \end{aligned}$$

So the inner product vanishes against all of $\mathcal{L}(Y)$. □

Remark 10.4. A useful consequence:

$$\mathbb{E}(X L(Y)) = \mathbb{E}(L(X|Y) L(Y)) \quad \forall L(Y) \in \mathcal{L}(Y).$$

Example 10.5. Let $Y_n = X + Z_n$ with $(Z_n)_{n \geq 1}$ i.i.d., zero mean, $\text{Var}(Z_1) = \sigma^2 < \infty$, and $X \in L^2$. Then

$$L(X|Y) = X.$$

Indeed $\frac{1}{n} \sum_{i=1}^n Y_i = X + \frac{1}{n} \sum_{i=1}^n Z_i \rightarrow X$ in L^2 , and each partial average is a finite affine combination of the Y_i , so X itself lies in the closure $\mathcal{L}(Y)$. Infinitely many independent noisy looks recover X exactly.

Theorem 10.6 (Vector case). *Let Y be a vector with $\Sigma_Y \succ 0$. Then*

$$L(X|Y) = \mu_X + \Sigma_{XY} \Sigma_Y^{-1} (Y - \mu_Y).$$

Proof. Check the two conditions of Theorem 10.3. First, $\mathbb{E}(L(X|Y)) = \mu_X$ ✓. Second, we want $\mathbb{E}((X - L(X|Y))Y^\top) = 0$:

$$\begin{aligned} \mathbb{E}((X - L(X|Y))Y^\top) &= \mathbb{E}((X - \mu_X)Y^\top) - \Sigma_{XY} \Sigma_Y^{-1} \mathbb{E}((Y - \mu_Y)Y^\top) \\ &= \Sigma_{XY} - \Sigma_{XY} \Sigma_Y^{-1} \Sigma_Y = 0. \quad \checkmark \end{aligned}$$

□

Proposition 10.7. *The BLE is linear:*

$$L(aX_1 + bX_2|Y) = aL(X_1|Y) + bL(X_2|Y).$$

Proof. Check conditions (1) and (2) of Theorem 10.3 for the right-hand side. □

10.3 Uncorrelated observations and recursion

In Theorem 10.6, if the Y_i are uncorrelated — so Σ_Y is a diagonal matrix — the formula simplifies. Writing $Y^n := (Y_1, \dots, Y_n)$,

$$L(X | Y^n) = \mathbb{E}(X) + \sum_{i=1}^n \frac{\text{Cov}(X, Y_i)}{\text{Var}(Y_i)} (Y_i - \mathbb{E}(Y_i)),$$

and therefore

$$L(X | Y^n) = L(X | Y^{n-1}) + \frac{\text{Cov}(X, Y_n)}{\text{Var}(Y_n)} (Y_n - \mathbb{E}(Y_n)). \quad (10.1)$$

Idea. Uncorrelated observations lead to a nice recursive structure for the BLE: each new observation contributes one additive correction term, and nothing already computed has to be revisited.

Q. Given generic observations $(Y_1, \dots, Y_n)$, how do we make them uncorrelated?

A. Gram–Schmidt.

Definition 10.8 (Linear innovation sequence). Given $(Y_n)_{n \geq 1} \subset L^2$, define the *linear innovations*

$$\hat{Y}_n = Y_n - L(Y_n | Y_1, \dots, Y_{n-1}), \quad n \geq 1,$$

so that in particular

$$\hat{Y}_1 = Y_1 - L(Y_1 | \emptyset) = Y_1 - \mathbb{E}(Y_1).$$

Remark 10.9. This is just an invertible change of basis,

$$(1, Y_1, \dots, Y_n) \longrightarrow (1, \hat{Y}_1, \dots, \hat{Y}_n),$$

so no information is lost:

$$L(X | Y_1, \dots, Y_n) = L(X | \hat{Y}_1, \dots, \hat{Y}_n).$$

The innovations are uncorrelated: for $k < n$,

$$\mathbb{E}(\hat{Y}_n \hat{Y}_k) = \mathbb{E}\left((Y_n - L(Y_n | Y_{n-1}, \dots, Y_1)) \hat{Y}_k\right) = 0,$$

because $\hat{Y}_k \in \mathcal{L}(Y_1, \dots, Y_k) \subset \mathcal{L}(Y_1, \dots, Y_{n-1})$ and the innovation is orthogonal to that span. So (10.1) applies with the $\hat{Y}_i$ in place of the Y_i :

$$\begin{aligned} L(X | Y^n) &= L(X | \hat{Y}^n) = \mathbb{E}(X) + \sum_{i=1}^n \frac{\text{Cov}(X, \hat{Y}_i)}{\text{Var}(\hat{Y}_i)} (\hat{Y}_i - \mathbb{E}(\hat{Y}_i)) \\ &= L(X | \hat{Y}^{n-1}) + \frac{\text{Cov}(X, \hat{Y}_n)}{\text{Var}(\hat{Y}_n)} \hat{Y}_n. \end{aligned}$$

10.4 Application: the Kalman filter

Setting. The state space model is

$$\left. \begin{aligned} X_{k+1} &= a_k X_k + U_k \\ Y_k &= X_k + V_k \end{aligned} \right\} \quad k \geq 0,$$

where U_k is the state (process) noise and V_k the observation noise. Here $X_0, (U_n)_{n \geq 0}, (V_n)_{n \geq 0}$ are uncorrelated and zero mean with known variances:

$$P_0 := \text{Var}(X_0), \quad Q_i := \text{Var}(U_i), \quad N_i := \text{Var}(V_i).$$

Goal. Recursively compute

$$\hat{X}_{k+1|k} := L(X_{k+1} | Y_0, \dots, Y_k)$$

in an efficient way. We also track the mean squared error

$$\sigma_{k+1|k}^2 = \mathbb{E}(|X_{k+1} - \hat{X}_{k+1|k}|^2).$$

Feeding the innovation recursion

$$L(X | Y^n) = L(X | Y^{n-1}) + \frac{\text{Cov}(X, \hat{Y}_n)}{\text{Var}(\hat{Y}_n)} \hat{Y}_n$$

through the state space model gives the following algorithm.

Theorem 10.10 (Kalman filter). *Initialize $\hat{X}_{0|-1} = 0$ and $\sigma_{0|-1}^2 = P_0$. For $k = 0, 1, \dots$ do*

$$\begin{aligned} \hat{X}_{k+1|k} &= a_k \hat{X}_{k|k-1} + K_k (Y_k - \hat{X}_{k|k-1}), \\ \sigma_{k+1|k}^2 &= a_k (a_k - K_k) \sigma_{k|k-1}^2 + Q_k, \end{aligned}$$

where

$$K_k := a_k \sigma_{k|k-1}^2 \left(\sigma_{k|k-1}^2 + N_k \right)^{-1}$$

is the Kalman gain.

The quantity $Y_k - \hat{X}_{k|k-1}$ is exactly the innovation at step k : the part of the new observation that the past could not predict. The result is a robust online linear estimator — constant work per time step, no growing history to store.

Lecture 11

The Kalman Filter and Second-Order Processes

Recall the innovations recursion for the best linear estimator:

$$L(X | Y_0, \dots, Y_{n+1}) = L(X | Y_0, Y_1, \dots, Y_n) + \frac{\text{Cov}(X, \hat{Y}_{n+1})}{\text{Var}(\hat{Y}_{n+1})} \hat{Y}_{n+1}.$$

11.1 The Kalman filter

Setting (State space model).

$$\left. \begin{aligned} X_{k+1} &= a_k X_k + U_k \\ Y_k &= X_k + V_k \end{aligned} \right\} \quad k \geq 0,$$

where $X_0, (U_k)_{k \geq 0}, (V_k)_{k \geq 0}$ are zero mean and uncorrelated, with

$$\text{Var}(X_0) = P_0, \quad \text{Var}(U_k) = Q_k, \quad \text{Var}(V_k) = N_k.$$

Theorem 11.1 (Kalman filter algorithm). *Initialize $\hat{X}_{0|-1} = 0$ and $\sigma_{0|-1}^2 = P_0$. For $k = 0, 1, \dots$ do*

$$\begin{aligned} \hat{X}_{k+1|k} &= a_k \hat{X}_{k|k-1} + K_k (Y_k - \hat{X}_{k|k-1}), \\ \sigma_{k+1|k}^2 &= a_k (a_k - K_k) \sigma_{k|k-1}^2 + Q_k, \end{aligned}$$

where the Kalman gain is

$$K_k = a_k \sigma_{k|k-1}^2 \left(\sigma_{k|k-1}^2 + N_k \right)^{-1}.$$

Derivation

Let $(\hat{Y}_i)$ be the linear innovation sequence for the observations. For simplicity take $a_k = 1$ for all k .

$$\begin{aligned}
\hat{X}_{k+1|k} &= L \left(\underbrace{X_{k+1}}_{=X_k+U_k} \middle| Y_0, \dots, Y_k \right) \\
&= L(X_k | Y_0, \dots, Y_k) + \underbrace{L(U_k | Y_0, \dots, Y_k)}_{=0} \quad (U \text{ and } Y \text{ uncorrelated}) \\
&= L(X_k | Y_0, \dots, Y_{k-1}) + \frac{\text{Cov}(X_k, \hat{Y}_k)}{\text{Var}(\hat{Y}_k)} \hat{Y}_k \\
&= \hat{X}_{k|k-1} + \frac{\text{Cov}(X_k, \hat{Y}_k)}{\text{Var}(\hat{Y}_k)} (Y_k - \hat{X}_{k|k-1}), \quad \textcircled{*}
\end{aligned}$$

where in the last line we used

$$\hat{Y}_k = Y_k - L(Y_k | Y_0, \dots, Y_{k-1}) = Y_k - \hat{X}_{k|k-1},$$

since $Y_k = X_k + V_k$ and V_k is uncorrelated with the past observations.

Now compute the two ingredients. First,

$$\begin{aligned}
\text{Cov}(X_k, \hat{Y}_k) &= \text{Cov}(X_k, Y_k - L(Y_k | Y_0, \dots, Y_{k-1})) \\
&= \text{Cov}(X_k, X_k + \underbrace{V_k}_{\rightarrow 0} - \hat{X}_{k|k-1}) \\
&= \text{Cov}(\underbrace{X_k - \hat{X}_{k|k-1}}_{\text{added 0 by } \perp}, X_k - \hat{X}_{k|k-1}) = \sigma_{k|k-1}^2,
\end{aligned}$$

where the third line uses the orthogonality principle: $X_k - \hat{X}_{k|k-1}$ is orthogonal to every linear estimator, in particular to $\hat{X}_{k|k-1}$ itself, so subtracting it inside the first slot changes nothing. Similarly,

$$\begin{aligned}
\text{Var}(\hat{Y}_k) &= \text{Cov}(Y_k - L(Y_k | Y_0, \dots, Y_{k-1}), Y_k - \hat{X}_{k|k-1}) \\
&= \text{Cov}(X_k + V_k, X_k + V_k - \hat{X}_{k|k-1}) \\
&= \text{Cov}(X_k, X_k - \hat{X}_{k|k-1}) + \text{Var}(V_k) = \sigma_{k|k-1}^2 + N_k.
\end{aligned}$$

Substituting both into $\textcircled{*}$,

$$\textcircled{*} = \hat{X}_{k|k-1} + \frac{\sigma_{k|k-1}^2}{\sigma_{k|k-1}^2 + N_k} (Y_k - \hat{X}_{k|k-1}),$$

which identifies the Kalman gain as $K_k = \sigma_{k|k-1}^2 / (\sigma_{k|k-1}^2 + N_k)$, matching Theorem 11.1 at $a_k = 1$.

For the error variance, using $X_{k+1} - \hat{X}_{k+1|k} = X_k + U_k - \hat{X}_{k|k-1} - K_k(Y_k - \hat{X}_{k|k-1})$:

$$\begin{aligned}
\sigma_{k+1|k}^2 &= \text{Cov}(X_{k+1} - \hat{X}_{k+1|k}, X_{k+1} - \hat{X}_{k+1|k}) \\
&= \text{Cov}\left(X_k + U_k, X_k + U_k - \hat{X}_{k|k-1} - K_k(Y_k - \hat{X}_{k|k-1})\right) \\
&= \text{Cov}\left(X_k, X_k - \hat{X}_{k|k-1} - K_k\left(\underbrace{Y_k}_{= X_k + V_k} - \hat{X}_{k|k-1}\right)\right) + \underbrace{\text{Var}(U_k)}_{= Q_k} \\
&= \text{Cov}\left(X_k, (X_k - \hat{X}_{k|k-1})(1 - K_k)\right) + Q_k \\
&= (1 - K_k) \sigma_{k|k-1}^2 + Q_k,
\end{aligned}$$

again matching Theorem 11.1 at $a_k = 1$.

11.2 Time series analysis

This is basically linear estimation when our observation is itself a process, $Y = (Y_t)_{t \in \mathcal{T}}$.

Definition 11.2 (Second-order process). A process $(X_n)_{n \in \mathbb{Z}}$ is a *second-order process* (SOP) if $\mathbb{E}(|X_n|^2) < \infty$ for all $n \in \mathbb{Z}$.

Remark 11.3. Since L^2 is a vector space, so is the set of second-order processes: if $X = (X_n)_{n \in \mathbb{Z}}$ and $Y = (Y_n)_{n \in \mathbb{Z}}$ are SOPs, then so is $aX + bY = (aX_n + bY_n)_{n \in \mathbb{Z}}$.

The statistics associated to SOPs are the mean and the covariance:

$$\mu_X(n) = \mathbb{E}(X_n), \quad R_{XY}(m, n) = \text{Cov}(X_m, Y_n), \quad m, n \in \mathbb{Z}.$$

Example 11.4. If X is a Gaussian process, its distribution is completely parametrized by μ_X and R_{XX} .

Example 11.5 (White noise). $(X_n)_{n \geq 1} \sim i.i.d. \mathcal{N}(0, \sigma^2)$ gives $\mu_X(n) = 0$ and $R_{XX}(m, n) = \delta_{mn} \sigma^2$. This is called a *white noise process*.

Example 11.6 (Random-phase sinusoid). Let $X_n = \alpha \sin(\omega_0 n + \Theta)$ with $\Theta \sim \text{Unif}(0, 2\pi)$. Then

$$\begin{aligned}
\mu_X(n) &= \frac{\alpha}{2\pi} \int_0^{2\pi} \sin(\omega_0 n + \theta) d\theta = 0, \\
R_{XX}(m, n) &= \frac{\alpha^2}{2\pi} \int_0^{2\pi} \sin(\omega_0 m + \theta) \sin(\omega_0 n + \theta) d\theta = \frac{\alpha^2}{2} \cos(\omega_0(m - n)).
\end{aligned}$$

In these last two examples R_{XX} is a function of $m - n$ alone.

Definition 11.7 (Wide sense stationary). A SOP is *wide sense stationary* (WSS) if $\mu_X(n) = \mu_X(0)$ for all n , and $R_{XX}(m, n)$ depends only on $m - n$. In that case we abuse notation and write $R_{XX}(m, n) = R_{XX}(m - n)$.

WSS processes are a good approximation to “steady state”.

Example 11.8 (Autoregressive process). Let $X_{n+1} = \alpha X_n + W_n$ for $n \geq 0$, where $(W_n)_{n \geq 0}$ are uncorrelated with unit variance and $X_0 = 0$. Then $\mu_X(n) = 0$ for all $n \geq 0$, and

$$R_{XX}(m, n) = \begin{cases} \alpha^{|m-n|} \frac{1 - \alpha^{2 \min(m, n)}}{1 - \alpha^2}, & m, n \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

If $|\alpha| < 1$, then

$$R_{XX}(m, n) \approx \frac{\alpha^{|m-n|}}{1 - \alpha^2} \quad \text{for } m, n \text{ large,}$$

so the process is asymptotically WSS — it forgets its initial condition and settles into a stationary regime.

Definition 11.9 (Jointly WSS). X, Y are *jointly WSS* (JWSS) processes if each is itself WSS and $R_{XY}(m, n) = \text{Cov}(X_m, Y_n)$ is a function of $m - n$ alone; we write $R_{XY}(m, n) = R_{XY}(m - n)$.

Example 11.10. Let X be WSS and $Y_n = \alpha X_{n-k} + W_n$, where W is zero-mean white noise uncorrelated with X . Then X, Y are JWSS with

$$\begin{aligned} R_{XY}(m, n) &= \text{Cov}(X_m, Y_n) = \text{Cov}(X_m, \alpha X_{n-k} + W_n) \\ &= \alpha \text{Cov}(X_m, X_{n-k}) = \alpha R_{XX}(m - n + k). \end{aligned}$$

11.3 The discrete-time Fourier transform

For $1 \leq p < \infty$, let $\ell^p(\mathbb{Z})$ denote the set of sequences $x = (x_n)_{n \in \mathbb{Z}}$ with

$$\|x\|_p := \left(\sum_{n \in \mathbb{Z}} |x_n|^p \right)^{1/p} < \infty.$$

Fact 11.11. ℓ^p spaces are complete with respect to their norms: if $(x^{(n)})_{n \geq 1} \subset \ell^p(\mathbb{Z})$ is Cauchy in the ℓ^p norm, i.e. $\|x^{(n)} - x^{(m)}\|_p \rightarrow 0$ as $n, m \rightarrow \infty$, then there is $x \in \ell^p(\mathbb{Z})$ with $\|x^{(n)} - x\|_p \rightarrow 0$ as $n \rightarrow \infty$.

Proposition 11.12 (Nesting). $\|x\|_q \leq \|x\|_p$ for $1 \leq p < q$, hence $\ell^p(\mathbb{Z}) \subset \ell^q(\mathbb{Z})$ for $p \leq q$. Moreover, by monotone convergence, $\ell^p(\mathbb{Z})$ is the ℓ^p -closure of $\ell^1(\mathbb{Z})$.

Remark 11.13 (Direction of the inequality). Note this runs *opposite* to the familiar L^p inequality on a probability space, where Jensen gives $\|X\|_p \leq \|X\|_q$ for $p < q$ and hence $L^q \subset L^p$. The difference is the underlying measure: counting measure on $\mathbb{Z}$ is infinite, and the total-mass-one step that powers Jensen is unavailable.

Lecture 12

Spectral Theory of WSS Processes

12.1 The discrete-time Fourier transform

For $p \geq 1$,

$$\ell^p(\mathbb{Z}) = \left\{ x : \left(\sum_{i \in \mathbb{Z}} |x_i|^p \right)^{1/p} < \infty \right\},$$

and ℓ^p is complete in its norm. Note the nesting

$$\ell^p(\mathbb{Z}) \subset \ell^q(\mathbb{Z}), \quad 1 \leq p \leq q.$$

Fact 12.1. $\ell^2(\mathbb{Z})$ is a Hilbert space with inner product $\langle x, y \rangle = \sum_{i \in \mathbb{Z}} x_i y_i$, and

$$L^2([-\pi, \pi)) = \left\{ f : [-\pi, \pi) \rightarrow \mathbb{C} ; \int_{-\pi}^{\pi} |f|^2 dx < \infty \right\}$$

is a Hilbert space with inner product $\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f g^* dx$.

Definition 12.2 (DTFT). For $x \in \ell^1(\mathbb{Z})$, define its *discrete-time Fourier transform* by

$$\hat{x}(\omega) = \sum_{n \in \mathbb{Z}} x(n) e^{-j\omega n}, \quad \omega \in [-\pi, \pi).$$

The series converges since $\|x\|_1 < \infty$; in fact $|\hat{x}(\omega)| \leq \|x\|_1$. This justifies the Fourier inversion identity

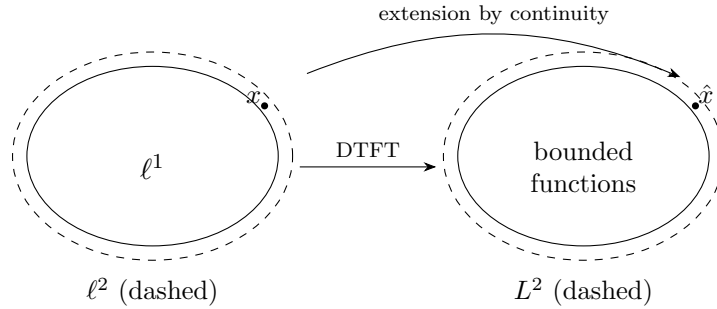
$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\omega) e^{j\omega n} d\omega,$$

and one can verify Parseval's identity:

$$\sum_{n \in \mathbb{Z}} x_n y_n = \langle x, y \rangle_{\ell^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\omega) \hat{y}(\omega)^* d\omega = \langle \hat{x}, \hat{y} \rangle_{L^2},$$

so in particular

$$\|x\|_{\ell^2}^2 = \|\hat{x}\|_{L^2}^2 \quad \forall x \in \ell^1.$$

Extension to ℓ^2 

Let $x \in \ell^2$ and pick $(x^{(n)})_{n \geq 1} \subset \ell^1$ with $\|x^{(n)} - x\|_{\ell^2} \rightarrow 0$. Then $(x^{(n)})_{n \geq 1}$ is Cauchy in ℓ^2 , and

$$\begin{aligned} & \|x^{(n)} - x^{(m)}\|_{\ell^2} \rightarrow 0 \text{ as } m, n \rightarrow \infty \\ \iff & \|\hat{x}^{(n)} - \hat{x}^{(m)}\|_{L^2} \rightarrow 0 \text{ as } m, n \rightarrow \infty \\ \iff & (\hat{x}^{(n)})_{n \geq 1} \text{ is Cauchy in } L^2, \end{aligned}$$

using Parseval on the ℓ^1 differences. By completeness there exists $\hat{x} \in L^2$ with $\|\hat{x}^{(n)} - \hat{x}\|_{L^2} \rightarrow 0$, and $\hat{x}$ is *defined* to be the Fourier transform of $x \in \ell^2$.

Theorem 12.3 (Parseval on ℓ^2).

$$\langle x, y \rangle_{\ell^2} = \langle \hat{x}, \hat{y} \rangle_{L^2} \quad \forall x, y \in \ell^2.$$

Remark 12.4. The DTFT is a generalized rotation matrix: it preserves inner products, hence lengths and angles.

Fact 12.5. If $x, y \in \ell^2$ and $\text{ess sup} |\hat{y}| < \infty$, then $z = x * y \in \ell^2$ and $\hat{z}(\omega) = \hat{x}(\omega)\hat{y}(\omega)$.

12.2 Spectral theory of WSS processes**Warm-up: the deterministic case**

Take $x \in \ell^1$, with $\hat{x}(\omega) = \sum_n x(n)e^{-j\omega n}$, and set

$$a(n) = \sum_k x(k)x(k-n) = x * x_-,$$

the autocorrelation (x_- is the flip $k \mapsto x(-k)$). Then

$$\hat{a}(\omega) = \hat{x}(\omega)\hat{x}^*(\omega) = |\hat{x}(\omega)|^2 \geq 0,$$

the *spectral density* of x .

We want to apply this idea to random processes. But we definitely do not have $\sum_n |X_n|^2 < \infty$ in general.

Idea. Truncate and see what happens.

Take $(X_n)_{n \in \mathbb{Z}}$ a zero-mean WSS process and define

$$X_n^N = \mathbf{1}_{\{|n| \leq N\}} X_n \implies (X_n^N)_{n \in \mathbb{Z}} \in \ell^1(\mathbb{Z}) \text{ a.s.}$$

Power spectral density

The energy spectral density of the truncation $(X_n^N)_{n \in \mathbb{Z}}$ is

$$A_N(\omega) = \sum_{|n| \leq N} X_n e^{-j\omega n} \left(\sum_{|n| \leq N} X_n e^{-j\omega n} \right)^* = \sum_{-N \leq m, n \leq N} X_n X_m e^{-j\omega(n-m)}.$$

Average energy normalized per unit time is power, so take expectations and divide by the window length:

$$\begin{aligned} \frac{1}{2N+1} \mathbb{E}(A_N(\omega)) &= \frac{1}{2N+1} \sum_{-N \leq m, n \leq N} R_{XX}(n-m) e^{-j\omega(n-m)} \\ &= \sum_{k=-2N}^{2N} R_{XX}(k) e^{-j\omega k} \left(1 - \frac{|k|}{2N+1} \right), \end{aligned} \quad (12.1)$$

collecting terms by the lag $k = n - m$, of which there are $2N + 1 - |k|$.

Definition 12.6 (Power spectral density). If $R_{XX} \in \ell^1$, then by dominated convergence

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \mathbb{E}(A_N(\omega)) = \hat{R}_{XX}(\omega),$$

and we write $\hat{R}_{XX}(\omega) =: S_{XX}(\omega)$, the *power spectral density* of the process X .

Remark 12.7. To see DCT applied cleanly, rewrite (12.1) as

$$\|R_{XX}\|_1 \sum_k \underbrace{\frac{|R_{XX}(k)|}{\|R_{XX}\|_1}}_{\text{prob. dist.}} \text{sgn}(R_{XX}(k)) e^{-j\omega k} \left(1 - \frac{|k|}{2N+1} \right),$$

i.e. as an expectation against a probability distribution on lags, with a bounded integrand converging pointwise to 1.

Definition 12.8 (Cross power spectrum). If $X \in L^2$ is a random variable and $Y = (Y_n)_{n \in \mathbb{Z}}$ is zero-mean WSS, define

$$S_{YX}(\omega) = \sum_{n \in \mathbb{Z}} \mathbb{E}(XY_n) e^{-j\omega n} = \hat{R}_{YX}(\omega),$$

provided the series converges in some suitable sense.

Example 12.9 (White noise). A zero-mean WSS white noise process with $\text{Var}(Z_0) = \sigma^2$ has $R_{ZZ}(n) = \delta(n)\sigma^2$, hence $S_{ZZ}(\omega) = \sigma^2$: flat across all frequencies.

Example 12.10 (Total power). By Parseval,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} S_{XX}(\omega) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{XX}(\omega) e^{j\omega \cdot 0} d\omega = R_{XX}(0) = \text{Var}(X_0),$$

the “power of X_0 ”. Also $S_{XX}(\omega) \geq 0$.

Definition 12.11 (Regular covariance). We say X has *regular covariance* if $R_{XX} \in \ell^2$ and

$$0 < \inf_{\omega} S_{XX}(\omega) \leq \sup_{\omega} S_{XX}(\omega) < \infty.$$

Think of this as the spectral analogue of $\Sigma_Y \succ 0$: it is the condition that keeps the covariance operator boundedly invertible.

Example 12.12. If X, Z are uncorrelated processes with $R_{XX} \in \ell^1$ and Z white noise, then $Y_n = X_n + Z_n$ has regular covariance: the white noise floor bounds S_{YY} away from 0.

12.3 Linear estimators as filters

Let $Y = (Y_n)_{n \in \mathbb{Z}}$ be a zero-mean WSS process with regular covariance, let $I \subset \mathbb{Z}$ be a collection of observation indices, and write $Y_I := (Y_n)_{n \in I}$.

Theorem 12.13. *For every zero-mean $X \in L^2$ there exists $h \in \ell^2(\mathbb{Z})$ such that*

$$L(X | Y_I) = \sum_{n \in I} h(n) Y_n \quad \text{in } L^2.$$

Remark 12.14 (Why regular covariance is needed). Before the proof, a counterexample showing the hypothesis cannot be dropped. Take $Y_n = X + Z_n$ with $Z_n \sim i.i.d. \mathcal{N}(0, \sigma^2)$. Then $L(X | Y) = X$ by Example 10.5, and X is *not* expressible as $\sum_n h(n) Y_n$ for any $h \in \ell^2$. Here Y has a constant common component, so S_{YY} is not bounded away from ∞ .

Proof. Consider any random variable $\sum_n a(n) Y_n$ with finitely many nonzero $a(n)$. Then

$$\begin{aligned} \mathbb{E} \left(\left| \sum_n a(n) Y_n \right|^2 \right) &= \sum_{n,k} a(n) a(k) R_{YY}(n-k) \\ &= \sum_{n,k} a(n) a(k) \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{YY}(\omega) e^{j\omega(n-k)} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{YY}(\omega) \left(\sum_n a(n) e^{j\omega n} \right) \left(\sum_k a(k) e^{-j\omega k} \right) d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{YY}(\omega) |\hat{a}(\omega)|^2 d\omega. \end{aligned}$$

Suppose $\lambda \leq S_{YY}(\omega) \leq \frac{1}{\lambda}$ for some $\lambda > 0$, which is exactly regular covariance. Then by Parseval,

$$\lambda \|a\|_2^2 \leq \mathbb{E} \left(\left| \sum_n a(n) Y_n \right|^2 \right) \leq \frac{1}{\lambda} \|a\|_2^2.$$

By definition of $L(X | Y_I)$ there are finite linear combinations $L_k = \sum_{n \in I} a_k(n) Y_n$ with $L_k \rightarrow L(X | Y_I)$ in L^2 , so $(L_k)_{k \geq 1}$ is Cauchy in L^2 . The two-sided bound above (applied to the difference $a_k - a_\ell$) gives

$$\|a_k - a_\ell\|_2^2 \asymp \mathbb{E}(|L_k - L_\ell|^2) \longrightarrow 0 \quad \text{as } k, \ell \rightarrow \infty,$$

equal up to the constants $\lambda, 1/\lambda$. Hence $(a_k)_{k \geq 1} \subset \ell^2(\mathbb{Z})$ is Cauchy, and by completeness there exists $h \in \ell^2(\mathbb{Z})$ with $\|a_k - h\|_2^2 \rightarrow 0$. $\square$

Lecture 13

The Wiener Filter

13.1 Wiener–Hopf equations

Setting. Let $(Y_n)_{n \in \mathbb{Z}}$ be a zero-mean WSS process, and write $Y_I := (Y_n)_{n \in I}$ for the observations indexed by $I \subset \mathbb{Z}$.

Theorem 13.1. *For every zero-mean $X \in L^2$ there exists $h \in \ell^2(\mathbb{Z})$ such that*

$$L(X | Y_I) = \sum_{n \in \mathbb{Z}} h(n) Y_n \quad \text{in } L^2.$$

Remark 13.2 (Important point). The best linear estimator can be expressed as a linear combination of the observations. This is what motivates moving from linear *estimation* to linear *filtering* of processes.

Theorem 13.3 (Wiener–Hopf equations). *In the setting above, h is the unique solution to the system of linear equations*

$$\mathbb{E}(XY_n) = (R_{YY} * h)[n] \quad \forall n \in I, \quad h[n] = 0 \quad n \notin I.$$

Proof. By the orthogonality principle, $\mathbb{E}(L(X | Y_I) Y_i) = \mathbb{E}(XY_i)$, and

$$\begin{aligned} \mathbb{E}(L(X | Y_I) Y_i) &= \mathbb{E} \left(\lim_{N \rightarrow \infty} \sum_{\substack{n \in \mathbb{Z} \\ |n| \leq N}} h(n) Y_n \cdot Y_i \right) \\ &= \lim_{N \rightarrow \infty} \sum_{\substack{n \in \mathbb{Z} \\ |n| \leq N}} h(n) \mathbb{E}(Y_n Y_i) \\ &= \sum_{n \in \mathbb{Z}} h(n) R_{YY}(i - n) = (R_{YY} * h)[i]. \end{aligned}$$

□

13.2 Estimating a random variable: the noncausal solution

Theorem 13.4. Let $(Y_n)_{n \in \mathbb{Z}} =: Y$ be zero-mean WSS with regular covariance, and let X be zero-mean in L^2 . Then

$$L(X | Y) = \sum_{n \in \mathbb{Z}} h(n) Y_n,$$

where h is defined by its Fourier transform

$$H(\omega) = \frac{S_{YX}(\omega)}{S_{YY}(\omega)}.$$

The estimation error is

$$\mathbb{E}((X - L(X | Y))^2) = \text{Var}(X) - \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{|S_{YX}(\omega)|^2}{S_{YY}(\omega)} d\omega.$$

Proof. Take the Fourier transform of the Wiener–Hopf equations and apply the convolution theorem to get $H = S_{YX}/S_{YY}$. For the error, orthogonality gives

$$\mathbb{E}((X - L(X | Y))^2) = \text{Var}(X) - \underbrace{\mathbb{E}(L(X | Y) X)}_{=: \textcircled{1}},$$

and

$$\textcircled{1} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} h(n) \mathbb{E}(Y_n X) = \sum_n h(n) \underbrace{\mathbb{E}(Y_n X)}_{= R_{YX}(n)} \stackrel{\text{Parseval}}{=} \frac{1}{2\pi} \int H^*(\omega) S_{YX}(\omega) d\omega.$$

□

13.3 Estimating a process: the Wiener filter

Next we estimate a whole process X given an observation process Y , where $(Y_n)_{n \in \mathbb{Z}}$ is JWSS with the zero-mean WSS process $(X_n)_{n \in \mathbb{Z}}$. The goal is to estimate X_n from the Y process, for every n at once.

Theorem 13.5 (Wiener filter). *The family $(L(X_n | Y))_{n \in \mathbb{Z}}$ can be realized by pushing Y through an LTI system with frequency response*

$$H(\omega) = \frac{S_{YX}^*(\omega)}{S_{YY}(\omega)}. \quad (13.1)$$

The estimation error is

$$\mathbb{E}((X_n - L(X_n | Y))^2) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(S_{XX}(\omega) - \frac{|S_{YX}(\omega)|^2}{S_{YY}(\omega)} \right) d\omega.$$

Proof. We need h with $L(X_n | Y) = \sum_k h(n-k) Y_k$ in L^2 . By wide-sense stationarity it suffices to take $n = 0$:

$$L(X_0 | Y) = \sum_k h(-k) Y_k \quad \text{in } L^2.$$

By Theorem 13.4, the Fourier transform of h_- (the reflection $k \mapsto h(-k)$) is $S_{YX}(\omega)/S_{YY}(\omega)$, hence the Fourier transform of h is $S_{YX}^*(\omega)/S_{YY}(\omega)$, where S_{YX} is the Fourier transform of $n \mapsto \mathbb{E}(X_0 Y_n)$. This gives (13.1). □

13.4 Linear filtering of WSS processes

Setting. X is a zero-mean WSS process $(X_n)_{n \in \mathbb{Z}}$ passed through an LTI system:

$$X \longrightarrow \boxed{g} \longrightarrow Y = X * g.$$

An LTI system is *stable* if it maps bounded inputs to bounded outputs, and

$$g \in \ell^1(\mathbb{Z}) \iff \text{the system is stable (BIBO)}.$$

Theorem 13.6 (Linear filtering of WSS processes). *Let $(X_n)_{n \in \mathbb{Z}} =: X$ be zero-mean WSS, and let g be the impulse response of a BIBO LTI system with frequency response G . Then there is a zero-mean WSS process $(Y_n)_{n \in \mathbb{Z}} =: Y$, JWSS with X , such that*

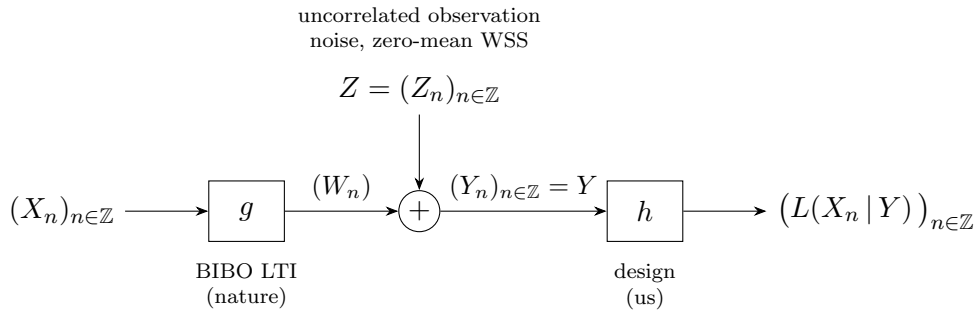
$$Y_n = \sum_k g(n-k)X_k \quad \text{a.s. and in } L^2, \quad \forall n \in \mathbb{Z}.$$

Moreover $R_{XX} \in \ell^2 \implies R_{YY} \in \ell^2$, in which case

$$S_{YY}(\omega) = |G(\omega)|^2 S_{XX}(\omega), \quad S_{YX}(\omega) = G(\omega) S_{XX}(\omega).$$

13.5 Putting it together: the engineering problem

Take $(X_n)_{n \in \mathbb{Z}}$ zero-mean WSS — the signal of interest. Nature filters it through some BIBO LTI system g and adds observation noise; we design h .



Suppose $R_{XX} \in \ell^2$ with $\sup S_{XX} < \infty$, and Z is a white noise process. Then $Y_k = W_k + Z_k$ with the two terms uncorrelated, so

$$R_{YY}(n) = \mathbb{E}(Y_0(W_n + Z_n)) = R_{WW}(n) + R_{ZZ}(n),$$

using $Y_0 = W_0 + Z_0$. Taking Fourier transforms and applying Theorem 13.6,

$$S_{YY}(\omega) = |G(\omega)|^2 S_{XX}(\omega) + S_{ZZ}(\omega), \quad S_{YX}(\omega) = G(\omega) S_{XX}(\omega),$$

and in particular Y has regular covariance. Substituting into (13.1):

$$H(\omega) = \frac{G^*(\omega) S_{XX}(\omega)}{|G(\omega)|^2 S_{XX}(\omega) + S_{ZZ}(\omega)} = \frac{1}{G(\omega) + \frac{S_{ZZ}(\omega)}{S_{XX}(\omega) G^*(\omega)}}.$$

Remark 13.7. With no noise ($S_{ZZ} \equiv 0$) this collapses to $H = 1/G$: pure deconvolution. The added term is what keeps the filter from blowing up where G is small.

Reading the second term as an inverse signal-to-noise ratio gives the compact form

$$H(\omega) = \frac{1}{G(\omega) + \frac{1}{(\text{signal-to-noise ratio}) \cdot G^*(\omega)}}.$$

Q. How do we approximate $G(\omega)$ in practice?

A. Send an impulse through the system, record the approximate impulse response, and transform it to get an approximation of $G(\omega)$.

Lecture 14

Discrete-Time Markov Chains

Second half of the course: common models for stochastic processes.

Markov chains are processes where the future is independent of the past given the present.

14.1 Definitions

Definition 14.1 (DTMC with discrete state space). A *discrete-time Markov chain* is a random process $(X_n)_{n \geq 0}$ taking values in a countable set $S := \{0, 1, \dots\}$ — the *state space* — with X_n the state of the process at time n , satisfying

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_1, \dots, X_0 = i_0\} = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$$

for all $n \geq 0$ and all states $i_0, i_1, \dots, i_{n-1}, i, j$.

Definition 14.2 (Temporal homogeneity). A Markov chain is *temporally homogeneous* if

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i\} = \mathbb{P}\{X_1 = j \mid X_0 = i\} \quad \forall n \geq 0, i, j \in S.$$

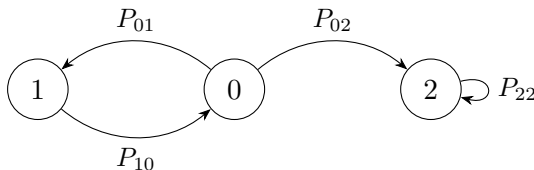
Definition 14.3. P_{ij} is the *transition probability from i to j* . The *transition matrix* collects these:

$$P := \begin{bmatrix} P_{00} & P_{01} & \cdots & P_{0k} & \cdots \\ P_{10} & P_{11} & & & \\ P_{20} & P_{21} & & \ddots & \\ \vdots & & & & \ddots \end{bmatrix}, \quad [P]_{ij} = P_{ij}.$$

The constraints say P is a *stochastic matrix*:

$$1. [P]_{ij} \geq 0, \quad 2. \sum_{j \in S} P_{ij} = 1.$$

A chain is equivalently represented by a state transition diagram.

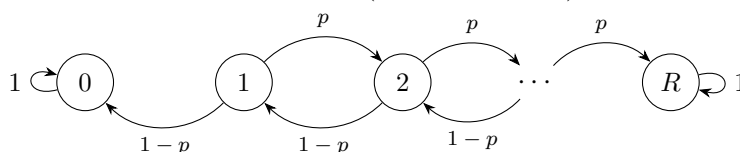


Example 14.4.

$$P := \begin{bmatrix} 0 & P_{01} & P_{02} \\ P_{10} & 0 & 0 \\ 0 & 0 & P_{22} \end{bmatrix}$$

Row sums force $P_{10} = 1$ and $P_{22} = 1$, so state 2 is absorbing — once the chain enters it, it never leaves.

Example 14.5 (Gambler's ruin).



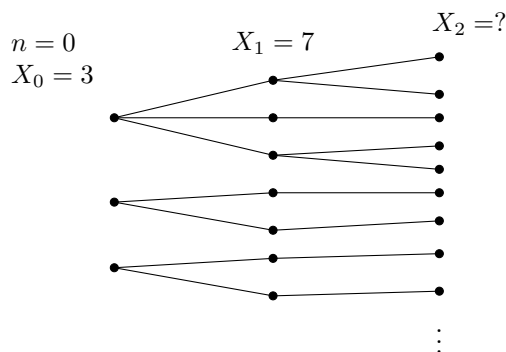
X_n is our wealth at time n . We play a game where we win one dollar with probability p and lose one dollar otherwise, stopping if bankrupt or if our wealth reaches R .

Example 14.6 (Branching process). Let (Z_n) be *i.i.d.* random variables supported on $\{0, 1, 2, \dots\}$, and set

$$P_{ij} = \mathbb{P}\{Z_1 + Z_2 + \dots + Z_i = j\}.$$

The idea: if the population at time n is X_n , each individual independently produces offspring, so the population at time $n + 1$ has distribution

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i\} = \mathbb{P}\{Z_1 + \dots + Z_i = j\}.$$



Example 14.7 (Multistage Markov chain). Suppose X_{n+1} is conditionally independent of the past given the most recent k states $(X_n, X_{n-1}, \dots, X_{n-k+1})$. Then the process $(Y_n)_{n \geq 0}$ is a Markov chain, where

$$Y_n = \begin{pmatrix} X_n \\ \vdots \\ X_{n-k+1} \end{pmatrix}.$$

Widening the state space buys back the Markov property.

14.2 Chapman–Kolmogorov

Definition 14.8. The n -step transition probabilities are $P_{ij}^n := \mathbb{P}\{X_n = j \mid X_0 = i\}$.

Theorem 14.9 (Chapman–Kolmogorov equations). For all $m, n \geq 0$ and $i, j \in S$,

$$P_{ij}^{m+n} = \sum_{k \in S} P_{ik}^m P_{kj}^n. \quad (14.1)$$

In particular $P_{ij}^n = [P^n]_{ij}$: writing $P^{(n)}$ for the matrix of n -step transition probabilities, (14.1) gives $P^{(n+1)} = P^{(n)}P$, and induction yields $P^{(n)} = P^n$. So the superscript really is a matrix power.

14.3 Classification of states

Definition 14.10 (Accessibility). State j is *accessible* from state i if $P_{ij}^n > 0$ for some $n \geq 0$; we write $i \rightarrow j$. Since $n = 0$ is allowed, $i \rightarrow i$ always. If $i \rightarrow j$ and $j \rightarrow i$, then i and j *communicate*, written $i \leftrightarrow j$.

Remark 14.11. $\leftrightarrow$ is an equivalence relation on S :

1. $i \leftrightarrow i$;
2. $i \leftrightarrow j \iff j \leftrightarrow i$;
3. $i \leftrightarrow j$ and $j \leftrightarrow k \implies i \leftrightarrow k$.

By the fundamental theorem of equivalence classes, $\leftrightarrow$ partitions S into disjoint *communicating classes*.

Example 14.12. For gambler's ruin the classes are

$$\{0\}, \quad \{R\}, \quad \{1, \dots, R-1\}.$$

Definition 14.13. A Markov chain is *irreducible* if it has only one class.

A lot of state properties turn out to be *class properties*: all states in the same class share the property.

Definition 14.14 (Period). For a state i , define its *period* $d(i) = \gcd\{n \geq 1 : P_{ii}^n > 0\}$.

Theorem 14.15. *Periodicity is a class property: if $i \leftrightarrow j$, then $d(i) = d(j)$.*

Proof. Since $i \leftrightarrow j$, pick $k, m \geq 1$ with $P_{ij}^k > 0$ and $P_{ji}^m > 0$. Chapman–Kolmogorov gives

$$P_{ii}^{k+m} \geq P_{ij}^k P_{ji}^m > 0 \implies d(i) \mid k+m.$$

Now take any ℓ with $P_{jj}^\ell > 0$. Then

$$P_{ii}^{k+m+\ell} \geq P_{ij}^k P_{jj}^\ell P_{ji}^m > 0 \implies d(i) \mid k+m+\ell \implies d(i) \mid \ell.$$

Since ℓ was arbitrary,

$$d(i) \mid \gcd\{\ell \geq 1 : P_{jj}^\ell > 0\} = d(j).$$

Exchanging the roles of i and j gives $d(j) \mid d(i)$, and two positive integers that divide each other are equal. $\square$

14.4 First passage times

Definition 14.16. Define

$$T_j := \inf\{n \geq 1 : X_n = j\}, \quad F_{ij}(\infty) = \mathbb{P}(T_j < \infty \mid X_0 = i).$$

Definition 14.17 (Recurrent / transient). A state i is *recurrent* if $F_{ii}(\infty) = 1$. If $F_{ii}(\infty) < 1$, then i is *transient*.

Lecture 15

Midterm Solutions

15.1 Problem 1: geometric to exponential

Problem 15.1. Let $X_p \sim \text{Geom}(p)$, so $\mathbb{P}\{X_p = k\} = p(1-p)^k$ for $k \in \mathbb{N}$. Find the distribution of pX_p as $p \downarrow 0$.

Approach 1: characteristic functions

$$\varphi_{pX_p}(t) = \mathbb{E}(e^{jtpX_p}) = \sum_{k \geq 0} p(1-p)^k e^{jtpk} = p \sum_{k \geq 0} z^k, \quad z := (1-p)e^{jtp}, \quad |z| < 1,$$

so the geometric series converges and

$$\varphi_{pX_p}(t) = \frac{p}{1 - (1-p)e^{jtp}}.$$

Dividing numerator and denominator by p ,

$$\lim_{p \downarrow 0} \frac{p}{1 - (1-p)e^{jtp}} = \lim_{p \downarrow 0} \frac{1}{\frac{1}{p} - \left(\frac{1}{p} - 1\right) e^{jtp}} = \frac{1}{1 - jt} = \varphi_{\text{Exp}(1)}(t),$$

using $e^{jtp} = 1 + jtp + O(p^2)$, which makes the denominator $1 - jt + O(p)$.

Approach 2: tail probabilities

For $t \geq 0$,

$$\mathbb{P}(pX_p > t) = \mathbb{P}(X_p > \lfloor t/p \rfloor) = \sum_{k=\lfloor t/p \rfloor + 1}^{\infty} p(1-p)^k = (1-p)^{\lfloor t/p \rfloor + 1}.$$

Since $\log(1-p) = -p + O(p^2)$,

$$(1-p)^{\lfloor t/p \rfloor + 1} \sim e^{-p(\lfloor t/p \rfloor + 1)} \xrightarrow{p \downarrow 0} e^{-t} = \mathbb{P}(\text{Exp}(1) > t).$$

Either way, $pX_p \xrightarrow{d} \text{Exp}(1)$.

15.2 Problem 2: a symmetric correlation bound

Problem 15.2. Show that

$$(A) \quad \forall \text{ bounded } f : \text{Var}(\mathbb{E}(f(X) | Y)) \leq \alpha \text{Var}(f(X))$$

implies

$$(B) \quad \forall \text{ bounded } g : \text{Var}(\mathbb{E}(g(Y) | X)) \leq \alpha \text{Var}(g(Y)).$$

Proof. Start from the tower property and Cauchy–Schwarz:

$$\begin{aligned} \text{Cov}(f(X), g(Y))^2 &= \text{Cov}(\mathbb{E}(f(X) | Y), g(Y))^2 \\ &\leq \text{Var}(\mathbb{E}(f(X) | Y)) \text{Var}(g(Y)) \\ &\leq \alpha \text{Var}(f(X)) \text{Var}(g(Y)), \end{aligned}$$

the last step by (A). Now specialize to $f(X) = \mathbb{E}(g(Y) | X)$. The left-hand side becomes, again by the tower property,

$$\text{Cov}(\mathbb{E}(g(Y) | X), g(Y))^2 = \text{Var}(\mathbb{E}(g(Y) | X))^2,$$

while the right-hand side is

$$\alpha \text{Var}(\mathbb{E}(g(Y) | X)) \text{Var}(g(Y)).$$

Cancelling one factor of $\text{Var}(\mathbb{E}(g(Y) | X))$ gives

$$\text{Var}(\mathbb{E}(g(Y) | X)) \leq \alpha \text{Var}(g(Y)). \quad \square$$

Proposition 15.3. $\alpha = 0$ if and only if X and Y are independent.

Proof. ($\Rightarrow$) Suppose $X \perp Y$. Then $\mathbb{E}(f(X) | Y) = \mathbb{E}(f(X))$ is a constant, so $\text{Var}(\mathbb{E}(f(X) | Y)) = 0$.

($\Leftarrow$) Suppose $\alpha = 0$, so $\text{Var}(\mathbb{E}(f(X) | Y)) = 0$ for every bounded f . Then $\mathbb{E}(f(X) | Y) = \mathbb{E}(f(X))$ a.s., and hence

$$\mathbb{E}(f(X)g(Y)) = \mathbb{E}(\mathbb{E}(f(X) | Y)g(Y)) = \mathbb{E}(f(X))\mathbb{E}(g(Y)),$$

for all bounded f, g , which is exactly independence. $\square$

15.3 Problem 3: density of polynomials

Problem 15.4. Show that if $\mathbb{E}(e^{t|X|}) < \infty$ for all $t \in \mathbb{R}$, then for any f with $f(X) \in L^2$ there is a sequence of polynomials $(P_n)_{n \geq 1}$ with $P_n(X) \rightarrow f(X)$ in L^2 .

(i) Reduction

It suffices to show:

$$\mathbb{E}(|\varphi(X)|^2) < \infty \text{ and } \mathbb{E}(X^k \varphi(X)) = 0 \quad \forall k \geq 0 \implies \varphi(X) = 0 \text{ a.s.} \quad (15.1)$$

Indeed, let $L(f(X) | X^0, X^1, \dots)$ be the best linear estimator of $f(X)$ given all powers of X — that is, the L^2 -closure of the polynomials in X . Note each X^k really is in L^2 , since $X^k = O(e^{|X|})$ and the moment generating function is finite everywhere, so the space is well defined. By the orthogonality principle,

$$\mathbb{E}\left(X^k(f(X) - L(f(X) | X^0, X^1, \dots))\right) = 0 \quad \forall k.$$

Applying (15.1) to $\varphi(X) := f(X) - L(f(X) | X^0, X^1, \dots)$ gives $f(X) = L(f(X) | X^0, \dots)$ a.s., which is precisely the statement that $f(X)$ is an L^2 -limit of polynomials.

(ii) Proving the reduction

We use the fact that $\mathbb{E}(e^{jtX} \varphi(X)) = 0$ for all t implies $\varphi(X) = 0$ a.s. — uniqueness of Fourier transforms, applied to the complex measure with density φ against the law of X .

Set $Z := e^{jtX} \varphi(X)$ and $Z_n := \sum_{k=0}^n \frac{(jtX)^k}{k!} \varphi(X)$, so $Z_n \rightarrow Z$ a.s. The partial sums are dominated:

$$|Z_n| = \left| \sum_{k=0}^n \frac{(jtX)^k}{k!} \varphi(X) \right| \leq \sum_{k=0}^n \frac{|t|^k |X|^k}{k!} |\varphi(X)| \leq e^{|t||X|} |\varphi(X)|,$$

and the dominating function is integrable by Cauchy–Schwarz:

$$\mathbb{E}\left(e^{|t||X|} |\varphi(X)|\right) \leq \sqrt{\mathbb{E}(e^{2|t||X|}) \mathbb{E}(|\varphi(X)|^2)} < \infty.$$

So dominated convergence applies, and

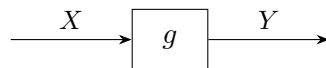
$$\mathbb{E}(e^{jtX} \varphi(X)) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(jt)^k}{k!} \mathbb{E}(X^k \varphi(X)) = 0,$$

every term vanishing by hypothesis. Hence $\varphi(X) = 0$ a.s.

15.4 Problem 4: a whitening filter

Problem 15.5. Let $X = (X_n)_{n \in \mathbb{Z}}$ be a zero-mean WSS white noise process with $\text{Var}(X_0) = 1$, passed through an LTI filter g to produce Y , where

$$R_{YY}(n) = \begin{cases} 1, & n = 0, \\ -\sqrt{3}/4, & |n| = 1, \\ 0, & \text{otherwise.} \end{cases}$$



Find g , and compute the error in estimating X_n from Y .

(i) Identifying the filter

Since $R_{XX} = \delta$,

$$R_{YY} = R_{XX} * g * g_- = g * g_-,$$

where $g_-(n) = g(-n)$. Matching coefficients for a two-tap filter $g = (g_1, g_2)$:

$$\begin{cases} g_1^2 + g_2^2 = 1, \\ g_1 g_2 = -\sqrt{3}/4. \end{cases}$$

The first equation reads 1, not 0 — it has to match $R_{YY}(0) = 1$. Solving, $g_1 = \frac{1}{2}$ and $g_2 = -\frac{\sqrt{3}}{2}$, i.e.

$$Y_n = \frac{1}{2}X_n - \frac{\sqrt{3}}{2}X_{n-1}, \quad n \in \mathbb{Z}.$$

(ii) The estimation error

X and Y are JWSS, and the noncausal Wiener error is

$$\mathbb{E}\left((X_n - L(X_n | Y))^2\right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(S_{XX}(\omega) - \frac{|S_{YX}(\omega)|^2}{S_{YY}(\omega)} \right) d\omega.$$

Here

$$S_{YY}(\omega) = 1 - \frac{\sqrt{3}}{2} \cos(\omega) \geq 1 - \frac{\sqrt{3}}{2} > 0,$$

so Y has regular covariance and the formula is valid. With $G(\omega) = \frac{1}{2} - \frac{\sqrt{3}}{2}e^{-j\omega}$ we have $S_{YX} = GS_{XX} = G$ and $S_{YY} = |G|^2 S_{XX} = |G|^2$, so

$$S_{XX} - \frac{|S_{YX}|^2}{S_{YY}} = 1 - \frac{|G|^2}{|G|^2} = 0,$$

and the error is exactly zero. The filter is invertible in the L^2 sense — its spectrum never vanishes — so X is perfectly recoverable from Y .

Lecture 16

Markov Processes: Recurrence and Limit Theorems

Throughout, $(X_n)_{n \geq 0}$ is a Markov chain on a countable state space S .

16.1 Recurrence and transience

Definition 16.1 (First passage time).

$$T_j := \inf\{n \geq 1 : X_n = j\}.$$

Definition 16.2 (Recurrent / transient). A state i is

1. *recurrent* if $\Pr\{T_i < \infty \mid X_0 = i\} = 1$;
2. *transient* if $\Pr\{T_i < \infty \mid X_0 = i\} < 1$.

Equivalently: (1) holds iff, given $X_0 = i$, the process revisits i infinitely often a.s.; (2) holds iff it revisits i only finitely often a.s.

Lemma 16.3. *State i is recurrent $\iff \sum_{n \geq 1} P_{ii}^n = \infty$.*

Proof. Let N be the number of reentries into state i given $X_0 = i$. By monotone convergence,

$$\mathbb{E}(N \mid X_0 = i) = \mathbb{E}\left(\sum_{n \geq 1} \mathbf{1}_{\{X_n = i\}} \mid X_0 = i\right) = \sum_{n \geq 1} P_{ii}^n.$$

□

Theorem 16.4. *Transience and recurrence are class properties.*

Proof. Assume $i \leftrightarrow j$, so there exist m, n with $P_{ij}^m > 0$ and $P_{ji}^n > 0$. For every $s \geq 1$,

$$P_{jj}^{m+n+s} \geq P_{ji}^m P_{ii}^s P_{ij}^n,$$

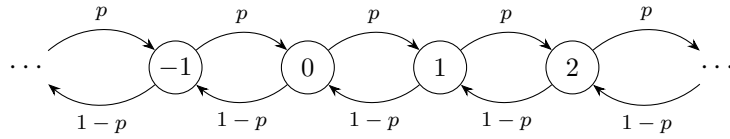
so summing over s ,

$$\sum_s P_{jj}^{m+n+s} \geq \underbrace{P_{ji}^m P_{ij}^n}_{>0} \sum_s P_{ii}^s.$$

By Lemma 16.3, i recurrent $\implies j$ recurrent. By symmetry j recurrent $\implies i$ recurrent, so i recurrent $\iff j$ recurrent. $\square$

Corollary 16.5. *If $i \leftrightarrow j$ and i is recurrent, then $\Pr\{T_j < \infty \mid X_0 = i\} = 1$.*

Example 16.6 (Random walk on $\mathbb{Z}$). Consider the walk that steps right with probability p and left with probability $1 - p$.



If $p \in (0, 1)$ the chain is irreducible. The chain has period 2, so state 0 is recurrent $\iff \sum_{n \geq 0} P_{00}^{2n} = \infty$. By Stirling,

$$\sum_{n \geq 0} P_{00}^{2n} = \sum_{n \geq 0} \binom{2n}{n} p^n (1-p)^n \approx \sum_{n \geq 0} \frac{(4p(1-p))^n}{\sqrt{\pi n}} = \begin{cases} +\infty & p = \frac{1}{2}, \\ < \infty & p \neq \frac{1}{2}, \end{cases}$$

since $4p(1-p) < 1$ unless $p = \frac{1}{2}$. So the walk is recurrent exactly in the symmetric case.

16.2 Markov limit theorems

These quantify the long-term behavior of Markov chains.

Notation.

$$N_j(n) := \#\{1 \leq k \leq n : X_k = j\}$$

is the number of reentries into state j up to time n , and

$$\mu_{jj} = \mathbb{E}(T_j \mid X_0 = j)$$

is the mean return time to j .

Theorem 16.7 (SLLN for Markov chains). *Let $(X_n)_{n \geq 0}$ be a Markov chain started in state $X_0 = i$. If $i \leftrightarrow j$, then*

$$\frac{N_j(n)}{n} \xrightarrow{\text{a.s.}} \frac{1}{\mu_{jj}}.$$

Proof. If j is transient, then $N_j(n) < \infty$ with probability 1, so $N_j(n)/n \rightarrow 0$ w.p. 1; and $\mu_{jj} = +\infty$, so $1/\mu_{jj} = 0$ as well.

If j is recurrent, the chain enters state j after a finite time w.p. 1, so without loss of generality we may start in state $X_0 = j$. Define $(\tau_k)_{k \geq 1}$ to be the round trip times of the k th tour $j \rightarrow j$. These are *i.i.d.* with

$$\tau_k \stackrel{d}{=} T_j \mid X_0 = j, \quad \mathbb{E}(\tau_k) = \mathbb{E}(T_j \mid X_0 = j) = \mu_{jj}.$$

By the SLLN, $\frac{1}{N} \sum_{k=1}^N \tau_k \rightarrow \mu_{jj}$ a.s., and since $N_j(n) \rightarrow \infty$ a.s.,

$$\frac{1}{N_j(n)} \sum_{k=1}^{N_j(n)} \tau_k \xrightarrow{\text{a.s.}} \mu_{jj}.$$

Now define

$$S_j(n) := \sum_{k=1}^{N_j(n)} \tau_k = \text{sum of round trip times completed by time } n,$$

so that $S_j(n) \leq n \leq S_{N_j(n)+1}$. Dividing through,

$$\mu_{jj} \xleftarrow{\text{a.s.}} \frac{S_{N_j(n)}}{N_j(n)} \leq \frac{n}{N_j(n)} \leq \underbrace{\frac{S_{N_j(n)+1}}{N_j(n)+1}}_{\rightarrow \mu_{jj} \text{ a.s.}} \cdot \underbrace{\frac{N_j(n)+1}{N_j(n)}}_{\rightarrow 1 \text{ a.s.}}.$$

Hence $n/N_j(n) \rightarrow \mu_{jj}$ a.s., which is equivalent to $N_j(n)/n \rightarrow 1/\mu_{jj}$ a.s. □

Corollary 16.8. *If $(X_n)_{n \geq 0}$ is irreducible, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P_{ij}^k = \frac{1}{\mu_{jj}}.$$

Proof.

$$\begin{aligned} \frac{1}{n} \sum_{k=1}^n P_{ij}^k &= \frac{1}{n} \sum_{k=1}^n \mathbb{E}(\mathbf{1}_{\{X_k=j\}} \mid X_0=i) \\ &= \mathbb{E} \left(\frac{1}{n} \sum_{k=1}^n \mathbf{1}_{\{X_k=j\}} \mid X_0=i \right) = \mathbb{E} \left(\frac{N_j(n)}{n} \mid X_0=i \right) =: S_n. \end{aligned}$$

Since $|N_j(n)/n| \leq 1$ for all n , dominated convergence and Theorem 16.7 give

$$\lim_{n \rightarrow \infty} S_n = \mathbb{E} \left(\lim_{n \rightarrow \infty} \frac{N_j(n)}{n} \mid X_0=i \right) = \frac{1}{\mu_{jj}}.$$

□

16.3 Stationary distributions

Definition 16.9 (Stationary distribution). A probability distribution $\pi = (\pi_i)_{i \in S}$ is a *stationary distribution* if, viewed as a row vector, $\pi P = \pi$; equivalently

$$\pi_j = \sum_{i \in S} \pi_i P_{ij}, \quad \forall j \in S.$$

Idea. If π is stationary and we start in a random state $X_0 \sim \pi$, then

$$\Pr\{X_1 = j\} = \sum_{i \in S} P_{ij} \pi_i = \pi_j \implies X_n \sim \pi \quad \forall n \geq 0.$$

The distribution is frozen in time even though the chain keeps moving.

Q. When does a stationary distribution exist, and when is it unique?

Definition 16.10 (Positive / null recurrence). A recurrent state j is

$$\begin{cases} \text{positive recurrent} & \text{if } \mu_{jj} < \infty, \\ \text{null recurrent} & \text{if } \mu_{jj} = \infty. \end{cases}$$

Fact 16.11. *Positive and null recurrence are class properties.*

Theorem 16.12. *An irreducible Markov chain satisfies exactly one of the following.*

1. *All states are transient or null recurrent. In this case*

$$\frac{1}{n} \sum_{k=1}^n P_{ij}^k \longrightarrow 0 \quad \forall i, j,$$

and no stationary distribution exists.

2. *All states are positive recurrent. In this case*

$$\pi_j = \frac{1}{\mu_{jj}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P_{ij}^k.$$

Example 16.13 (Ergodic theorem for Markov chains / Markov reward theorem). Let the chain be irreducible with stationary distribution π , and let $r: S \rightarrow \mathbb{R}$ be bounded. Then

$$\frac{1}{n} \sum_{k=1}^n r(X_k) \xrightarrow{\text{a.s.}} \underbrace{\sum_{j \in S} \pi_j r(j)}_{= \mathbb{E}_\pi(r)}.$$

The idea is just bookkeeping on top of Theorem 16.7:

$$\frac{1}{n} \sum_{k=1}^n r(X_k) = \sum_{j \in S} \frac{N_j(n)}{n} r(j),$$

and each occupation fraction converges to $\pi_j = 1/\mu_{jj}$.

Lecture 17

More Markov Chains and Reversibility

17.1 Recap: the classification theorem

Theorem 17.1 (Final theorem on Markov chains). *An irreducible Markov chain satisfies exactly one of the following.*

1. *All states are transient or null recurrent. No stationary distribution exists, and*

$$\frac{1}{n} \sum_{k=1}^n P_{ij}^k \longrightarrow 0 \quad \text{as } n \rightarrow \infty, \quad \forall i, j \in S.$$

2. *All states are positive recurrent. A unique stationary distribution exists, given by*

$$\pi_j = \frac{1}{\mu_{jj}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P_{ij}^k.$$

The key ingredient throughout is irreducibility.

Theorem 17.2 (Ergodic theorem). *Let $r: S \rightarrow \mathbb{R}$ be bounded and let the chain be irreducible with stationary distribution π . Then*

$$\frac{1}{n} \sum_{k=1}^n r(X_k) \longrightarrow \mathbb{E}_\pi(r) \quad \text{a.s.}$$

Example 17.3 (Random walk on a graph). Let $G(V, E)$ be a finite connected graph, and take

$$P_{ij} = \begin{cases} \frac{1}{\deg(i)} & \text{if } (i, j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The chain is positive recurrent by the pigeonhole principle (finitely many states), so the stationary distribution exists and is unique. It is

$$\pi_i = \frac{\deg(i)}{2|E|},$$

which one checks directly:

$$\pi_j = \frac{\deg(j)}{2|E|} \stackrel{?}{=} \sum_i P_{ij} \pi_i = \sum_{i: i \sim j} \frac{1}{\deg(i)} \cdot \frac{\deg(i)}{2|E|} = \frac{\deg(j)}{2|E|}. \quad \checkmark$$

Q. What is the average round trip time from j back to j ?

A. $\mu_{jj} = \frac{1}{\pi_j} = \frac{2|E|}{\deg(j)}.$

Well-connected vertices are revisited often; leaves are revisited rarely.

17.2 Convergence to stationarity by coupling

Theorem 17.4. Let $(X_n)_{n \geq 0}$ be an irreducible aperiodic Markov chain with stationary distribution π . Then

$$\lim_{n \rightarrow \infty} \sum_j |P_{ij}^n - \pi_j| = 0 \quad \forall i \in S,$$

that is, $\lim_{n \rightarrow \infty} \|P_{i \cdot}^n - \pi\|_1 = 0$.

Remark 17.5. The same theorem essentially holds for periodic chains, in the sense that if j has period d , then $P_{jj}^{dn} \rightarrow d\pi_j$ for all $j \in S$.

Proof (introduces the new idea of coupling). Let $(X_n)_{n \geq 0}$ and $(Y_n)_{n \geq 0}$ be independent copies of the chain with the same transition matrix P . Then $(X_n, Y_n)_{n \geq 0}$ is a Markov chain on $S \times S$ with transition probabilities

$$P_{(i,i'),(j,j')} = P_{ij}P_{i'j'}.$$

The product chain is irreducible. By irreducibility and aperiodicity of the individual chain, for each j there is $n_0(j)$ with $P_{jj}^n > 0$ for all $n \geq n_0(j)$. By irreducibility there exist k, L with $P_{ij}^k > 0$ and $P_{i'j'}^L > 0$. Then

$$P_{(i,i'),(j,j')}^{k+L+M} = P_{ij}^{k+L+M} P_{i'j'}^{k+L+M} \geq P_{ij}^k P_{jj}^{L+M} \cdot P_{i'j'}^L P_{j'j'}^{k+M} > 0$$

for all $M \geq \max(n_0(j), n_0(j'))$. $\checkmark$

The product chain has a stationary distribution. $Q_{jj'} = \pi_j \pi_{j'}$ is stationary for the product chain. Since the product chain is irreducible, this stationary distribution is unique and all states are positive recurrent.

Coupling time. For each state j define

$$T_j = \inf\{n \geq 0 : X_n = j, Y_n = j\}, \quad T = \inf\{n \geq 0 : X_n = Y_n\}.$$

Note $T \leq T_j$, and by positive recurrence of the product chain $\Pr\{T_j < \infty\} = 1$, so $\Pr\{T < \infty\} = 1$.

After coupling, the two chains agree in law. Given $\{T \leq n\}$, X_n and Y_n are equal in distribution:

$$\begin{aligned}
 \mathbb{P}\{X_n = j, T \leq n\} &= \sum_{m=0}^n \sum_i \mathbb{P}\{X_n = j, T = m, X_m = i\} \\
 &= \sum_{m=0}^n \sum_i \mathbb{P}\{X_n = j \mid X_m = i, T = m\} \mathbb{P}\{X_m = i, T = m\} \\
 &= \sum_{m=0}^n \sum_i \mathbb{P}\{Y_n = j \mid Y_m = i\} \mathbb{P}\{Y_m = i, T = m\} \\
 &= \mathbb{P}\{Y_n = j, T \leq n\},
 \end{aligned}$$

using that on $\{T = m\}$ we have $X_m = Y_m = i$ and both chains evolve with the same transition matrix afterwards.

Splitting on the coupling event.

$$\begin{aligned}
 \mathbb{P}\{X_n = j\} &= \mathbb{P}\{X_n = j, T \leq n\} + \mathbb{P}\{X_n = j, T > n\} \\
 &= \mathbb{P}\{Y_n = j, T \leq n\} + \mathbb{P}\{X_n = j, T > n\} \\
 &\leq \mathbb{P}\{Y_n = j\} + \mathbb{P}\{X_n = j, T > n\},
 \end{aligned}$$

so

$$\mathbb{P}\{X_n = j\} - \mathbb{P}\{Y_n = j\} \leq \mathbb{P}\{X_n = j, T > n\}.$$

By symmetry in X and Y ,

$$|\mathbb{P}\{X_n = j\} - \mathbb{P}\{Y_n = j\}| \leq \mathbb{P}\{X_n = j, T > n\} + \mathbb{P}\{Y_n = j, T > n\},$$

and summing over j ,

$$\sum_j |\mathbb{P}\{X_n = j\} - \mathbb{P}\{Y_n = j\}| \leq \sum_j \left(\mathbb{P}\{X_n = j, T > n\} + \mathbb{P}\{Y_n = j, T > n\} \right) = 2\mathbb{P}(T > n).$$

Conclusion. Start with $Y_0 \sim \pi$ (so $Y_n \sim \pi$ for all n) and $X_0 = i$. Then

$$\sum_j |P_{ij}^n - \pi_j| \leq 2\mathbb{P}(T > n),$$

and since $\Pr\{T < \infty\} = 1$,

$$\limsup_{n \rightarrow \infty} \sum_j |P_{ij}^n - \pi_j| \leq \limsup_{n \rightarrow \infty} 2\mathbb{P}\{T > n\} = 0.$$

□

17.3 Reversibility

Definition 17.6 (Reversible chain). A Markov chain with transition matrix P and stationary distribution π is *reversible* if

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \forall i, j \in S \quad (\text{detailed balance equations}).$$

In this case we say (P, π) is reversible.

Q. Why the term *reversible*?

A. Because if $X_0 \sim \pi$, then

$$(X_0, \dots, X_n) \stackrel{d}{=} (X_n, X_{n-1}, \dots, X_0) :$$

the chain looks the same run backwards as forwards.

Lecture 18

Spectral Gap and First Step Analysis

18.1 Reversibility, continued

Definition 18.1. We say (P, π) is *reversible* if

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \forall i, j \in S.$$

Remark 18.2. Detailed balance forces π to be stationary:

$$\sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \quad (\iff \pi P = \pi),$$

since the rows of P sum to 1.

If (π, P) is reversible, $(X_n)_{n \geq 0}$ is a Markov chain with transition matrix P , and $X_0 \sim \pi$, then for all $k \geq 0$,

$$(X_0, \dots, X_k) \stackrel{d}{=} (X_k, \dots, X_0).$$

Example 18.3 (Random walk on a graph, again). With $P_{ij} = 1/\deg(i)$ on edges, detailed balance reads

$$\pi_i \frac{1}{\deg(i)} = \pi_j \frac{1}{\deg(j)} \implies \pi_j \propto \deg(j).$$

Normalizing, with $\pi_j = \alpha \deg(j)$,

$$1 = \sum_j \pi_j = \alpha \sum_j \deg(j) = \alpha \cdot 2|E| \implies \alpha = \frac{1}{2|E|}. \quad \checkmark$$

Detailed balance recovers Example 17.3 in one line, without solving $\pi P = \pi$.

18.2 Convergence to equilibrium

Definition 18.4 (Total variation distance).

$$\|\mu - \nu\|_{\text{TV}} = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)| = \frac{1}{2} \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)| = \frac{1}{2} \|\mu - \nu\|_{\ell^1},$$

where the last two expressions require $|\Omega| \leq |\mathbb{N}|$.

Last time: $\|P_i^n - \pi\|_{\text{TV}} \rightarrow 0$ as $n \rightarrow \infty$, for a Markov chain that is irreducible, aperiodic, and positive recurrent.

The question now is *how fast*.

For $f: S \rightarrow \mathbb{R}$, define

$$\text{Var}_\pi(f) = \text{Var}(f(X)), \quad X \sim \pi, \quad (Pf)(i) = \sum_j P_{ij}f(j) = \mathbb{E}(f(X_1) \mid X_0 = i).$$

Definition 18.5 (Spectral gap). For reversible (P, π) , define the *spectral gap*

$$\gamma := 1 - \lambda_2,$$

where $\lambda_2 \geq 0$ is the smallest number λ satisfying

$$\text{Var}_\pi(Pf) \leq \lambda \text{Var}_\pi(f) \quad \forall f: S \rightarrow \mathbb{R} \text{ with } \text{Var}_\pi(f) < \infty.$$

Unpacked, this says

$$\text{Var}\left(\mathbb{E}(f(X_1) \mid X_0)\right) \leq \lambda \text{Var}(f(X_0)) :$$

one step of the chain contracts variance by a factor λ .

Where the terminology comes from

View f as a random variable on $L^2(S, 2^S, \pi)$. Then

$$\begin{aligned} \langle g, Pf \rangle &= \sum_i g(i) \left(\sum_j P_{ij}f(j) \right) \pi_i \\ &\stackrel{\text{DBE}}{=} \sum_i g(i) \sum_j \pi_j P_{ji}f(j) \\ &= \sum_j f(j) \underbrace{\left(\sum_i P_{ji}g(i) \right)}_{(Pg)(j)} \pi_j = \langle Pg, f \rangle, \end{aligned}$$

so P is a *self-adjoint* linear transformation on $L^2(S, 2^S, \pi)$.

Compare the finite-dimensional picture. With $\langle x, y \rangle = x^\top y$ for $x, y \in \mathbb{R}^n$, suppose A is a linear transformation with $\langle x, Ay \rangle = \langle Ax, y \rangle$ for all x, y . Then

$$x^\top Ay = (Ax)^\top y = x^\top A^\top y \implies A = A^\top,$$

so A is diagonalizable with real spectrum. Correspondingly the spectrum of P satisfies

$$\Lambda(P) \subset [-\lambda_2, \lambda_2] \cup \{1\},$$

with the eigenvalue 1 belonging to the constant function, and $\gamma = 1 - \lambda_2$ measuring the gap between 1 and the rest of the spectrum.

18.3 Mixing bound from the spectral gap

Theorem 18.6. *If (P, π) has spectral gap γ , then*

$$\|P_i^n - \pi\|_{\text{TV}}^2 \leq \frac{(1 - \gamma)^n}{\pi_i} \quad \forall i \in S, \quad n \geq 0.$$

Let μ, π be probability measures on $(S, 2^S)$ with $\pi(i) = 0 \implies \mu(i) = 0$. We say μ has *density* h with respect to π if

$$\mu_j = h(j)\pi_j \quad (d\mu = h d\pi).$$

Lemma 18.7. *Let (P, π) be reversible. If $d\mu = h d\pi$, then $P^n h$ is the density of μP^n with respect to π .*

Proof of Theorem 18.6. First, $P^n h$ has mean 1 under π : by self-adjointness and $P\mathbf{1} = \mathbf{1}$,

$$\langle \mathbf{1}, P^n(h - \mathbf{1}) \rangle = \langle P^n \mathbf{1}, h - \mathbf{1} \rangle = \langle \mathbf{1}, h - \mathbf{1} \rangle = \sum_j (\mu_j - \pi_j) = 0.$$

Hence

$$\langle P^n h - \mathbf{1}, P^n h - \mathbf{1} \rangle = \|P^n h - \mathbf{1}\|^2 = \text{Var}_\pi(P^n h) \leq (1 - \gamma)^n \text{Var}_\pi(h),$$

iterating the spectral gap inequality n times, where

$$\text{Var}_\pi(P^n h) = \sum_j ((P^n h)(j))^2 \pi_j - \left(\sum_j (P^n h)(j) \pi_j \right)^2.$$

Now compute the total variation distance:

$$\begin{aligned} \|\mu P^n - \pi\|_{\text{TV}}^2 &= \left(\frac{1}{2} \sum_i \pi_i \left| \frac{(\mu P^n)_i}{\pi_i} - 1 \right| \right)^2 = \left(\frac{1}{2} \sum_i \pi_i |(P^n h)(i) - 1| \right)^2 \\ &\stackrel{\text{Jensen}}{\leq} \frac{1}{4} \sum_i \pi_i |(P^n h)(i) - 1|^2 = \frac{1}{4} \|P^n h - \mathbf{1}\|^2 \leq \frac{1}{4} (1 - \gamma)^n \text{Var}_\pi(h). \end{aligned}$$

Taking $\mu = \delta_i$ gives $h(j) = \mathbf{1}_{\{j=i\}}/\pi_i$ and $\text{Var}_\pi(h) = 1/\pi_i - 1$, which yields the stated bound (with room to spare). $\square$

Remark 18.8. For an irreducible, positive recurrent, finite Markov chain, the rate of convergence is determined by the second largest eigenvalue in absolute value.

18.4 First step analysis

Often the quantity of interest is a hitting time,

$$T_A := \inf\{n \geq 0 : X_n \in A\}.$$

Example 18.9 (Gambler's ruin). $T_{\{0,R\}}$ is the time until the game ends.

Expected hitting time

Condition on the first step. Writing $g(i) = \mathbb{E}(T_A \mid X_0 = i)$,

$$\mathbb{E}(T_A \mid X_0 = i) = \begin{cases} 1 + \sum_{j \in S} P_{ij} \mathbb{E}(T_A \mid X_0 = j) & \forall i \notin A, \\ 0 & \text{otherwise,} \end{cases}$$

that is,

$$g(i) = \begin{cases} 1 + \sum_j P_{ij} g(j) & i \notin A, \\ 0 & \text{otherwise.} \end{cases}$$

Hitting probabilities

For states $a \neq b$, define

$$h(i) := \mathbb{P}(T_a < T_b \mid X_0 = i).$$

Then

$$h(a) = 1, \quad h(b) = 0, \quad h(i) = \sum_j P_{ij} h(j) \quad \forall i \notin \{a, b\}.$$

No problem — linearly solve. :)

Lecture 19

Martingales and Stopping Times

19.1 Definition

Definition 19.1 (Martingale). Let $(X_n)_{n \geq 0}$ be a stochastic process. A process $(M_n)_{n \geq 0}$ is said to be a *martingale with respect to X* if

1. $(M_n)_{n \geq 0}$ is adapted to $(X_n)_{n \geq 0}$;
2. $\mathbb{E}(|M_n|) < \infty$ for all $n \geq 0$;
3. $\mathbb{E}(M_{n+1} \mid X_0, \dots, X_n) = M_n$ for all $n \geq 0$. ⊙

Remark 19.2. Think of X_n as just more and more information — “info revealed up to time n ”. That M is adapted to X means precisely that M_n is a measurable function of $X_0, \dots, X_n$ for every $n \geq 0$.

Remark 19.3. Condition ⊙ says the game is fair: knowing everything up to time n , your expected position at time $n + 1$ is exactly where you already are.

Definition 19.4 (Sub- and supermartingales). If “=” in condition 3 is replaced by “ $\geq$ ”, we say M is a *submartingale*; “ $\leq$ ” gives a *supermartingale*.

Remark 19.5. If M is a martingale, then it is both a sub- and a supermartingale.

19.2 Examples

Example 19.6. The symmetric random walk $(S_n)_{n \geq 0}$ on $\mathbb{Z}$ is a martingale with respect to itself.

Example 19.7. Gambler’s ruin with fair odds is a martingale with respect to itself.

Example 19.8. Let $(X_n)_{n \geq 0}$ be independent and integrable. Then

$$M_n = \sum_{i=0}^n (X_i - \mathbb{E}(X_i))$$

is a martingale with respect to both itself and X .

Example 19.9. Let $(X_n)_{n \geq 0}$ be independent and integrable with $\mathbb{E}(X_n) = 1$. Then $M_n = \prod_{i=0}^n X_i$ is a martingale:

$$\mathbb{E}(M_{n+1} \mid X_0, \dots, X_n) = \mathbb{E}(M_n X_{n+1} \mid X_0, \dots, X_n) = M_n \mathbb{E}(X_{n+1}) = M_n. \quad \checkmark$$

The middle step pulls out M_n , which is a function of $X_0, \dots, X_n$, and then uses independence of X_{n+1} from $X_0, \dots, X_n$.

Example 19.10 (Branching process). Let $(X_n)_{n \geq 0}$ be a branching process, $X_n :=$ the population at the n th generation. Let $\mu :=$ the expected number of offspring per individual, and take X_0 integrable. *Claim:* $M_n = \frac{X_n}{\mu^n}$ is a martingale.

Proof. Let $Z_1, \dots, Z_{X_n}$ be the numbers of offspring of individuals $1, \dots, X_n$ in generation n , so that

$$X_{n+1} = \sum_{i=1}^{X_n} Z_i.$$

Then

$$\begin{aligned} \mathbb{E}(M_{n+1} \mid X_0, \dots, X_n) &= \frac{1}{\mu^{n+1}} \mathbb{E}(X_{n+1} \mid X_0, \dots, X_n) \\ &= \frac{1}{\mu^{n+1}} \mathbb{E}\left(\sum_{i=1}^{X_n} Z_i \mid X_0, \dots, X_n\right) = \frac{\mu X_n}{\mu^{n+1}} = M_n. \end{aligned}$$

□

Example 19.11 (Doob martingale). Let $(X_n)_{n \geq 0}$ be any process and Y any integrable random variable. Then

$$M_n = \mathbb{E}(Y \mid X_0, \dots, X_n)$$

is a martingale with respect to X — immediately from the tower property.

Example 19.12. Take $Y = f(X_0, \dots, X_N)$ for integrable f . In this case $M_n \rightarrow Y$ a.s.; indeed $M_n = Y$ exactly once $n \geq N$, since Y is then already a function of the conditioning variables.

19.3 Conditioning further into the future

Proposition 19.13. If $(M_n)_{n \geq 0}$ is a martingale (resp. sub-, supermartingale) with respect to $(X_n)_{n \geq 0}$, then

$$\mathbb{E}(M_m \mid X_0, \dots, X_n) = M_n \quad (\text{resp. } \geq, \leq) \quad \forall m > n.$$

Proof. It suffices to consider M a submartingale; for the supermartingale case apply the result to $-M$. Iterate the tower property: for $k \geq 1$,

$$\begin{aligned} \mathbb{E}(M_{n+k} \mid X_0, \dots, X_n) &= \mathbb{E}\left(\mathbb{E}(M_{n+k} \mid X_0, \dots, X_{n+k-1}) \mid X_0, \dots, X_n\right) \\ &\geq \mathbb{E}(M_{n+k-1} \mid X_0, \dots, X_n) \geq \dots \geq M_n, \end{aligned}$$

each inequality using the submartingale property together with monotonicity of conditional expectation. □

Remark 19.14. Taking expectations, $\mathbb{E}(M_n) = \mathbb{E}(M_0)$ if M is a martingale ($\leq$ if supermartingale, $\geq$ if submartingale).

19.4 Stopping times and the stopped process

Definition 19.15 (Stopping time). A random variable T taking values in $\{0, 1, 2, \dots\} \cup \{+\infty\}$ is a *stopping time* with respect to $(X_n)_{n \geq 0}$ if for every $n \geq 0$ the event $\{T \leq n\}$ is determined by $(X_0, \dots, X_n)$, i.e. $\mathbf{1}_{\{T \leq n\}}$ is a measurable function of $(X_0, \dots, X_n)$.

Example 19.16. Let $(X_n)_{n \geq 0}$ be the stock market outcome and T the time at which I sell my portfolio. Whether I have sold by time n can only depend on what I have seen by time n .

Proposition 19.17 (Stopped process). Let $(M_n)_{n \geq 0}$ be a (sub-, super-) martingale with respect to $(X_n)_{n \geq 0}$ and let T be a stopping time. Then the stopped process $(M_{n \wedge T})_{n \geq 0}$ is also a martingale (resp. sub-, supermartingale). Here $a \wedge b = \min(a, b)$.

Proof. It suffices to assume M is a submartingale.

Integrability. Splitting on the value of T ,

$$\begin{aligned} \mathbb{E}(|M_{T \wedge n}|) &= \sum_{i=0}^{n-1} \mathbb{E}(|M_{T \wedge n}| \mathbf{1}_{\{T=i\}}) + \mathbb{E}(|M_{T \wedge n}| \mathbf{1}_{\{T \geq n\}}) \\ &= \sum_{i=0}^{n-1} \mathbb{E}(|M_i| \mathbf{1}_{\{T=i\}}) + \mathbb{E}(|M_n| \mathbf{1}_{\{T \geq n\}}) < \infty, \end{aligned}$$

a finite sum of terms each dominated by $\mathbb{E}(|M_i|) < \infty$.

Adaptedness. Both $(M_i)_{0 \leq i \leq n}$ and $(\mathbf{1}_{\{T=i\}})_{0 \leq i \leq n}$ are functions of $(X_0, \dots, X_n)$, and

$$M_{T \wedge n} = \underbrace{M_n \mathbf{1}_{\{T > n\}}}_{M_n(1 - \mathbf{1}_{\{T \leq n\}})} + \underbrace{M_T \mathbf{1}_{\{T \leq n\}}}_{\text{fn of } (X_0, \dots, X_n)},$$

so $M_{T \wedge n}$ is a function of $(X_0, \dots, X_n)$ as well.

Submartingale property. Split according to whether the process has stopped, using that $T \wedge (n+1) = n+1$ on $\{T \geq n+1\}$ and $T \wedge (n+1) = T \wedge n$ on $\{T \leq n\}$:

$$\begin{aligned} \mathbb{E}(M_{T \wedge (n+1)} \mid X_0, \dots, X_n) &= \mathbb{E}(M_{T \wedge (n+1)} \mathbf{1}_{\{T \geq n+1\}} \mid X_0, \dots, X_n) \\ &\quad + \mathbb{E}(M_{T \wedge (n+1)} \mathbf{1}_{\{T \leq n\}} \mid X_0, \dots, X_n) \\ &= \mathbb{E}(M_{n+1} \mid X_0, \dots, X_n) \mathbf{1}_{\{T \geq n+1\}} + \mathbb{E}(M_{T \wedge n} \mathbf{1}_{\{T \leq n\}} \mid X_0, \dots, X_n) \\ &\geq M_n \mathbf{1}_{\{T \geq n+1\}} + M_{T \wedge n} \mathbf{1}_{\{T \leq n\}} \\ &= M_{n \wedge T} \mathbf{1}_{\{T \geq n+1\}} + M_{T \wedge n} \mathbf{1}_{\{T \leq n\}} = M_{T \wedge n}. \end{aligned}$$

Both indicators are functions of $(X_0, \dots, X_n)$, which is what lets them come outside the conditional expectation. $\square$

Example 19.18. My portfolio value $(M_n)_{n \geq 0}$ is a martingale and the time at which I sell is T . Then $\mathbb{E}(M_{T \wedge n}) = \mathbb{E}(M_0)$: no selling rule based only on the past changes the expected payoff.

Proposition 19.19. If $(M_n)_{n \geq 0}$ is a submartingale and T a stopping time, both with respect to $(X_n)_{n \geq 0}$, then

$$\mathbb{E}(M_0) \stackrel{(a)}{\leq} \mathbb{E}(M_{T \wedge n}) \stackrel{(b)}{\leq} \mathbb{E}(M_n).$$

Proof. We just saw that $(M_{T \wedge n})_{n \geq 0}$ is a submartingale, which gives ③ by Remark 19.14.

For ④, we claim $M'_n := M_n - M_{T \wedge n}$ is a submartingale. Granting the claim,

$$\mathbb{E}(M'_n) \geq \mathbb{E}(M'_0) = 0, \quad \text{and} \quad \mathbb{E}(M'_n) = \mathbb{E}(M_n) - \mathbb{E}(M_{T \wedge n}),$$

which is ④. To prove the claim, note

$$M'_{n+1} = M_{n+1} - M_{T \wedge (n+1)} = \mathbf{1}_{\{T \leq n\}}(M_{n+1} - M_n) + \underbrace{(M_n - M_{T \wedge n})}_{M'_n},$$

since on $\{T \leq n\}$ we have $T \wedge (n+1) = T = T \wedge n$, while on $\{T \geq n+1\}$ both sides vanish. Therefore

$$\begin{aligned} \mathbb{E}(M'_{n+1} \mid X_0, \dots, X_n) &= \mathbb{E}(\mathbf{1}_{\{T \leq n\}}(M_{n+1} - M_n) \mid X_0, \dots, X_n) + M'_n \\ &= \mathbf{1}_{\{T \leq n\}} \underbrace{\mathbb{E}(M_{n+1} - M_n \mid X_0, \dots, X_n)}_{\geq 0} + M'_n \\ &\geq M'_n. \end{aligned}$$

□

Lecture 20

Optional Stopping and Concentration

20.1 Recap

Last time: $(M_n)_{n \geq 0}$ is a submartingale with respect to $(X_n)_{n \geq 0}$ if

1. M is adapted to X ;
2. each M_n is integrable;
3. $\mathbb{E}(M_{n+1} \mid X_0, \dots, X_n) \geq M_n$ for all $n \geq 0$.

T is a stopping time with respect to X if $\{T \leq n\} \in \sigma(X_0, \dots, X_n)$ for all $n \geq 0$. The stopped process $(M_{T \wedge n})_{n \geq 0}$ is then a submartingale and satisfies

$$\mathbb{E}(M_0) \leq \mathbb{E}(M_{T \wedge n}) \leq \mathbb{E}(M_n) \quad \forall n \geq 0.$$

Replace all inequalities by “=” for martingales, and by “ $\geq$ ” for supermartingales.

Idea. Submartingales on average like to increase, so you cannot lose much on average by betting on a submartingale.

Goal. Today: the optional stopping theorem and applications to concentration. Throughout, M is a martingale and T a stopping time, both with respect to X .

20.2 The optional stopping theorem

Theorem 20.1 (Optional stopping). *Suppose any one of the following holds.*

- (i) *There is $k < \infty$ such that $T \leq k$ a.s.*
- (ii) *There is an integrable Z such that $|M_{n \wedge T}| \leq Z$ a.s. for all $n \geq 0$, and $T < \infty$ a.s.*
- (iii) *$\mathbb{E}(T) < \infty$, and there is $k < \infty$ such that $|M_n - M_{n-1}| \leq k$ a.s. for all $n \geq 1$.*

Then

$$\mathbb{E}(M_0) \leq \mathbb{E}(M_T)$$

for submartingales, with equality if M is a martingale.

Proof. (i). Already proved: taking $n = k$ in $\mathbb{E}(M_0) \leq \mathbb{E}(M_{T \wedge n}) \leq \mathbb{E}(M_n)$ gives $M_{T \wedge k} = M_T$ a.s.

(ii). Since $T < \infty$ a.s., $M_{T \wedge n} \xrightarrow{\text{a.s.}} M_T$ as $n \rightarrow \infty$. The dominating random variable Z lets us apply the dominated convergence theorem, so $\mathbb{E}(M_{T \wedge n}) \rightarrow \mathbb{E}(M_T)$; combined with $\mathbb{E}(M_0) \leq \mathbb{E}(M_{T \wedge n})$ this gives $\mathbb{E}(M_0) \leq \mathbb{E}(M_T)$.

(iii). Telescoping and bounding each increment,

$$|M_{T \wedge n} - M_0| \leq \sum_{i=1}^{T \wedge n} |M_i - M_{i-1}| \leq (T \wedge n)k \leq Tk \implies |M_{T \wedge n}| \leq |M_0| + Tk.$$

Since $\mathbb{E}(T) < \infty$ we have $T < \infty$ a.s., and $Z := |M_0| + Tk$ is integrable. So (iii) $\implies$ (ii), and we are done. $\square$

Example 20.2 (Gambler's ruin with fair odds). Consider the symmetric random walk $(S_n)_{n \geq 0}$ on $\mathbb{Z}$ with $S_0 = k$, and the stopped process $(S_{n \wedge T})_{n \geq 0}$, where

$$T := \inf\{n \geq 0 : S_n \in \{0, R\}\}.$$

What is the ruin probability — or rather its complement, $\mathbb{P}(S_T = R)$?

By optional stopping,

$$k = \mathbb{E}(S_0) \stackrel{\text{OST}}{=} \mathbb{E}(S_T) = \mathbb{P}(S_T = 0) \cdot 0 + \mathbb{P}(S_T = R) \cdot R \implies \mathbb{P}(S_T = R) = \frac{k}{R}.$$

Remark 20.3. The hypothesis being used is (ii): the stopped walk never leaves $[0, R]$, so $|S_{n \wedge T}| \leq R$, a constant and hence integrable. That $T < \infty$ a.s. is the standard fact that the symmetric walk exits a bounded interval in finite time. Note T itself is unbounded, so (i) does not apply.

20.3 Wald's identity

Theorem 20.4 (Wald's identity). *Let $(Y_n)_{n \geq 1}$ be adapted to $(X_n)_{n \geq 1}$, with Y_{n+1} independent of $(X_1, \dots, X_n)$ for every $n \geq 1$,*

$$\sup_{n \geq 1} \mathbb{E}|Y_n| < \infty, \quad \text{and} \quad \mathbb{E}(Y_n) = \mu \quad \forall n \geq 1.$$

Let $T \geq 1$ be a stopping time with respect to X with $\mathbb{E}(T) < \infty$. Then

$$\mathbb{E}\left(\sum_{n=1}^T Y_n\right) = \mu \mathbb{E}(T).$$

Proof. Key observation. The event $\{T < n\} = \{T \leq n-1\}$ is determined by $(X_1, \dots, X_{n-1})$, hence is independent of Y_n ; so is its complement:

$$\{T \geq n\} \perp\!\!\!\perp Y_n.$$

A dominating variable. Put $Z := \sum_{n=1}^T |Y_n - \mu|$. Then

$$\begin{aligned} \mathbb{E}(Z) &= \mathbb{E} \left(\sum_{n=1}^T |Y_n - \mu| \right) = \mathbb{E} \left(\sum_{n \geq 1} |Y_n - \mu| \mathbf{1}_{\{T \geq n\}} \right) \\ &= \sum_{n \geq 1} \underbrace{\mathbb{E}(|Y_n - \mu|)}_{\leq C \in \mathbb{R}} \mathbb{E}(\mathbf{1}_{\{T \geq n\}}) \leq C \sum_{n \geq 1} \mathbb{P}(T \geq n) = C \mathbb{E}(T) < \infty, \end{aligned}$$

using the key observation to split the expectation, and $\mathbb{E}|Y_n - \mu| \leq \sup_n \mathbb{E}|Y_n| + |\mu| =: C$.

Conclusion. Define

$$M_n = \sum_{i=1}^n (Y_i - \mu), \quad M_0 = 0,$$

which is a martingale with respect to X , since $Y_{n+1} \perp\!\!\!\perp (X_1, \dots, X_n)$ gives $\mathbb{E}(Y_{n+1} - \mu \mid X_1, \dots, X_n) = 0$. Hence $(M_{T \wedge n})_{n \geq 0}$ is a martingale, and

$$|M_{T \wedge n}| \leq \sum_{i=1}^T |Y_i - \mu| = Z.$$

So Theorem 20.1(ii) applies and gives

$$\mathbb{E}(M_T) = \mathbb{E}(M_0) = 0 \quad \Longleftrightarrow \quad \mathbb{E} \left(\sum_{n=1}^T Y_n \right) - \mu \mathbb{E}(T) = 0.$$

□

20.4 Concentration inequalities

Theorem 20.5 (Doob's martingale inequality). *Let $(M_n)_{n \geq 0}$ be a submartingale. For every $\alpha > 0$ and $n \geq 0$,*

$$\mathbb{P} \left(\max_{0 \leq k \leq n} M_k \geq \alpha \right) \leq \frac{\mathbb{E}(M_n^+)}{\alpha}, \quad \text{where } M_n^+ = \max\{M_n, 0\}.$$

Proof. Let $T = \inf\{n \geq 0 : M_n^+ \geq \alpha\}$, and use the fact that $(M_n^+)_{n \geq 0}$ is a submartingale — conditional Jensen applied to the convex nondecreasing map $x \mapsto x^+$. Since $\alpha > 0$, the maximum of M_k exceeds α exactly when the maximum of M_k^+ does, and on that event $T \leq n$ with $M_{T \wedge n}^+ = M_T^+ \geq \alpha$. Hence

$$\mathbb{P} \left(\max_{0 \leq k \leq n} M_k^+ \geq \alpha \right) = \mathbb{P}(M_{T \wedge n}^+ \geq \alpha) \leq \frac{\mathbb{E}(M_{T \wedge n}^+)}{\alpha} \leq \frac{\mathbb{E}(M_n^+)}{\alpha},$$

the first inequality by Markov and the second by $\mathbb{E}(M_{T \wedge n}^+) \leq \mathbb{E}(M_n^+)$ for the submartingale M^+ . □

Example 20.6 (Maximal inequality for nonnegative martingales). If $(M_n)_{n \geq 0}$ is a nonnegative martingale, then for every $\alpha > 0$,

$$\mathbb{P} \left(\sup_{n \geq 0} M_n \geq \alpha \right) \leq \frac{\mathbb{E}(M_0)}{\alpha}.$$

Proof. Since $M \geq 0$ we have $M_n^+ = M_n$ and $\mathbb{E}(M_n) = \mathbb{E}(M_0)$ for every n . Fix $0 < \beta < \alpha$. The events $A_n := \{\max_{0 \leq k \leq n} M_k \geq \beta\}$ increase in n and $\{\sup_n M_n \geq \alpha\} \subset \bigcup_n A_n$, so by Theorem 20.5,

$$\mathbb{P}\left(\sup_{n \geq 0} M_n \geq \alpha\right) \leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \frac{\mathbb{E}(M_0)}{\beta}.$$

Letting $\beta \uparrow \alpha$ gives the claim. □

Lecture 21

Martingale Bounds and the Convergence Theorem

21.1 Doob's maximal inequality

Theorem 21.1 (Doob's (sub)martingale inequality). *If $(M_n)_{n \geq 1}$ is a submartingale, then for all $\alpha > 0$ and $n \geq 0$,*

$$\Pr \left\{ \max_{1 \leq k \leq n} M_k \geq \alpha \right\} \leq \frac{\mathbb{E}(M_n^+)}{\alpha}.$$

Example 21.2 (Nonnegative martingales). If M is a nonnegative martingale, then $\mathbb{E}(M_n^+) = \mathbb{E}(M_n) = \mathbb{E}(M_0)$, so

$$\Pr \left(\sup_{k \geq 0} M_k \geq \alpha \right) \leq \frac{\mathbb{E}(M_0)}{\alpha}.$$

The bound is uniform in n , hence extends to the supremum over all time.

Example 21.3 (Branching processes). Let $M_n = X_n/\mu^n$, where $(X_n)_{n \geq 0}$ is a branching process with expected offspring μ . Then M is a nonnegative martingale, so

$$\mathbb{P} \left(\sup_{k \geq 0} \frac{X_k}{\mu^k} \geq \alpha \right) \leq \frac{\mathbb{E}(X_0)}{\alpha}.$$

Example 21.4 (Brownian motion). For $\lambda > 0$, $e^{\lambda B_t - \lambda^2 t/2}$ is a nonnegative martingale, and optimizing over λ gives

$$\mathbb{P} \left(\sup_{0 \leq t \leq 1} B_t \geq c \right) = \mathbb{P} \left(\sup_{0 \leq t \leq 1} e^{\lambda B_t - \lambda^2 t/2} \geq e^{\lambda c} \right) \leq e^{-c^2/2}.$$

21.2 Azuma–Hoeffding

Theorem 21.5 (Azuma–Hoeffding). *Let $(M_n)_{n \geq 0}$ be a submartingale with respect to $(X_n)_{n \geq 0}$, and let $(L_k)_{k \geq 1}$, $(U_k)_{k \geq 1}$ be predictable with respect to X — i.e. L_{k+1} and U_{k+1} are functions of $(X_0, \dots, X_k)$ for all $k \geq 0$ — with*

$$L_k \leq M_{k-1} - M_k \leq U_k \quad \forall k \geq 1, \quad U_k - L_k \leq C_k \text{ a.s. } \forall k \geq 1,$$

for some constants $(C_k)_{k \geq 1} \subset \mathbb{R}$. Then

$$\Pr\{M_n \leq M_0 - t\} \leq \exp\left(\frac{-2t^2}{\sum_{k=1}^n C_k^2}\right) \quad \forall t > 0, n \geq 0.$$

Corollary 21.6. *If $(M_n)_{n \geq 0}$ is a martingale with $|M_k - M_{k-1}| \leq C_k$ a.s. for all $k \geq 1$, then*

$$\mathbb{P}(|M_n - M_0| \geq t) \leq 2 \exp\left(\frac{-t^2}{2 \sum_{k=1}^n C_k^2}\right).$$

Lemma 21.7 (Hoeffding). *If Y is real valued with $\mathbb{E}(Y) \leq 0$ and $a \leq Y \leq b$ a.s. for $a, b \in \mathbb{R}$, then*

$$\mathbb{E}(e^{\alpha Y}) \leq \exp\left(\frac{\alpha^2}{8}(b-a)^2\right).$$

Proof of Theorem 21.5. Let $Y_k = M_{k-1} - M_k$ for $k \geq 1$. The submartingale property gives $\mathbb{E}(Y_k | X_0, \dots, X_{k-1}) \leq 0$, and by hypothesis $L_k \leq Y_k \leq U_k$ with $U_k - L_k \leq C_k$. Conditioning on the last step and applying Lemma 21.7 to the innermost expectation,

$$\begin{aligned} \mathbb{E}(e^{\alpha(M_0 - M_n)}) &= \mathbb{E}\left(\mathbb{E}\left(\prod_{k=1}^n e^{\alpha Y_k} \mid X_0, \dots, X_{n-1}\right)\right) \\ &= \mathbb{E}\left(\prod_{k=1}^{n-1} e^{\alpha Y_k} \mathbb{E}(e^{\alpha Y_n} | X_0, \dots, X_{n-1})\right) \\ &\leq \exp\left(\frac{\alpha^2}{8} C_n^2\right) \mathbb{E}\left(\prod_{k=1}^{n-1} e^{\alpha Y_k}\right) \leq \exp\left(\frac{\alpha^2}{8} \sum_{k=1}^n C_k^2\right), \end{aligned}$$

iterating the bound down to $k = 1$. (The bracketing variables L_n, U_n are predictable, so they are constants under the conditioning, which is what lets the lemma apply.) By Markov's inequality,

$$\mathbb{P}(M_0 - M_n \geq t) = \mathbb{P}(e^{\alpha(M_0 - M_n)} \geq e^{\alpha t}) \leq e^{-\alpha t} \exp\left(\frac{\alpha^2}{8} \sum_{k=1}^n C_k^2\right).$$

Minimizing over α (at $\alpha = 4t / \sum_k C_k^2$) gives the stated bound. $\square$

21.3 McDiarmid's inequality

Theorem 21.8 (McDiarmid). *Let $X_1, \dots, X_n$ be independent random variables and consider $f: (X_1, \dots, X_n) \mapsto f(X_1, \dots, X_n) \in \mathbb{R}$ with bounded differences:*

$$|f(X_1, \dots, X_k, \dots, X_n) - f(X_1, \dots, X'_k, \dots, X_n)| \leq C_k$$

for all $k \in \{1, \dots, n\}$ and all $X_1, \dots, X_n, X'_k$. Then

$$\mathbb{P}\{f(X_1, \dots, X_n) \geq \mathbb{E}(f(X_1, \dots, X_n)) + t\} \leq \exp\left(\frac{-2t^2}{\sum_{k=1}^n C_k^2}\right).$$

Proof. First, f is bounded: swapping coordinates one at a time,

$$|f(X_1, \dots, X_n) - f(0, \dots, 0)| \leq C_1 + |f(0, X_2, \dots, X_n) - f(0, \dots, 0)| \leq \dots \leq \sum_{k=1}^n C_k.$$

Now build the Doob martingale

$$M_k = \mathbb{E}(f(X_1, \dots, X_n) \mid X_1, \dots, X_k), \quad k \geq 1, \quad M_0 = \mathbb{E}(f(X_1, \dots, X_n)),$$

and set

$$\left. \begin{aligned} L_k &= \inf_{x_k} \mathbb{E}(f(X_1, \dots, X_n) \mid X_0, \dots, X_k = x_k) - M_{k-1} \\ U_k &= \sup_{x_k} \mathbb{E}(f(X_1, \dots, X_n) \mid X_0, \dots, X_k = x_k) - M_{k-1} \end{aligned} \right\} \text{predictable.}$$

Then $L_k \leq M_k - M_{k-1} \leq U_k$ and $U_k - L_k \leq C_k$ by bounded differences. Apply Azuma–Hoeffding. $\square$

Example 21.9 (Chromatic number of an Erdős–Rényi graph). Let $G \sim G(n, p)$ be an Erdős–Rényi random graph on n vertices, where edges are placed independently with probability p . Let $\chi(G)$ be the *chromatic number* of G : the minimum number of colors needed to paint the vertices so that no edge is monochromatic. Computing it is NP-complete.

Claim 21.10. If $G \sim G(n, p)$, then

$$\mathbb{P}(|\chi(G) - \mathbb{E}(\chi(G))| \geq t\sqrt{n}) \leq 2e^{-2t^2}.$$

Proof. Label the vertices $1, \dots, n$ and let X_i be the set of edges from vertex i to $\{1, \dots, i-1\}$. These are independent, and $\chi(G) = \chi(X_1, \dots, X_n)$. Changing one X_k changes the chromatic number by at most one — give the affected vertex its own color — so

$$|\chi(X_1, \dots, X_n) - \chi(X_1, \dots, X'_k, \dots, X_n)| \leq 1.$$

Invoke McDiarmid with $C_k = 1$, so $\sum_k C_k^2 = n$. $\square$

Note that we get sharp concentration without knowing $\mathbb{E}(\chi(G))$ at all — the quantity is hard to compute, but it barely fluctuates.

21.4 The martingale convergence theorem

Motivation (gambler’s ruin). We have $M_n \rightarrow M_\infty$ a.s., with

$$M_\infty = \begin{cases} 0 & \text{with probability } p, \\ R & \text{with probability } 1 - p. \end{cases}$$

Theorem 21.11 (Martingale convergence theorem). Let $(M_n)_{n \geq 0}$ be a submartingale with $\sup_{n \geq 0} \mathbb{E}(|M_n|) < \infty$. Then

$$M := \lim_{n \rightarrow \infty} M_n$$

exists a.s. and is integrable.

Example 21.12 (Critical and subcritical branching). Let $(X_n)_{n \geq 0}$ be a branching process with mean offspring $\mu \leq 1$. Then $(X_n)_{n \geq 0}$ is a nonnegative supermartingale (a martingale exactly when $\mu = 1$), and since $\mathbb{E}(X_n) = \mathbb{E}(X_0)\mu^n$,

$$\sup_{n \geq 0} \mathbb{E}(|X_n|) = \sup_{n \geq 0} \mathbb{E}(X_n) = \mathbb{E}(X_0) < \infty.$$

So $X_\infty = \lim_{n \rightarrow \infty} X_n$ exists. Since the X_n are integer valued, the limit forces the process to be eventually constant, leaving two possibilities:

1. the offspring distribution is deterministically 1, so $X_n \equiv X_0$; or
2. the process dies out almost surely.

Lecture 22

Martingale Convergence and Poisson Processes

22.1 The martingale convergence theorem

Theorem 22.1 (Martingale convergence). *Let $(M_n)_{n \geq 0}$ be a submartingale with $\sup_{n \geq 0} \mathbb{E}(|M_n|) < \infty$. Then $M = \lim_{n \rightarrow \infty} M_n$ exists a.s. and is integrable.*

Example 22.2 (Opinion dynamics on a graph). Let $G = (V, E)$ be an undirected connected graph. At time $n \geq 0$ each vertex $v \in V$ holds a belief $S_n(v) \in \{0, 1\}$. At the start of the next round each vertex independently, and with probability $p \in (0, 1)$, adopts the belief of one of its neighbors chosen uniformly at random.

Fact 22.3.

$$M_n = \sum_{v \in V} \deg(v) S_n(v), \quad n \geq 0,$$

is a martingale with respect to $X_n = (S_n(1), S_n(2), \dots, S_n(|V|))$, $n \geq 0$.

Proof. For a single vertex,

$$\mathbb{E}[S_{n+1}(v) \mid X_n] = (1 - p)S_n(v) + \frac{p}{\deg(v)} \sum_{u \sim v} S_n(u).$$

Multiplying by $\deg(v)$ and summing, the double sum can be reorganized by counting each edge from the other end, $\sum_v \sum_{u \sim v} S_n(u) = \sum_u \deg(u) S_n(u)$, so

$$\mathbb{E}[M_{n+1} \mid X_n] = (1 - p)M_n + p \sum_{u \in V} \deg(u) S_n(u) = (1 - p)M_n + pM_n = M_n. \quad \square$$

The two consensus configurations (all beliefs 0, all beliefs 1) are absorbing, and from any other configuration there is positive probability of reaching one of them in a bounded number of steps. Since the state space is finite, all beliefs eventually polarize to all 0 or all 1 with probability 1. Let

$$T = \inf\{n \geq 0 : \text{all beliefs agree}\}, \quad \mathbb{P}(T < \infty) = 1.$$

Now $M_T = 2|E|$ on the event that all beliefs polarize to 1, and $M_T = 0$ on the event that they polarize to 0, since $\sum_v \deg(v) = 2|E|$. So

$$\mathbb{E}(M_T) = \mathbb{P}\{\text{all beliefs} \rightarrow 1\} \sum_{v \in V} \deg(v) + \mathbb{P}\{\text{all beliefs} \rightarrow 0\} \cdot 0 = 2|E| \mathbb{P}\{\text{all beliefs} \rightarrow 1\}.$$

The martingale is bounded and T is a.s. finite, so optional stopping gives $\mathbb{E}(M_T) = \mathbb{E}(M_0)$ and therefore

$$\mathbb{P}\{\text{all beliefs} \rightarrow 1\} = \frac{1}{2|E|} \mathbb{E}(M_T) = \frac{1}{2|E|} \mathbb{E}(M_0) = \frac{\sum_v \deg(v) S_0(v)}{\sum_v \deg(v)}.$$

The chance of the graph settling on belief 1 is the degree-weighted fraction of vertices that started there — high-degree vertices count for more.

22.2 Poisson processes

The Poisson process is the canonical *point* process, or *counting* process. Its basic building block is the exponential distribution.

Proposition 22.4 (Memorylessness). *For $s, t \geq 0$,*

$$\mathbb{P}\{T > t + s \mid T > t\} = \mathbb{P}\{T > s\}.$$

Proof. The condition says $\mathbb{P}(T > t + s) = \mathbb{P}\{T > t\} \mathbb{P}\{T > s\}$, i.e. writing F^c for the complementary distribution function,

$$F^c(t + s) = F^c(t) F^c(s) \quad \forall t, s \geq 0.$$

Since F^c is monotone (hence measurable), the only solutions of this Cauchy functional equation are

$$F^c(t) = e^{-\lambda t} \quad \text{for some } \lambda > 0$$

(together with the degenerate cases $F^c \equiv 0$ and $F^c \equiv 1$). □

If $T \sim \text{Exp}(\lambda)$ then

$$\mathbb{P}\{T \leq t\} = 1 - e^{-\lambda t}, \quad f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \quad \mathbb{E}(T) = \frac{1}{\lambda}, \quad \text{Var}(T) = \frac{1}{\lambda^2}.$$

Idea (Basic intuition). If buses arrive at a bus stop at rate λ according to a memoryless arrival process, then I have to wait T seconds for the first bus to show up — and how long I have already been waiting tells me nothing about how much longer I will wait.

Proposition 22.5.

1. If $T_i \sim \text{Exp}(\lambda_i)$ are independent random variables, then $\min(T_1, \dots, T_n) \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$.
2. If $\tau_1, \dots, \tau_k \sim \text{Exp}(\lambda)$ are i.i.d., then $T = \tau_1 + \dots + \tau_k$ has the Erlang distribution

$$f_T(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{k-1}}{(k-1)!} \mathbf{1}_{\{t \geq 0\}}.$$

Definition 22.6 (Poisson process). Let $(\tau_k)_{k \geq 1}$ be *i.i.d.* $\text{Exp}(\lambda)$ and set

$$T_0 = 0, \quad T_n = \sum_{k=1}^n \tau_k, \quad n \geq 1.$$

For $t \geq 0$ define

$$N_t = \sup\{n \geq 0 : T_n \leq t\}.$$

The process $(N_t)_{t \geq 0}$ is called a *rate- λ Poisson process*.

Example 22.7 (Counting process). For each ω :

1. $t \mapsto N_t(\omega)$ is right-continuous, with $N_t \in \mathbb{Z}$;
2. $t \mapsto N_t(\omega)$ is monotone (non-decreasing);
3. $t \mapsto N_t(\omega)$ is *simple*: every jump has size exactly 1, so no two arrivals occur at the same instant.

Q. Is it Poisson?

Recall $X \sim \text{Poisson}(\lambda)$ means $\mathbb{P}(X = k) = e^{-\lambda} \lambda^k / k!$ for $k \in \{0, 1, \dots\}$.

Proposition 22.8. *If $(N_t)_{t \geq 0}$ is a rate- λ Poisson process, then $N_t \sim \text{Poisson}(\lambda t)$ for every $t \geq 0$.*

Proof. Condition on the two independent pieces — T_n , which is Erlang, and $\tau_{n+1} \sim \text{Exp}(\lambda)$, which is independent of it:

$$\mathbb{P}\{N_t = n\} = \mathbb{P}\{T_n \leq t < T_n + \tau_{n+1}\} = \int_0^t \lambda e^{-\lambda u} \frac{(\lambda u)^{n-1}}{(n-1)!} e^{-\lambda(t-u)} du = \frac{e^{-\lambda t} (\lambda t)^n}{n!}. \quad \square$$

By the memorylessness of the exponential distribution, $(N_{t+s} - N_s)_{t \geq 0}$ is independent of $(N_t)_{0 \leq t \leq s}$ and is itself a rate- λ Poisson process.

Corollary 22.9. *Poisson processes have independent, stationary increments.*

This leads to an abstract characterization of the Poisson process.

Theorem 22.10 (Abstract characterization). *If $(N_t)_{t \geq 0}$ is a rate- λ Poisson process, then*

1. $N_0 = 0$;
2. $(N_t)_{t \geq 0}$ has independent, stationary increments;
3. $N_t \sim \text{Poisson}(\lambda t)$ for all $t \geq 0$.

Conversely, if these three properties hold for a counting process, then it is a rate- λ Poisson process.

22.3 Transformations of Poisson processes

Superposition

Theorem 22.11. If $(N_t^{(i)})_{t \geq 0}$, $i = 1, \dots, k$, are independent Poisson processes of rates λ_i , then

$$N_t = \sum_{i=1}^k N_t^{(i)}$$

is a Poisson process of rate $\lambda = \sum_{i=1}^k \lambda_i$.

Thinning

The reverse direction corresponds to thinning.

Theorem 22.12. Let $(N_t)_{t \geq 0}$ be a rate- λ Poisson process, and independently mark arrivals “0” or “1” with probabilities p and $1 - p$ respectively. Let $(N_t^{(i)})_{t \geq 0}$ be the process counting arrivals marked “ i ”, $i \in \{0, 1\}$. Then the marked processes $(N_t^{(0)})_{t \geq 0}$ and $(N_t^{(1)})_{t \geq 0}$ are independent Poisson processes of rates $p\lambda$ and $(1 - p)\lambda$ respectively.

Remark 22.13. The independence is the surprising part. For a fixed time horizon the two counts are built from a common total, which one would expect to couple them — but the total is itself Poisson, and a Poisson count split by independent coin flips gives independent Poisson pieces.

22.4 Inhomogeneous Poisson processes

These allow for a varying intensity of arrivals.

Definition 22.14 (IhPP). An *inhomogeneous Poisson process* with rate function $\lambda_t \geq 0$ (its *intensity*) is a counting process $(N_t)_{t \geq 0}$ satisfying

1. $N_0 = 0$;
2. $(N_t)_{t \geq 0}$ has independent increments;
3. $N_t - N_s \sim \text{Poisson}\left(\int_s^t \lambda(u) du\right)$ for all $t \geq s \geq 0$.

Construction for bounded intensity

Suppose $\lambda(t) \leq \Lambda$ for all t , and let $(N_t)_{t \geq 0}$ be a Poisson process of rate Λ . For an arrival at time t , “keep” it with probability $\lambda(t)/\Lambda$, independently of all other arrivals. The resulting process of kept arrivals is an inhomogeneous Poisson process with rate function $\lambda(t)$.

Remark 22.15. This is Theorem 22.12 with a time-varying retention probability: the thinning argument never used the fact that p was constant.

Lecture 23

Poisson Processes and CTMCs

23.1 Recap: inhomogeneous Poisson processes

Definition 23.1 (IHPP). A counting process $(N_t)_{t \geq 0}$ is an *inhomogeneous Poisson process* with rate function $\lambda(t)$ if

1. $N_0 = 0$;
2. $(N_t)_{t \geq 0}$ has independent increments;
3. $N_t - N_s \sim \text{Poisson}\left(\int_s^t \lambda(u) du\right)$ for all $0 \leq s \leq t$.

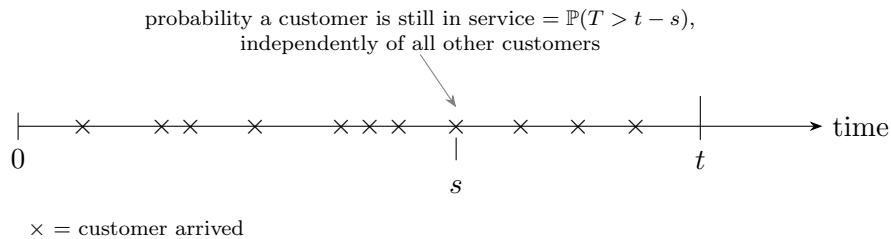
Remark 23.2 (Thinning). One can construct an IHPP with rate function $\lambda(t) = p(t)\Lambda$, where $p: (0, \infty) \rightarrow [0, 1]$, by *thinning* a rate- Λ Poisson process: accept an arrival at time t with probability $p(t)$, independently of everything else.

23.2 Application: the $M/G/\infty$ queue

Setting. The notation reads: M for memoryless arrivals, G for a general service time distribution, ∞ for the number of servers.

Customers arrive at a service counter according to a rate- λ Poisson process. There are infinitely many cashiers, so each customer instantly enters service upon arrival and spends time $T \sim G$ in service, independently of all other customers.

Q. If X_t is the number of people in the system at time t , what is the distribution of X_t ?



Idea. Each customer i from the λ -Poisson process is still in service at time t with probability $\mathbb{P}(T > t - s_i)$, where s_i is their arrival time. So let $(N_s^p)_{0 \leq s \leq t}$ be the IHPP obtained by accepting an arrival at time s with probability $p_G(T > t - s)$: this is exactly the process counting customers still in service at time t .

By Remark 23.2,

$$X_t \sim \text{Poisson}\left(\lambda \int_0^t \mathbb{P}(T > t - s) ds\right) = \text{Poisson}\left(\lambda \int_0^t \mathbb{P}(T > s) ds\right),$$

and therefore, as $t \rightarrow \infty$,

$$X_t \xrightarrow{d} \text{Poisson}(\lambda \mathbb{E}_G(T)).$$

The steady-state occupancy depends on the service distribution only through its mean — this is Little's law in disguise.

23.3 Conditioning on the number of arrivals

Let $(N_t)_{t \geq 0}$ be a homogeneous Poisson process with rate λ , and suppose $\{N_t = n\}$.

Q. Given $\{N_t = n\}$, what is the joint distribution of the arrival times $(T_1, \dots, T_n)$?

A. (Somewhat surprising.) The arrivals look like n *i.i.d.* $\text{Unif}([0, t])$ random variables, sorted.

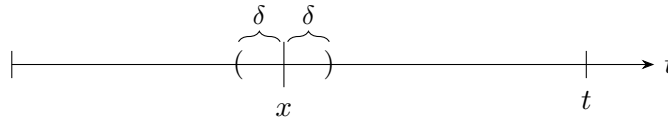
Definition 23.3 (Order statistics). For random variables $(X_i)_{i=1}^n$, the sorted values

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

are called the *order statistics* of $(X_1, \dots, X_n)$.

Theorem 23.4. Conditioned on $\{N_t = n\}$, the vector $(T_1, \dots, T_n)$ is equal in distribution to the order statistics of n *i.i.d.* $\text{Unif}([0, t])$ random variables.

Example 23.5 (Distance to the nearest arrival). Given a fixed point $x \in (0, t)$ and $\{N_t = n\}$, what is the distance from x to the nearest arrival?



By Theorem 23.4, each of the n arrivals independently misses the window of width 2δ with probability $(t - 2\delta)/t$, so

$$\Pr(\text{dist. to nearest arrival} > \delta) = \left(\frac{t - 2\delta}{t}\right)^n \approx e^{-2\delta n/t} \quad \text{for } t \gg 1, \delta = O(1).$$

That is, the distance is approximately $\text{Exp}(2n/t)$.

Example 23.6 (Paying for arrivals). Suppose $f: [0, \infty] \rightarrow [0, \infty)$ is measurable and we pay $\$f(t)$ for an arrival at time t . If arrivals follow a $\text{Poisson}(\lambda)$ process, what do we pay on average? Take $f(s) = 0$ for all $s > t$. Then

$$\begin{aligned} \mathbb{E} \left(\sum_{n \geq 1} f(T_n) \right) &= \mathbb{E} \left(\mathbb{E} \left(\sum_{n \geq 1} f(T_n) \mid N_t \right) \right) \\ &= \mathbb{E} \left(\sum_{n=1}^{N_t} \mathbb{E}(f(T_n) \mid N_t) \right) \\ &= \mathbb{E} \left(\sum_{n=1}^{N_t} \mathbb{E}(f(U)) \right), \quad U \sim \text{Unif}([0, t]) \\ &= \mathbb{E} \left(N_t \cdot \frac{1}{t} \int_0^t f(s) ds \right) = \lambda \int_0^t f(s) ds. \end{aligned}$$

Letting $t \rightarrow \infty$,

$$\mathbb{E} \left(\sum_{n \geq 1} f(T_n) \right) = \lambda \int_0^\infty f(s) ds.$$

END OF POISSON PROCESSES :)

23.4 Continuous-time Markov chains

Let S be a countable state space (usually $\mathbb{N}$).

Definition 23.7 (CTMC). A process $(X_t)_{t \geq 0}$ is a (temporally homogeneous) *continuous-time Markov chain* if, for every initial starting state:

1. the sample paths $t \mapsto X_t(\omega)$, $t \geq 0$, are a.s. right-continuous;
2. for all $0 \leq t_1 < t_2 < \dots < t_k < t \leq s$, all $k \in \mathbb{N}$, and all states $i_0, i_1, \dots, i_k, i, j \in S$,

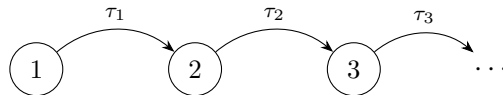
$$\mathbb{P}(X_s = j \mid X_t = i, X_0 = i_0, \dots, X_{t_k} = i_k) = \mathbb{P}(X_s = j \mid X_t = i).$$

Example 23.8. A rate- λ Poisson process $(N_t)_{t \geq 0}$ is a CTMC, by the memoryless nature of the process.

Explosions

Some strange things can happen with CTMCs that cannot happen with discrete-time chains.

Example 23.9 (An explosive chain). Consider $(X_t)_{t \geq 0}$ taking values in $\{1, 2, \dots\}$, and assume the process spends time τ_j in state j before advancing to state $j + 1$, where $\tau_j \sim \text{Exp}(j^2)$.



If T_n is the time of the n th transition, then

$$\mathbb{E}(T_n) = \mathbb{E}\left(\sum_{j=1}^n \tau_j\right) = \sum_{j=1}^n \frac{1}{j^2} < 2.$$

By Markov's inequality $\mathbb{P}(T_n < 4) \geq \frac{1}{2}$ for every n , and since the events $\{T_n \leq 4\}$ are decreasing in n ,

$$\mathbb{P}\left(\sup_{n \geq 1} T_n < 4\right) \geq \frac{1}{2}.$$

The left-hand side is the probability of infinitely many jumps before $t = 4$: the chain runs through all of its states in finite time.

The standard way to deal with this is to introduce a new state Δ and set $X_t(\omega) = \Delta$ if the chain exploded before time t on the sample path $X_t(\omega)$.

Transition probabilities

Definition 23.10.

$$P_{ij}^t := \mathbb{P}(X_t = j \mid X_0 = i), \quad \forall t \geq 0,$$

collected into

$$P^t := \begin{bmatrix} P_{00}^t & P_{01}^t & \cdots \\ P_{10}^t & P_{11}^t & \\ \vdots & & \ddots \end{bmatrix}.$$

Remark 23.11. Here t is an *index*, not an exponent — unlike the DTMC case, where P^n really is the n th matrix power.

Theorem 23.12 (Markov semigroup). *The transition probabilities $(P^t)_{t \geq 0}$ form a Markov semigroup:*

$$P^{s+t} = P^s P^t, \quad s, t \geq 0, \quad \text{and} \quad \lim_{t \downarrow 0} P^t = I.$$

Definition 23.13 (Absorbing state). A state i is *absorbing* if $P_{ii}^t = 1$ for all $t \geq 0$; otherwise it is non-absorbing.

Theorem 23.14 (Holding times). *Let $(X_t)_{t \geq 0}$ be a CTMC starting in a non-absorbing state $X_0 = i$. The holding time*

$$T = \inf\{t \geq 0 : X_t \neq i\}$$

is exponentially distributed, $T \sim \text{Exp}(\lambda_i)$, where λ_i satisfies

$$P_{ii}^h = 1 - h\lambda_i + o(h).$$

Moreover the next state X_T is independent of T , with distribution

$$P_{ij} := \mathbb{P}(X_T = j \mid X_0 = i) = \lim_{h \downarrow 0} \frac{P_{ij}^h}{1 - P_{ii}^h}, \quad j \neq i.$$

So a CTMC decomposes into two independent pieces: exponential holding times governed by the rates λ_i , and an embedded discrete-time jump chain with transition matrix (P_{ij}) .

Lecture 24

Generators and the Kolmogorov Equations

24.1 The constructive description of a CTMC

Theorem 24.1 (Holding times). *Let $(X_t)_{t \geq 0}$ be a CTMC starting in a non-absorbing state $X_0 = i$. The holding time*

$$T = \inf\{t \geq 0 : X_t \neq i\}$$

has distribution $T \sim \text{Exp}(\lambda_i)$, where λ_i satisfies

$$P_{ii}^h = 1 - h\lambda_i + o(h).$$

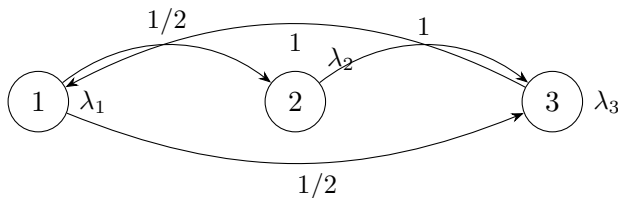
Moreover the next state X_T is independent of T , with distribution

$$P_{ij} := \mathbb{P}\{X_T = j \mid X_0 = i\} = \lim_{h \downarrow 0} \frac{P_{ij}^h}{1 - P_{ii}^h}, \quad j \neq i \quad (P_{ii} = 0 \text{ by definition}).$$

Idea. A CTMC starting in state i holds for $T \sim \text{Exp}(\lambda_i)$ time before jumping to state j with probability P_{ij} .

Theorem 24.2 (Strong Markov property at a jump). *Let $(X_t)_{t \geq 0}$ and T be as above. Conditioned on T and on $X_T = j$, the process $(X_{t+T})_{t \geq 0}$ is a CTMC starting in state j with transition probabilities $(P^t)_{t \geq 0}$.*

Conclusion (constructive definition of a CTMC, up to explosions). Each state i has an associated rate λ_i , and there is a DTMC on the same space with transition probabilities $(P_{ij})_{i,j \in S}$, where $P_{ii} = 0$ for all $i \in S$ — the *embedded* or *jump* chain. The CTMC makes transitions according to the jump chain, and holds in state i between transitions for $\text{Exp}(\lambda_i)$ time, independent of everything else.



Remark 24.3. Exponential timers, and jumps according to the jump chain. Altogether, one can characterize a CTMC — up to the time of explosion, i.e. the minimal construction — entirely by the parameters $(\lambda_i)_{i \in S}$ and $(P_{ij})_{i,j \in S}$.

24.2 The infinitesimal generator

Definition 24.4 (Infinitesimal generator). The *infinitesimal generator* of a CTMC is the matrix Q with entries

$$q_{ij} = [Q]_{ij} := \begin{cases} \lambda_i P_{ij}, & i \neq j, \\ -\lambda_i, & i = j. \end{cases}$$

Q. Why “infinitesimal generator”?

A. Because it is exactly the first-order behaviour of the semigroup:

$$P_{ii}^h = 1 - h\lambda_i + o(h), \quad P_{ij}^h = hq_{ij} + o(h) \quad (i \neq j).$$

In matrix form, $P^h = I + hQ + o(h)$.

Theorem 24.5 (Kolmogorov equations). *The map $t \mapsto P^t$ is continuously differentiable on $[0, \infty)$ and solves*

$$\frac{d}{dt} P^t = Q P^t \quad (\text{Kolmogorov backward equation}).$$

Moreover, if $\sup_i \lambda_i < \infty$, then

$$\frac{d}{dt} P^t = P^t Q \quad (\text{Kolmogorov forward equation}),$$

and $P^t = e^{tQ}$ is the unique solution.

Proof. For the forward equation, use the semigroup property with P^h on the right:

$$P^{t+h} = P^t P^h = P^t (I + hQ + o(h)) = P^t + hP^t Q + o(h),$$

so

$$\frac{P^{t+h} - P^t}{h} = \frac{hP^t Q + o(h)}{h} \xrightarrow{h \downarrow 0} \frac{d}{dt} P^t = P^t Q.$$

This step is where $\sup_i \lambda_i < \infty$ is needed: pushing P^t through the $o(h)$ requires that error term to be *uniformly* controllable across the (possibly infinitely many) states.

For the backward equation, put P^h on the left instead:

$$P^{t+h} = P^h P^t = (I + hQ + o(h)) P^t = P^t + hQ P^t + o(h) \implies \frac{d}{dt} P^t = Q P^t,$$

and here no uniformity is needed, since P^t has entries in $[0, 1]$ with rows summing to one. This is why the backward equation holds in general and the forward equation does not. $\square$

Remark 24.6 (Placement of the boundedness hypothesis). In the handwritten notes the remark “if $\sup_i \lambda_i < \infty$, $o(h)$ is uniformly controllable” sits between the two computations. It belongs with the *first* one — the forward equation — which is the one the theorem statement guards with that hypothesis.

24.3 The modeling problem

Given a description of a system, form the Q matrix.

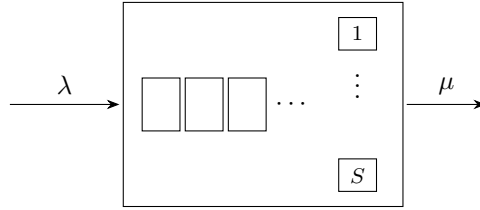
Example 24.7 (Poisson process). For a rate- λ Poisson process $(N_t)_{t \geq 0}$,

$$q_{ii} = -\lambda_i = -\lambda, \quad q_{i,i+1} = \lambda.$$

Example 24.8 ($M/M/S$ queue). Memoryless arrivals (rate λ), memoryless service distribution (rate μ), and S servers. The state space is $S = \mathbb{N}$, and

$$q_{i,i+1} = \lambda, \quad q_{i,i-1} = (i \wedge S)\mu.$$

The $i \wedge S$ is the number of servers actually busy: once the queue is longer than S , adding customers no longer speeds up departures.



Example 24.9 ($M/M/S$ with balking). As before, except that a customer who sees a long line may simply leave. If there are n people in the system, a new arrival joins with probability a_n . Everything is as before except

$$q_{i,i+1} = \lambda a_i,$$

which is a thinned Poisson process: keep each arrival with probability a_i .

Example 24.10 (Birth–death process). Individuals are born at rate λ and die at rate μ , per individual. With $S = \mathbb{N}$,

$$q_{i,i+1} = i\lambda, \quad q_{i,i-1} = i\mu.$$

Remark 24.11. For $i \geq 1$ the jump chain has transition probabilities

$$P_{i,i+1} = \frac{i\lambda}{i\lambda + i\mu} = \frac{\lambda}{\lambda + \mu}, \quad P_{i,i-1} = \frac{\mu}{\lambda + \mu},$$

independent of i — the population size affects how *fast* the chain moves, not *where* it goes. State 0 is absorbing, since $q_{0,1} = q_{0,-1} = 0$.

24.4 CTMC limit theorems

Definition 24.12. A CTMC with at least two states is *irreducible* if its embedded chain is irreducible.

Definition 24.13 (Stationary distribution). A probability vector ρ is a *stationary distribution* for a CTMC if $\rho = \rho P^t$ for all $t \geq 0$.

Remark 24.14. Unlike the discrete-time case, we cannot just check $\rho = \rho P$ for a single one-step matrix — the condition is required for every t simultaneously. This is what makes the generator characterization below so useful.

Theorem 24.15. *A probability vector ρ satisfying $\sum_i \rho_i \lambda_i < \infty$ is a stationary distribution if and only if $\rho Q = 0$. Moreover, if the chain is irreducible, ρ is unique.*

Proof. By the backward equation and dominated convergence (justified by $\sum_i \lambda_i \rho_i < \infty$),

$$\frac{d}{dt} \rho P^t = \rho Q P^t.$$

($\Leftarrow$) If $\rho Q = 0$ then $\frac{d}{dt} \rho P^t = 0$ for all $t \geq 0$. But $\rho P^0 = \rho$, so $\rho P^t = \rho$ for all $t \geq 0$.

($\Rightarrow$) If $\rho P^t = \rho$ for all t , then $\frac{d}{dt} \rho P^t = 0$, so $\rho Q P^t = 0$ for all $t \geq 0$. Evaluating at $t = 0$ gives $\rho Q = \rho Q P^0 = 0$. $\square$

Lecture 25

CTMC Limit Theorems and Reversibility

Recall. For a CTMC with semigroup $(P^t)_{t \geq 0}$ and generator Q :

1. ρ is a *stationary distribution* (SD) if $\rho = \rho P^t$ for all $t \geq 0$;
2. if $\sum_i \rho_i \lambda_i < \infty$, then ρ is an SD $\iff \rho Q = 0$.

25.1 Stationary distributions and the embedded chain

Theorem 25.1. *The following are equivalent.*

1. There is an SD ρ for the CTMC with $\sum_i \rho_i \lambda_i < \infty$.
2. There is an SD π for the jump chain with $\sum_i \pi_i / \lambda_i < \infty$.

If either holds, then $\pi_k \propto \rho_k \lambda_k$ for all $k \in S$.

Proof. (1 $\Rightarrow$ 2). Statement 1 gives $\rho Q = 0$, i.e. for every j ,

$$\sum_i \rho_i q_{ij} = 0 \iff \sum_{i \neq j} \rho_i \lambda_i P_{ij} = \underbrace{\rho_j \lambda_j}_{=: \alpha^{-1} \pi_j}, \quad \text{where } \alpha = \frac{1}{\sum_i \rho_i \lambda_i}.$$

Multiplying through by α ,

$$\sum_{i \neq j} \alpha \pi_i P_{ij} = \alpha \pi_j \implies \pi P^{\text{emb}} = \pi. \quad \square$$

25.2 Long-term behavior

Definition 25.2 (Return time).

$$T_j = \inf\{t \geq 0 : X_t = j, \exists s < t : X_s \neq j\},$$

the first return to j after leaving it. The mean return time is $m_j = \mathbb{E}(T_j \mid X_0 = j)$.

Transience and recurrence are the same as before: j is transient if the chain returns finitely often with probability 1, and recurrent if it returns infinitely often with probability 1. A recurrent state j is

$$\begin{cases} \text{null recurrent} & \text{if } m_j = \infty, \\ \text{positive recurrent} & \text{if } m_j < \infty. \end{cases}$$

All of these are class properties.

Theorem 25.3. *An irreducible CTMC satisfies exactly one of the following.*

1. *All states are transient, or all are null recurrent; no SD exists.*
2. *All states are positive recurrent. An SD exists, is unique, and satisfies*

$$\rho_j = \frac{1}{m_j \lambda_j} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t P_{ij}^s ds \quad \forall i, j \in S.$$

Remark 25.4. The factor λ_j is what distinguishes this from the discrete-time statement $\pi_j = 1/\mu_{jj}$. In continuous time we care not only about how often the chain returns to j but about how long it stays once it gets there, and the mean holding time is $1/\lambda_j$.

Corollary 25.5. *If $\sum_j \rho_j \lambda_j = \sum_j 1/m_j < \infty$, then the embedded chain is also positive recurrent, with*

$$\pi_j = \frac{\rho_j \lambda_j}{\sum_i \rho_i \lambda_i}, \quad m_j = \frac{\sum_i \pi_i / \lambda_i}{\pi_j}.$$

Theorem 25.6 (Ergodic theorem). *Let the CTMC be irreducible with SD ρ , and let $r: S \rightarrow \mathbb{R}$ be bounded. Then*

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t r(X_s) ds = \mathbb{E}_{X \sim \rho}(r(X)).$$

Theorem 25.7 (Convergence to equilibrium). *Let $(P^t)_{t \geq 0}$ be irreducible and admit an SD ρ . Then*

$$\|P_{i\cdot}^t - \rho\|_{\text{TV}} \rightarrow 0 \quad \text{as } t \rightarrow \infty, \quad \forall i \in S.$$

Remark 25.8. Unlike the discrete-time case, no aperiodicity hypothesis is needed: the exponential holding times smear out any would-be periodicity.

25.3 Reversibility

Definition 25.9 (Reversible). A CTMC with generator Q is *reversible* if there exists a distribution ρ with $\sum_i \rho_i \lambda_i < \infty$ and

$$\rho_i q_{ij} = \rho_j q_{ji} \quad \forall i, j \in S. \quad (\text{DBE})$$

Such a ρ is automatically an SD, and we say (Q, ρ) is *reversible*.

Remark 25.10. Detailed balance passes to the jump chain:

$$\left. \begin{aligned} \rho_i q_{ij} &= \rho_i \lambda_i P_{ij} = \left(\sum_k \rho_k \lambda_k \right) \pi_i P_{ij} \\ \rho_j q_{ji} &= \rho_j \lambda_j P_{ji} = \left(\sum_k \rho_k \lambda_k \right) \pi_j P_{ji} \end{aligned} \right\} \implies \pi_i P_{ij} = \pi_j P_{ji}.$$

Remark 25.11 (Why “reversible”). If $X_0 \sim \rho$, then

$$(X_t)_{t=0}^T \stackrel{d}{=} (X_{T-t})_{t=0}^T :$$

the process run backwards has the same law as the process run forwards.

Example 25.12 (Birth–death process with immigration). Take

$$\left. \begin{aligned} q_{n,n+1} &= \lambda n + \gamma \\ q_{n,n-1} &= \mu n \end{aligned} \right\} n \geq 0, \quad \begin{aligned} \lambda &= \text{birth rate per individual,} \\ \gamma &= \text{immigration rate,} \\ \mu &= \text{death rate per individual.} \end{aligned}$$

The detailed balance equations $\rho_n q_{n,n+1} = \rho_{n+1} q_{n+1,n}$ for $n \geq 0$ read

$$\rho_n (\lambda n + \gamma) = \rho_{n+1} \mu (n+1) \quad \implies \quad \rho_{n+1} = \rho_n \left(\frac{\lambda n + \gamma}{\mu (n+1)} \right),$$

which telescopes to

$$\rho_{n+1} = \rho_0 \frac{\gamma}{\lambda} \frac{1}{n+1} \left(\frac{\lambda}{\mu} \right)^{n+1} \prod_{k=1}^n \left(1 + \frac{\gamma}{k\lambda} \right).$$

If $\gamma > 0$ and $\lambda < \mu$, the right-hand side is summable over n . Indeed

$$\prod_{k=1}^n \left(1 + \frac{\gamma}{k\lambda} \right) \leq \exp \left(\frac{\gamma}{\lambda} \sum_{k=1}^n \frac{1}{k} \right) \leq \exp \left(\frac{\gamma}{\lambda} (1 + \log n) \right) = e^{\gamma/\lambda} n^{\gamma/\lambda},$$

so

$$\rho_{n+1} \leq \rho_0 e^{\gamma/\lambda} \left(\frac{\gamma}{\lambda(n+1)} \right) \left(\frac{\lambda}{\mu} \right)^{n+1} (n+1)^{\gamma/\lambda},$$

which is summable over all n : the geometric factor $(\lambda/\mu)^{n+1} < 1$ beats the polynomial growth $(n+1)^{\gamma/\lambda}$.

Example 25.13 ($M/M/\infty$ queue). Arrival rate γ , service rate μ . This is the previous example with $\lambda = 0$:

$$\rho_{n+1} = \rho_n \frac{\gamma}{(n+1)\mu} \quad \forall n \geq 0 \quad \implies \quad \rho_n = e^{-\gamma/\mu} \frac{(\gamma/\mu)^n}{n!} \quad \implies \quad \rho \sim \text{Poisson}(\gamma/\mu).$$

25.4 Spectral gap

Write the law of the chain at time t as $\rho_t = \rho_0 e^{tQ}$. By reversibility Q is self-adjoint on $L^2(\rho)$, so it has a spectral decomposition

$$Q = \sum_i \theta_i u_i u_i^\top, \quad \text{giving} \quad \rho_t = \sum_i e^{t\theta_i} \langle \rho_0, u_i \rangle u_i^\top.$$

Remark 25.14 (Notation). The eigenvalues θ_i here are non-positive and have nothing to do with the jump rates λ_i . The largest is $\theta_0 = 0$, whose eigenvector recovers the stationary distribution; everything else decays exponentially.

Definition 25.15 (Dirichlet form).

$$\mathcal{E}(f, f) := \frac{1}{2} \sum_{i,j} |f(i) - f(j)|^2 q_{ij} \rho_i.$$

Definition 25.16 (Spectral gap). The *spectral gap* λ is the best constant in the inequality

$$\text{Var}_\rho(f) \leq \frac{1}{\lambda} \mathcal{E}(f, f). \quad (\text{Poincaré inequality})$$

Remark 25.17. λ is the modulus of the smallest nonzero eigenvalue of Q , which is self-adjoint on $L^2(\rho)$ by reversibility.

Theorem 25.18. *If a reversible CTMC (Q, ρ) admits spectral gap λ , then*

$$\|P_i^t - \rho\|_{\text{TV}}^2 \leq \frac{e^{-2\lambda t}}{\rho_i} \quad \forall i \in S, t \geq 0.$$

Remark 25.19. Theorem 25.7 says the chain converges; this says *how fast*, with a rate that is entirely a spectral quantity. The $1/\rho_i$ prefactor is the price of starting from a deterministic state: the more improbable the starting state under ρ , the longer the burn-in.

Lecture 26

Binary Hypothesis Testing

26.1 Setup

Probability: given a model $(\Omega, \mathcal{F}, \mathbb{P})$, what does the data look like?

Statistics: the other direction — given the data, which model best fits?

Given a measurable space $(\Omega, \mathcal{F})$, suppose someone reveals a sample $\omega \in \Omega$. How do we decide whether

$$\underbrace{(\Omega, \mathcal{F}, P_0)}_{H_0 := \text{null hypothesis}} \quad \text{or} \quad \underbrace{(\Omega, \mathcal{F}, P_1)}_{H_1 := \text{alternative hypothesis}}$$

is the better model?

Definition 26.1 (Test). A *test* is an $\mathcal{F}$ -measurable function $\hat{H}: \Omega \rightarrow \{H_0, H_1\}$, i.e. $\hat{H}^{-1}(H_0) \in \mathcal{F}$.

The two ways a test can be wrong:

$$P_0\{\hat{H} = H_1\} = \text{under the null, the probability of a mistake} \quad = \text{Type 1 error } (\alpha),$$

the *false positive* rate;

$$P_1\{\hat{H} = H_0\} = \text{under the alternative, likewise} \quad = \text{Type 2 error } (\beta),$$

the *false negative* rate.

Definition 26.2 (Power). The *power* of a test is

$$P_1\{\hat{H} = H_1\} = 1 - P_1\{\hat{H} = H_0\} = 1 - \beta,$$

the true positive rate.

26.2 The likelihood ratio

Assume $P_1 \ll P_0$, i.e. for $A \in \mathcal{F}$, $P_0(A) = 0 \implies P_1(A) = 0$.

Theorem 26.3 (Radon–Nikodym, Ch. 1). *There exists an a.s. unique measurable*

$$\Lambda := \frac{dP_1}{dP_0}$$

which satisfies

$$\mathbb{E}_{P_1}[\mathbf{1}_A] = \mathbb{E}_{P_0}[\Lambda \mathbf{1}_A] \quad \forall A \in \mathcal{F}.$$

$\Lambda(\omega)$ should be thought of as the relative likelihood of observing ω under H_1 versus H_0 .

Example 26.4 (Densities). If P_0, P_1 have densities p_0, p_1 on $\mathbb{R}$, then

$$\Lambda(\omega) = \frac{p_1(\omega)}{p_0(\omega)},$$

since

$$\mathbb{E}_{P_1}[\mathbf{1}_A] = \int_A p_1(x) dx = \int_A \frac{p_1(x)}{p_0(x)} p_0(x) dx = \mathbb{E}_{P_0}[\Lambda(\omega) \mathbf{1}_A].$$

Example 26.5 (Discrete). If P_0, P_1 are discrete distributions with pmfs p_0, p_1 , then again $\Lambda(\omega) = p_1(\omega)/p_0(\omega)$.

Example 26.6 (Shifted Gaussians). If $P_i = \mathcal{N}(\mu_i, \sigma^2)$, then

$$\Lambda(\omega) = \exp\left(\frac{(\omega - \mu_0)^2 - (\omega - \mu_1)^2}{2\sigma^2}\right) = \exp\left(\frac{2\omega(\mu_1 - \mu_0) + \mu_0^2 - \mu_1^2}{2\sigma^2}\right).$$

Note this is monotone in ω : the likelihood ratio test reduces to thresholding the sample itself.

Example 26.7 (An atom under the null). Let

$$P_1 = \mathcal{N}(0, 1), \quad P_0 = \frac{1}{2}\mathcal{N}(0, 1) + \frac{1}{2}\delta_0,$$

so that $P_1 \ll P_0$. Then

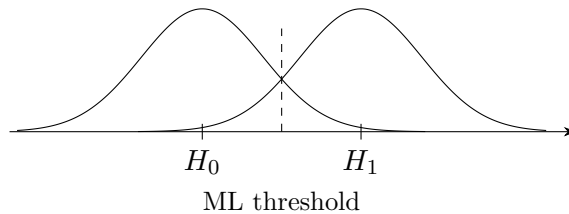
$$\Lambda(\omega) = \begin{cases} 2 & \omega \neq 0, \\ 0 & \omega = 0. \end{cases}$$

26.3 Threshold tests and the error curve

Definition 26.8 (Threshold test). For $\eta \geq 0$, the *threshold test* with threshold η is

$$\hat{H}_\eta(\omega) = \begin{cases} H_1 & \text{if } \Lambda(\omega) \geq \eta, \\ H_0 & \text{otherwise.} \end{cases}$$

Example 26.9 (Maximum likelihood test). Take $\eta = 1$.



Example 26.10 (Maximum a posteriori test). Suppose nature chooses H_0 with probability π_0 and H_1 with probability π_1 . By Bayes' rule,

$$\begin{aligned}\mathbb{P}(H = H_0 \mid \omega) &= \frac{\mathbb{P}(H = H_0, \omega)}{\mathbb{P}(\omega)} = \frac{\overbrace{\mathbb{P}(\omega \mid H = H_0)}^{p_0(\omega)} \overbrace{\mathbb{P}(H = H_0)}^{\pi_0}}{\mathbb{P}(\omega)}, \\ \mathbb{P}(H = H_1 \mid \omega) &= \frac{\overbrace{\mathbb{P}(\omega \mid H = H_1)}^{p_1(\omega)} \overbrace{\mathbb{P}(H = H_1)}^{\pi_1}}{\mathbb{P}(\omega)}.\end{aligned}$$

Comparing the two posteriors,

$$\mathbb{P}(H = H_1 \mid \omega) \geq \mathbb{P}(H = H_0 \mid \omega) \iff p_1(\omega)\pi_1 \geq p_0(\omega)\pi_0 \iff \Lambda(\omega) \geq \frac{\pi_0}{\pi_1}.$$

So the MAP test is a threshold test:

$$\hat{H}_{\text{MAP}}(\omega) = \hat{H}_\eta(\omega) \quad \text{with } \eta = \frac{\pi_0}{\pi_1}.$$

Claim 26.11. *The MAP test minimizes the total probability of error over all tests, assuming the prior π .*

Proof. Consider any test $\hat{H}$. Then

$$\begin{aligned}\mathbb{P}(\hat{H} \text{ errs}) &= \pi_0\alpha + \pi_1\beta \\ &= \pi_0\mathbb{E}_{P_0}[\mathbf{1}_{\{\hat{H}=H_1\}}] + \pi_1\mathbb{E}_{P_1}[\mathbf{1}_{\{\hat{H}=H_0\}}] \\ &= \pi_0\mathbb{E}_{P_0}[1 - \mathbf{1}_{\{\hat{H}=H_0\}}] + \pi_1\mathbb{E}_{P_0}[\Lambda \mathbf{1}_{\{\hat{H}=H_0\}}] \\ &= \pi_0 + \pi_1\mathbb{E}_{P_0}\left[\left(\Lambda - \frac{\pi_0}{\pi_1}\right) \mathbf{1}_{\{\hat{H}=H_0\}}\right].\end{aligned}$$

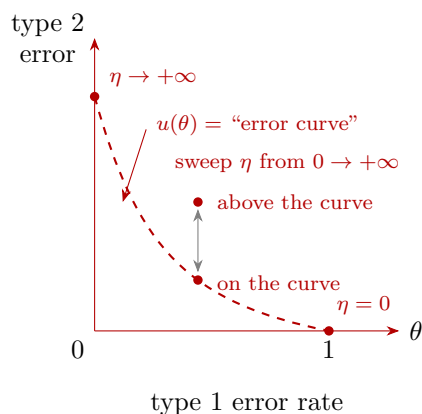
The MAP test sets $\hat{H} = H_0$ exactly on $\{\Lambda < \pi_0/\pi_1\}$, i.e. where the bracket is negative, so pointwise

$$\left(\Lambda - \frac{\pi_0}{\pi_1}\right) \mathbf{1}_{\{\hat{H}=H_0\}} \geq \left(\Lambda - \frac{\pi_0}{\pi_1}\right) \mathbf{1}_{\{\hat{H}_{\text{MAP}}=H_0\}},$$

and taking expectations gives $\mathbb{P}(\hat{H} \text{ errs}) \geq \mathbb{P}(\hat{H}_{\text{MAP}} \text{ errs})$. □

26.4 The error curve

Threshold tests trace out a so-called *error curve*: sweeping η from 0 to $+\infty$ moves the operating point from the bottom right to the top left.



Definition 26.12 (Error curve). The *error curve* $u: [0, 1] \rightarrow [0, 1]$ is defined by

$$u(\theta) = \sup_{\eta \geq 0} \left\{ P_1\{\hat{H}_\eta = H_0\} + \eta(P_0\{\hat{H}_\eta = H_1\} - \theta) \right\}, \quad \theta \in [0, 1].$$

This is always a convex function.

Remark 26.13. $u(\theta)$ is a Legendre transform.

We say a test $\hat{H}$ lies *above* the error curve if

$$P_1\{\hat{H} = H_0\} \geq u(P_0\{\hat{H} = H_1\}),$$

and *on* the error curve if

$$P_1\{\hat{H} = H_0\} = u(P_0\{\hat{H} = H_1\}).$$

Theorem 26.14 (Neyman–Pearson lemma). *All tests $\hat{H}$ lie above the error curve, and all threshold tests $(\hat{H}_\eta)_{\eta \geq 0}$ lie on the error curve.*

In other words, no test can beat the trade-off traced by the likelihood ratio thresholds, and every point on that trade-off is achieved by some threshold test.

Lecture 27

The Neyman–Pearson Lemma and Sufficient Statistics

Recall from Lecture 26. The *threshold test* at level η is

$$\hat{H}_\eta(\omega) = \begin{cases} H_1 & \text{if } \Lambda(\omega) \geq \eta, \\ H_0 & \text{otherwise,} \end{cases}$$

and the MAP test is the threshold test at $\eta = \pi_0/\pi_1$ for a prior π .

Remark 27.1. The MAP test minimizes the probability of error over all tests.

Definition 27.2 (Error curve).

$$u(\theta) = \sup_{\eta \geq 0} \left\{ P_1\{\hat{H}_\eta = H_0\} + \eta(P_0\{\hat{H}_\eta = H_1\} - \theta) \right\}, \quad 0 \leq \theta \leq 1.$$

27.1 The Neyman–Pearson lemma

Theorem 27.3 (Neyman–Pearson). *All tests lie above the error curve, and all threshold tests lie on the error curve.*

Proof. All tests lie above. Fix $\eta \in [0, \infty)$ and choose $\pi_1 \in [0, 1]$ with $\eta = \pi_0/\pi_1$. Take any test $\hat{H}$. The minimum error property of the MAP test (Remark 27.1) says

$$\pi_1 P_1\{\hat{H} = H_0\} + \pi_0 P_0\{\hat{H} = H_1\} \geq \pi_1 P_1\{\hat{H}_\eta = H_0\} + \pi_0 P_0\{\hat{H}_\eta = H_1\}.$$

Dividing through by π_1 and using $\pi_0/\pi_1 = \eta$,

$$P_1\{\hat{H} = H_0\} + \eta P_0\{\hat{H} = H_1\} \geq P_1\{\hat{H}_\eta = H_0\} + \eta P_0\{\hat{H}_\eta = H_1\},$$

that is,

$$\underbrace{P_1\{\hat{H} = H_0\}}_{= \beta(\hat{H})} \geq P_1\{\hat{H}_\eta = H_0\} + \eta(P_0\{\hat{H}_\eta = H_1\} - P_0\{\hat{H} = H_1\}).$$

This holds for every $\eta \geq 0$, so we may take the supremum on the right:

$$\beta(\hat{H}) \geq u(P_0\{\hat{H} = H_1\}) = u(\alpha(\hat{H})). \quad \checkmark$$

All threshold tests lie on. Fix η_0 and put $\theta = P_0\{\hat{H}_{\eta_0} = H_1\}$. Taking $\eta = \eta_0$ in the supremum defining $u(\theta)$ kills the bracketed term, so

$$u(\theta) \geq P_1\{\hat{H}_{\eta_0} = H_0\}.$$

The first part gives the reverse inequality, so the two are equal and the threshold test lies exactly on the error curve. $\square$

27.2 Achievability: the curve is not all reached

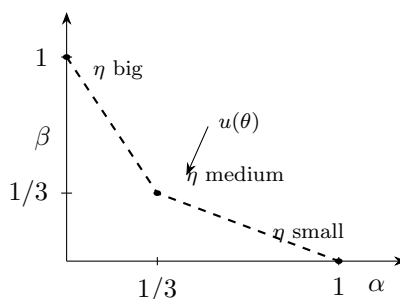
Remark 27.4. Neyman–Pearson does *not* guarantee that every point on the error curve is achieved by a threshold test.

Example 27.5. Let $\Omega = \{0, 1\}$, $P_0 = \text{Bern}(1/3)$, $P_1 = \text{Bern}(2/3)$. Then

$$\Lambda(\omega) = \begin{cases} 2, & \omega = 1, \\ 1/2, & \omega = 0, \end{cases} \quad \implies \quad \hat{H}_\eta(\omega) = \begin{cases} H_1, & \eta \leq 1/2, \\ H_\omega, & 1/2 < \eta \leq 2, \\ H_0, & 2 < \eta. \end{cases}$$

So as η sweeps $[0, \infty)$ the threshold tests realize only *three* points (α, β) :

$$\eta \text{ big} \rightarrow (0, 1), \quad \eta \text{ medium} \rightarrow \left(\frac{1}{3}, \frac{1}{3}\right), \quad \eta \text{ small} \rightarrow (1, 0).$$



A short computation gives the error curve explicitly:

$$u(\theta) = \begin{cases} 1 - 2\theta, & 0 \leq \theta \leq 1/3, \\ (1 - \theta)/2, & 1/3 \leq \theta \leq 1. \end{cases}$$

It is a supremum of finitely many affine functions of θ , hence piecewise linear and convex, and every point on the two open segments is unachieved by any deterministic threshold test. They are achieved by *randomizing* between the adjacent vertices.

Neyman–Pearson lemma, restated. Subject to a type 1 error constraint, randomized threshold tests are most powerful.

27.3 Sufficient statistics

Definition 27.6 (Sufficient statistic). In the context of binary hypothesis testing, a *sufficient statistic* $T = T(\omega)$ is any function of ω such that $\Lambda(\omega)$ can be computed from $T(\omega)$.

Since every threshold test depends on ω only through $\Lambda(\omega)$, a sufficient statistic is all you ever need to keep.

Example 27.7 (Gaussians with a common covariance). Let $P_i = \mathcal{N}(\mu_i, \Sigma)$ with $\mu_i \in \mathbb{R}^n$.

Claim 27.8. $T(\omega) := \omega^\top \Sigma^{-1}(\mu_1 - \mu_0)$ is a sufficient statistic.

Proof. The normalizing constants cancel in the ratio:

$$\Lambda(\omega) = \frac{dP_1(\omega)}{dP_0(\omega)} = \frac{\exp\left(-\frac{(\omega - \mu_1)^\top \Sigma^{-1}(\omega - \mu_1)}{2}\right)}{\exp\left(-\frac{(\omega - \mu_0)^\top \Sigma^{-1}(\omega - \mu_0)}{2}\right)}.$$

Expanding both quadratic forms, the $\omega^\top \Sigma^{-1} \omega$ terms cancel against each other:

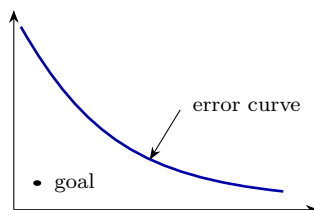
$$\begin{aligned} \Lambda(\omega) &= \exp\left(-\cancel{\frac{\omega^\top \Sigma^{-1} \omega}{2}} + \omega^\top \Sigma^{-1} \mu_1 - \frac{\mu_1^\top \Sigma^{-1} \mu_1}{2} \right. \\ &\quad \left. + \cancel{\frac{\omega^\top \Sigma^{-1} \omega}{2}} - \omega^\top \Sigma^{-1} \mu_0 + \frac{\mu_0^\top \Sigma^{-1} \mu_0}{2}\right) \\ &= \exp\left(\omega^\top \Sigma^{-1}(\mu_1 - \mu_0) + \text{const}\right), \end{aligned}$$

which depends on ω only through $T(\omega)$. □

Remark 27.9. Λ is monotone increasing in T , so the likelihood ratio test reduces to thresholding the scalar $T(\omega)$ — an n -dimensional observation collapses to one number.

27.4 Looking ahead: sequential testing

The picture so far is that the error curve is a hard floor: a target pair $(\varepsilon_1, \varepsilon_2)$ sitting below it cannot be achieved by *any* test on a fixed sample. The way out is to stop insisting on a fixed sample and instead take more and more samples, deciding when to stop as the data arrive.

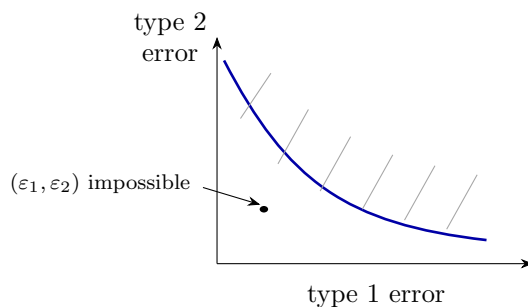


That is the subject of Lecture 28.

Lecture 28

Sequential Analysis

Last time: any test must have type 1 and type 2 error rates lying on or above the error curve.



Q. What if we have target type 1 and type 2 error rates $(\varepsilon_1, \varepsilon_2)$ that lie *below* the curve?

A. Get more data!

28.1 Setup

Let $(X_n)_{n \geq 1}$ be *i.i.d.* Q , and test

$$H_0 : Q = P_0 \quad \text{against} \quad H_1 : Q = P_1.$$

We work on $(\Omega, \mathcal{F}, Q)^{\otimes \mathbb{N}}$, so that $\omega \in \Omega$ is a sequence $(\omega_1, \omega_2, \dots)$ and $X_n(\omega) = \omega_n$. Let $(\hat{H}_n)_{n \geq 1}$ be an adapted sequence of tests, i.e. $\hat{H}_n$ is a function of $X_1, \dots, X_n$ for every $n \geq 1$.

Definition 28.1 (Sequential test). Let T be a stopping time. The *sequential test* is $\hat{H}_T$, with type 1 and type 2 error rates

$$\alpha = P_{H_0}\{\hat{H}_T = H_1\}, \quad \beta = P_{H_1}\{\hat{H}_T = H_0\}.$$

Remark 28.2. We can always assume $\alpha + \beta \leq 1$: otherwise flipping the test's output does better.

Goal. Find the minimum number of samples on average required to achieve $(\alpha, \beta) \leq (\varepsilon_1, \varepsilon_2)$.

28.2 Relative entropy

Definition 28.3 (Relative entropy). For $\mu \ll \nu$ with $\Lambda = \frac{d\mu}{d\nu}$, define

$$D(\mu \parallel \nu) := \mathbb{E}_\mu(\log \Lambda) = \mathbb{E}_\nu[\Lambda \log \Lambda],$$

and set $D(\mu \parallel \nu) = +\infty$ if $\mu \not\ll \nu$.

Remark 28.4. $x \log x$ is convex, so by Jensen

$$\mathbb{E}_\nu[\Lambda \log \Lambda] \geq \underbrace{\mathbb{E}_\nu[\Lambda]}_{=1} \log \mathbb{E}_\nu[\Lambda] = 0.$$

Hence $D(\mu \parallel \nu) \geq 0$, with equality if and only if $\mu = \nu$.

Notation. For $p, q \in [0, 1]$ we abbreviate the binary relative entropy

$$D(p \parallel q) \equiv D(\text{Bern}(p) \parallel \text{Bern}(q)) = p \log\left(\frac{p}{q}\right) + (1-p) \log\left(\frac{1-p}{1-q}\right).$$

Note $D(p \parallel q) = D(1-p \parallel 1-q)$.

28.3 A lower bound on the sample size

Theorem 28.5. For every sequential test with type 1 and type 2 error rates satisfying $\alpha + \beta \leq 1$,

$$\mathbb{E}_{H_0}(T) \geq \frac{D(\alpha \parallel 1-\beta)}{D(P_0 \parallel P_1)}, \quad \mathbb{E}_{H_1}(T) \geq \frac{D(1-\beta \parallel \alpha)}{D(P_1 \parallel P_0)}. \quad (\star)$$

Proof sketch. By symmetry it is enough to prove $(\star)$. Write, for each n ,

$$\begin{aligned} \textcircled{A_n} &:= P_{H_1}\{\hat{H}_n = H_1 \mid T = n\} \log\left(\frac{P_{H_1}\{\hat{H}_n = H_1 \mid T = n\}}{P_{H_0}\{\hat{H}_n = H_1 \mid T = n\}}\right) \\ &\quad + P_{H_1}\{\hat{H}_n = H_0 \mid T = n\} \log\left(\frac{P_{H_1}\{\hat{H}_n = H_0 \mid T = n\}}{P_{H_0}\{\hat{H}_n = H_0 \mid T = n\}}\right), \end{aligned}$$

and let $\textcircled{B_n}$ be the same quantity with $\hat{H}_n$ replaced by $\hat{H}_T$. Then $\textcircled{A_n} \geq \textcircled{B_n}$, since conditioning on $\{T = n\}$ and passing from $\hat{H}_n$ to $\hat{H}_T$ can only coarsen the partition, and relative entropy decreases under coarsening. With some effort,

$$\mathbb{E}_{H_1}\left(\sum_{i=1}^T \log \Lambda(X_i) \mid T = n\right) \geq \textcircled{A_n} \geq \textcircled{B_n}.$$

Averaging over n and using convexity of relative entropy,

$$\begin{aligned} \mathbb{E}_{H_1}\left(\sum_{i=1}^T \log \Lambda(X_i)\right) &\geq \sum_{n \geq 0} P_{H_1}(T = n) \textcircled{B_n} \\ &\geq (1-\beta) \log\left(\frac{1-\beta}{\alpha}\right) + \beta \log\left(\frac{\beta}{1-\alpha}\right) = D(1-\beta \parallel \alpha). \end{aligned}$$

The left-hand side is handled by Wald's identity:

$$\mathbb{E}_{H_1}\left(\sum_{i=1}^T \log \Lambda(X_i)\right) = \mathbb{E}_{H_1}(T) \underbrace{\mathbb{E}_{H_1}(\log \Lambda(X_1))}_{=D(P_1 \parallel P_0)}.$$

□

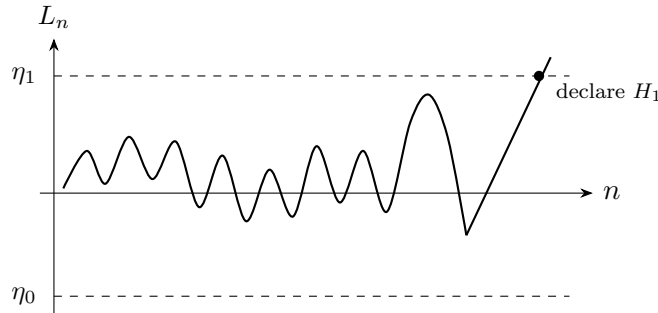
28.4 The sequential probability ratio test

Definition 28.6 (SPRT). For thresholds η_0, η_1 , define the stopping time

$$T = \inf\{n \geq 1 : L_n \notin (\eta_0, \eta_1)\}, \quad L_n := \prod_{i=1}^n \Lambda(X_i),$$

and the test

$$\hat{H}_T = \begin{cases} H_0 & \text{if } L_T \leq \eta_0, \\ H_1 & \text{if } L_T \geq \eta_1. \end{cases}$$



Q. How do we choose the thresholds?

Proposition 28.7. For $0 \leq \eta_0 < \eta_1$, the error rates (α, β) of the SPRT satisfy

$$\frac{\alpha}{1 - \beta} \leq \eta_1^{-1}, \quad \frac{\beta}{1 - \alpha} \leq \eta_0.$$

Example 28.8. Choosing $\eta_1^{-1} = \varepsilon_1$ and $\eta_0 = \varepsilon_2$ ensures $\alpha \leq \varepsilon_1$ and $\beta \leq \varepsilon_2$.

Proof of Proposition 28.7. Take $\hat{H}_T$ to be the SPRT and change measure using the likelihood ratio:

$$\begin{aligned} \beta &= P_{H_1}\{\hat{H}_T = H_0\} = \mathbb{E}_{H_1}[\mathbf{1}_{\{\hat{H}_T = H_0\}}] = \mathbb{E}_{H_0}(L_T \mathbf{1}_{\{\hat{H}_T = H_0\}}) \\ &= \sum_{n \geq 1} P_{H_0}(T = n) \mathbb{E}_{H_0}(L_n \mathbf{1}_{\{\hat{H}_T = H_0\}} \mid T = n) \\ &\leq \sum_{n \geq 1} P_{H_0}(T = n) \eta_0 \mathbb{E}_{H_0}(\mathbf{1}_{\{\hat{H}_T = H_0\}} \mid T = n) = \eta_0(1 - \alpha), \end{aligned}$$

the inequality because $L_T \leq \eta_0$ on the event $\{\hat{H}_T = H_0\}$. The bound on α is symmetric. $\square$

28.5 Near-optimality of the SPRT

Claim 28.9. The SPRT is “nearly” optimal — qualitatively, not quantitatively. Let the SPRT have error rates α, β . If $D(P_0 \parallel P_1), D(P_1 \parallel P_0) < \infty$, then

$$\mathbb{E}_{H_0}[T] \approx \frac{D(\alpha \parallel 1 - \beta)}{D(P_0 \parallel P_1)}, \quad \mathbb{E}_{H_1}[T] \approx \frac{D(\beta \parallel 1 - \alpha)}{D(P_1 \parallel P_0)}.$$

Proof sketch. Take $\hat{H}_T$ to be the SPRT; one can show $\mathbb{E}_{H_1}[T] < \infty$ (this requires an argument). Wald's identity then gives

$$\mathbb{E}_{H_1}(\log L_T) = \mathbb{E}_{H_1}(T) \mathbb{E}_{P_1}(\log \Lambda) = \mathbb{E}_{H_1}(T) D(P_1 \| P_0).$$

On the other hand, splitting on which way the test decides,

$$\begin{aligned} \beta \mathbb{E}_{H_1}(\log L_T | \hat{H}_T = H_0) &+ (1 - \beta) \mathbb{E}_{H_1}(\log L_T | \hat{H}_T = H_1) \\ \Downarrow &\Downarrow \\ \beta \log \eta_0 &+ (1 - \beta) \log \eta_1 \\ \Downarrow &\Downarrow \\ \beta \log \left(\frac{\beta}{1 - \alpha} \right) &+ (1 - \beta) \log \left(\frac{1 - \beta}{\alpha} \right) = D(\beta \| 1 - \alpha). \end{aligned}$$

The first pair of approximations ignores the *overshoot* past the threshold at the stopping time; the second treats the inequalities of Proposition 28.7 as equalities. Combining with Theorem 28.5,

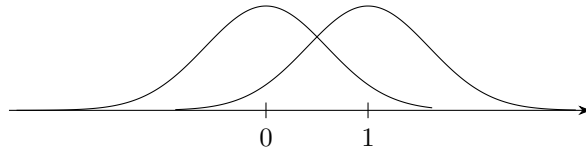
$$\frac{D(\beta \| 1 - \alpha)}{D(P_1 \| P_0)} \leq \mathbb{E}_{H_1}[T] \lesssim \frac{D(\beta \| 1 - \alpha)}{D(P_1 \| P_0)}. \quad \square$$

28.6 Worked example: shifted Gaussians

Example 28.10. Let $(X_n)_{n \geq 1}$ be *i.i.d.* Q with

$$H_0 : Q = \mathcal{N}(0, 1) \quad \text{against} \quad H_1 : Q = \mathcal{N}(1, 1),$$

and suppose we require $\alpha, \beta \leq 0.01$.



Fixed sample size. By sufficiency of the sample mean, the test thresholds $\frac{1}{n} \sum_i X_i$ at $\frac{1}{2}$, giving

$$\beta = \alpha = P_{H_0} \left\{ \frac{1}{n} \sum_i X_i \geq \frac{1}{2} \right\} = Q \left(\frac{\sqrt{n}}{2} \right) \underset{n=22}{\approx} 0.0095,$$

so $n = 22$ samples are required.

Sequential test. Here $D(P_0 \| P_1) = \frac{1}{2}$, so Theorem 28.5 gives

$$\mathbb{E}_{H_0}(T) \geq \frac{D(0.01 \| 0.99)}{D(P_0 \| P_1)} \approx 9 \text{ samples.}$$

Simulation, 100k trials.

- $\eta_0 = \eta_1^{-1} = 0.01 \implies \alpha, \beta \leq 0.01$; the SPRT uses ≈ 10.5 samples on average (and actually achieves $\alpha, \beta = 0.0055$, so the thresholds are conservative).
- $\eta_0 = \eta_1^{-1} = 0.0175 \implies \alpha, \beta \lesssim 0.01$; requires ≈ 9.3 samples on average.

Against 22 for the fixed-sample test, and against the information-theoretic floor of 9. Stopping when you have seen enough is worth better than a factor of two here.